

Yield-Curve Control Policy under Inelastic Financial Markets*

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Abstract

We develop a New Keynesian model that incorporates the term structure of financial markets, emphasizing the role of government and central bank balance sheet composition in monetary policy transmission. Our model accounts for microfounded market segmentation across asset classes and maturities based on finite and estimable asset demand elasticities. We show that unconventional policy interventions, such as large-scale asset purchase programs and yield-curve control policies, effectively stabilize the economy during normal periods and at the zero lower bound, albeit by extending ZLB episodes and reducing the efficacy of future short-term rate adjustments.

Keywords: Term Structure, Yield-Curve Control, Market Segmentation

JEL codes: E32, E43, E44, E52, E58

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Note: The ordering of the authors’ names is not indicative of relative contributions; it was chosen arbitrarily (or by mutual agreement) for practical reasons.

1 Introduction

Since the 2007–2008 Global Financial Crisis, unconventional monetary policies have become mainstream. Constrained by the zero lower bound (ZLB) on short-term rates, policymakers adopted strategies—such as expanding central bank balance sheets and increasing government debt issuance—to lower long-term rates, stimulate aggregate demand, and mitigate recessionary pressures. The COVID-19 pandemic further intensified this dynamic, prompting the Federal Reserve to implement another round of unconventional interventions as the policy rate once again reached the ZLB.¹

This paper develops a tractable New Keynesian framework that incorporates an endogenous term structure of interest rates in bond markets, enabling an analysis of both conventional and unconventional monetary and fiscal policies. In contrast, standard log-linearized models typically include only a single policy rate, neglecting both the term structure of interest rates and heterogeneous asset returns. This limitation arises from the absence of frictions in asset substitution (e.g., asset demands are perfectly elastic), which leads to the equalization of expected returns across assets and maturities.² As a result, additional assets become redundant in monetary policy analysis.

To address these limitations, we develop a quantitative framework that underscores the role of market segmentation and inelastic demand across bonds of varying maturities in shaping the effectiveness of unconventional monetary policy. While a large existing literature incorporates preferred-habitat investors—who specialize in particular maturities—and arbitrageurs who absorb residual risk and help sustain an upward-sloping yield curve (Ray, 2019; Vayanos and Vila, 2021; Droste et al., 2021), we abstract from arbitrageurs entirely. Instead, our model features households that endogenously adjust their portfolio allocations across bond maturities and private loans, subject to finite own- and cross-elasticities of demand, in the spirit of Jansen et al. (2024).³

Given these portfolio allocations, households encounter an effective return on their

¹In March 2020, the Federal Reserve reduced its policy rate to a range of 0%–0.25%. It also pledged to keep interest rates at these low levels until the economy reached full employment and sustained inflation at 2%. At the same time, the unprecedented CARES Act provided nearly \$500 billion in support of the Federal Reserve.

²This outcome also follows from the absence of a price of risk under log-linear approximation, resulting in the well-known *expectations hypothesis*, which posits that long-term bond returns equal the average of expected future short-term rates.

³This asset demand system with finite elasticities, which gives rise to market segmentation, builds on the recent demand system literature (Kojen and Yogo, 2019) and the inelastic markets literature (Gabaix and Kojen, 2022).

aggregate savings, which enters directly into the Euler equation and serves as a sufficient statistic for monetary policy transmission. Balance-sheet policies affect the real economy through this channel, as households rebalance both along the yield curve and across asset classes, thereby adjusting the effective savings rate that governs consumption dynamics. This framework allows us to study how the size and composition of government and central bank balance sheets influence output, inflation, and welfare within a unified and tractable New Keynesian setting.

Within this environment, a demand shock to a particular bond maturity—such as quantitative easing targeting long-term bonds—spills over to other maturities with a positive but incomplete pass-through, a phenomenon referred to as the *localization effect* in [Droste et al. \(2021\)](#). In particular, our framework delivers a positive term premium response to contractionary monetary policy shocks: as households shift from long-term to short-term bonds in search of higher yields, long-term bond prices decrease and term premia increase.⁴ This contrasts with models based on [Vayanos and Vila \(2021\)](#), such as [Droste et al. \(2021\)](#), which predict the opposite pattern.⁵

Under conventional monetary policy, the central bank directly sets the short-term interest rate—typically following a Taylor rule—while long-term yields are largely determined by household demand and the supply of government bonds. In this environment, substitution across maturities and asset classes dampens the impact of monetary policy rate changes on the broader economy. For instance, a contractionary policy shock prompts households to rebalance away from long-term bonds toward short-term ones, thereby muting the effect on the effective savings rate and the business cycle.⁶ If portfolio shares are sufficiently responsive to relative returns, the effective savings rate may even decline, potentially leading to an expansion. Under such conditions, a reduction in bond supply lowers long-term yields and encourages households to shift toward higher-yielding private loans, increasing the effective savings rate and triggering a downturn in aggregate demand. This outcome—where a decline in bond supply leads to a recession—aligns with the safe-asset shortage literature

⁴In this context, the mechanism resembles [Hanson and Stein \(2015\)](#), who emphasize the importance of yield-seeking investors in driving positive term premium responses to monetary policy tightening.

⁵Typically, monetary tightening reduces preferred-habitat investors' demand for their targeted maturity, forcing arbitrageurs to absorb more risk and thereby raising term premia. This discrepancy has been addressed by incorporating arbitrageur wealth ([Kekre et al., 2023](#)). Under contractionary policy, a decline in arbitrageur wealth increases their effective risk aversion, leading to a higher required term premium.

⁶This occurs because short-term interest rates are generally lower than long-term rates. As households reallocate their portfolios toward short-term bonds, the overall composition shifts in a way that lowers the effective savings rate.

(Caballero and Farhi, 2017; Caballero et al., 2021), though in our framework it arises from household portfolio rebalancing.

A yield-curve control (YCC) policy—in which the central bank directly shapes the entire yield curve—can be particularly effective. Recall that when short-term rates decline, households tend to shift their portfolios toward long-term bonds, which dampens the overall effect on the effective savings rate. By contrast, YCC policies influence both short- and long-term rates simultaneously, for example by guiding them through specific policy rules, thereby limiting substitution across maturities and reducing its dampening effect on the business cycle. Moreover, because the entire yield curve adjusts, YCC strengthens the responsiveness of monetary policy to economic conditions. As a result, YCC plays a dual role: it curbs portfolio rebalancing while enhancing overall policy effectiveness, leading to improved stabilization of the business cycle. Overall, YCC improves household welfare compared with conventional policies focused solely on short-term interest rates.

However, YCC policy also has side effects, including the potential to prolong zero lower bound (ZLB) episodes. By directly easing long-term yields, YCC induces additional downward pressure on short-term bond returns through household portfolio reallocation. Lower long-term rates lead households to reallocate savings away from long-term bonds toward: (i) short-maturity bonds, which further depress short-term yields; and (ii) private loan markets, reducing firms’ borrowing costs and, in turn, lowering consumption prices through reduced production costs.⁷ When the ZLB is binding, YCC operates mainly through its impact on long-term yields, which can delay the economy’s exit from the ZLB. In this way, household portfolio reallocation generates a feedback loop between ZLB duration and reliance on YCC: the policy extends the time spent at the ZLB while the economy becomes increasingly dependent on its stabilizing effects. To the best of our knowledge, this result is novel.

Our asset demand system allows for finite own- and cross-elasticities across both asset classes and maturities, so households adjust their portfolios in response to changes in relative returns and other fundamentals, albeit less frictionlessly than in standard log-linearized models. Compared with the literature (Ray, 2019; Droste et al., 2021; Jansen et al., 2024),⁸

⁷Declines in the price index further strengthen downward pressure on the short-term policy rate under an inflation-targeting rule, thereby extending the duration of ZLB episodes.

⁸Although Jansen et al. (2024) estimate maturity-specific demands with investor heterogeneity and arbitrageurs—also featuring finite own- and cross-elasticities—we instead embed a tractable finite-elasticity household demand system within a quantitative New Keynesian model that explicitly incorporates government and central bank balance sheets.

our framework captures market segmentation through a parsimonious household demand structure that governs substitution both across maturities and between government bonds and private loans. A central advantage of this approach is that the demand elasticity of each asset class serves as a sufficient statistic for its degree of market segmentation, thereby facilitating empirical testing and estimation of the segmented markets hypothesis. Within this framework, we estimate the degree of segmentation in bond markets—equivalently, their demand elasticities—based on the structure implied by the model. We find our estimated elasticities align with the results documented in [Jansen et al. \(2024\)](#).

Finally, we develop a microfoundation for inelastic asset demand based on imperfect information about asset returns. We assume that each household consists of a continuum of families and members, each holding a different and imperfect information set about future asset returns, while consumption is fully insured within the household. Since there is no common signal, the household allocates its total savings evenly across members, who then invest their individual portfolios in the asset they perceive to be most profitable. This structure generates an inelastic portfolio demand function, where cross-sectional dispersion in expected returns governs the degree of market segmentation across asset classes.

To facilitate aggregation of individual portfolio decisions, we model differences in expected returns as Fréchet-distributed shocks around rationally anticipated values. Borrowing this aggregation approach from the international trade literature ([Eaton and Kortum, 2002](#)), the framework naturally accommodates new asset types and varying degrees of segmentation across maturities and asset classes, while yielding tractable expressions for portfolio shares as functions of relative expected returns. Our specification nests the standard expectations hypothesis as a special case, while also allowing for deviations driven by imperfect information and behavioral frictions.

Related Literature We contribute to several strands of the macroeconomics and finance literature. Early work emphasizes the role of macroeconomic factors in explaining the term structure of interest rates ([Ang and Piazzesi, 2003](#); [Rudebusch and Wu, 2008](#); [Bekaert et al., 2010](#)).⁹ Part of this literature employs an affine term structure representation ([Duffie and Kan, 1996](#)). A related literature studies time-varying term premia in a New Keynesian model without market segmentation represented by a finite-elasticity portfolio demand system ([Rudebusch and Swanson, 2012](#); [Buncic and Lentner, 2016](#); [Kliem and Meyer-Gohde,](#)

⁹By examining the joint dynamics of bond yields and macroeconomic variables in a VAR setting with a no-arbitrage restriction, [Ang and Piazzesi \(2003\)](#) show that models incorporating business cycle factors yield better forecasts than those relying solely on unobservable factors.

2022; Mertens and Zhang, 2025). As emphasized by Gürkaynak and Wright (2012), a central challenge for this literature is to maintain tractability while generating time variation in term premia. We instead model the term structure of interest rates within a New Keynesian framework that includes multiple asset classes, such as government bonds and private loans, and a finite-elasticity portfolio demand system. By explicitly accounting for the balance sheets of the government and central bank, alongside endogenous portfolio decisions of households across the yield curve, we generate time-varying term premia that depend on macroeconomic shocks, household portfolio choices and the central bank and government balance sheets while keeping the model tractable and interpretable.

The preferred-habitat theory of the term structure has been developed in seminal works such as Modigliani and Sutch (1966), Vayanos and Vila (2021), and Kekre et al. (2023).¹⁰ Building on Vayanos and Vila (2021), Ray (2019) incorporates preferred-habitat investors and limits to arbitrage within a New Keynesian framework,¹¹ while Droste et al. (2021) analyze quantitative easing shocks that operate through these channels in a model grounded in Ray (2019).¹² In contrast, Jansen et al. (2024) focus on cross-substitution across Treasury maturities within a detailed market structure featuring heterogeneous investors and arbitrageurs.

Our approach departs from this work in both the source of market segmentation and the broader macroeconomic setting. Rather than modeling cross-substitution through investor clienteles and arbitrageurs, we adopt a representative-household demand system in which portfolio allocations across maturities, as well as between government bonds and private loans, respond endogenously to relative expected returns, subject to finite own- and cross-elasticities. Unlike Ray (2019) and Droste et al. (2021), we abstract from both preferred-habitat investors and arbitrageurs, and instead capture segmentation directly through these finite elasticities. This formulation allows market segmentation to be summarized by estimable parameters, interprets the effective savings rate as an equilibrium portfolio return, and enables us to analyze how government and central bank balance-sheet policies transmit jointly through the yield curve, output, and inflation. Even in a log-linearized version, the

¹⁰In international macroeconomics, Gourinchas et al. (2022) and Greenwood et al. (2022) examine how preferred-habitat mechanisms jointly shape exchange rates and the term structure of interest rates.

¹¹In Ray (2019), the arbitrageur's capacity to bear risk shapes term premia across the yield curve, which in turn determines the effective savings rate households face when making intertemporal consumption decisions.

¹²Costa (2019) shows by embedding a preferred-habitat theory of the term structure in a DSGE model that monetary stabilization is more difficult to achieve when the central bank considers time-varying term premia in the yield curve, which is consistent with our results. We also show that YCC policy is a powerful tool for stabilizing the business cycle in those cases.

model generates positive term premia both in and out of steady state. In contrast, the existing literature derives term premia from arbitrageurs’ risk aversion, implying that no term premium emerges once those models are log-linearized. Applying a log-linear approximation also allows us to examine the zero lower bound (ZLB), which binds only occasionally. Moreover, the endogenous portfolio rebalancing channel—also explored by [Alpanda and Kabaca \(2020\)](#) in the context of LSAP spillovers—gives rise to distinctive dynamic responses (see Section 3.1).

A related literature studies how the maturity structure and composition of public liabilities matter when debt provides liquidity services, price is subject to issuance, or short-term asset markets are segmented. Work on public debt management emphasizes incomplete markets and maturity structure ([Bhandari et al., 2017](#); [Aparisi de Lannoy et al., 2022](#)).¹³ [d’Avernas and Vandeweyer \(2024\)](#) emphasize Treasury bill scarcity and segmented short-term asset markets from a large amount of excess reserves and a binding regulatory leverage ratio, and [Gómez Cram et al. \(2025\)](#) link Treasury risk premia to monetary-fiscal interactions and limited market participation. These papers complement our analysis by showing that the pricing and allocation of public debt depend on maturity structure, market segmentation, and the joint balance sheets of fiscal and monetary authorities.

Another strand of the literature ([Gertler and Karadi, 2011](#); [Cúrdia and Woodford, 2011](#); [Christensen and Krogstrup, 2018, 2019](#); [Karadi and Nakov, 2021](#); [Del Negro et al., 2017](#); [Iovino and Sergeyev, 2018](#); [Eggertsson et al., 2024](#)) studies balance-sheet policy transmission and liquidity frictions more directly.¹⁴ Most of this work emphasizes overall balance-sheet expansion, crisis-specific market disruptions, or the banking channel rather than substitution across bonds of different maturities within a unified term-structure environment. In contrast, our framework shows how the central bank can rebalance bond holdings across the yield curve to enhance stabilization. Our result that active management of long-term bonds improves welfare is consistent with [Sims and Wu \(2021\)](#).¹⁵

While our treatment of ZLB recessions builds on earlier work ([Swanson and Williams, 2014](#); [Dordal i Carreras et al., 2016](#); [Caballero and Farhi, 2017](#); [Caballero et al., 2021](#)), we place additional emphasis on how household portfolio reallocation—under finite elasticities—

¹³[Bigio et al. \(2023\)](#) show how liquidity costs shape optimal debt-maturity management.

¹⁴[Gertler and Karadi \(2011\)](#) note that compared with private market intermediaries, the central bank faces fewer balance sheet constraints, and as private intermediaries’ constraints tighten during crises, central bank intermediation becomes more beneficial; [Cúrdia and Woodford \(2011\)](#) find targeted asset purchases effective under exogenous market disruptions.

¹⁵[Sims and Wu \(2021\)](#) assume that a wholesale firm and fiscal authorities issue perpetuities with declining coupon payments, whereas we consider bonds with varying maturities.

ties—shapes outcomes during such episodes. We also highlight the advantages of jointly managing both the size and the maturity composition of the central bank’s balance sheet when the ZLB is binding. To our knowledge, this is among the first general equilibrium models to integrate the term structure of interest rates, a binding ZLB, multiple financial assets, yield-curve control policies, and the portfolio balance channel with demand functions featuring finite elasticities.

Layout Section 2 presents a simplified model that illustrates the key mechanisms of our framework. Section 3 introduces the full model. Section 4 examines the steady-state implications of various policies and calibration choices. Section 5 explores the short-run cyclical responses to different shocks under alternative monetary policy regimes and economic conditions, including the ZLB. Section 6 concludes. The Appendix contains additional figures and tables. Online Appendix A provides detailed derivations and proofs; Online Appendix B outlines our calibration and estimation strategies; and Online Appendix C derives the second-order approximation to welfare.¹⁶

2 Illustrative Model

This section presents a simplified framework designed to highlight the model’s core mechanisms, with a particular focus on how endogenous portfolio rebalancing and yield-curve control (YCC) shape the transmission of monetary policy. The analysis shows that substitution across maturities and asset classes weakens the effect of changes in the policy rate on the broader economy. For example, an expansionary shock leads households to shift their portfolios away from short-term bonds toward higher-yielding long-term bonds. This substitution dampens the policy’s accommodative impact on the effective savings rate, which controls the intertemporal consumption decision, and, in turn, on the business cycle.

Within this setting, YCC policies can be especially powerful because they enable the central bank to influence both short- and long-term interest rates simultaneously. By doing so, YCC limits substitution across maturities—and across asset classes when multiple assets are available. It also enhances the overall effectiveness of monetary policy by allowing the central bank to move the entire yield curve in response to economic conditions.

For tractability, the simplified model is log-linearized and features two assets: short-term and long-term bonds, though they can be interpreted more broadly as different asset

¹⁶Supplementary Material in Online Appendices D-G includes further figures and explanations.

classes. It abstracts from inflation dynamics by assuming perfectly rigid prices.¹⁷ The representative household allocates its portfolio based on the relative expected returns of these assets, with finite own- and cross-return elasticities. The central bank follows a Taylor-type rule to set short-term interest rates, while long-term rates depend on the household's allocation to long-term bonds and government bond supply. In the full model presented later, conventional policy leads to *incomplete* pass-through to long-term rates—for instance, higher short-term rates can push long-term rates up as households rebalance toward short-term assets. In contrast, the simplified model abstracts from this mechanism by assuming that long-term rates remain fixed under conventional policy. YCC is introduced through a separate Taylor-type rule for long-term rates, allowing the central bank to directly manage the entire yield curve.

Setting The representative household faces the following Euler equation:

$$\hat{c}_t = \mathbb{E}_t [\hat{c}_{t+1} - \hat{r}_{t+1}^S], \quad (1)$$

where $\hat{c}_t$ and $\hat{r}_{t+1}^S$ are the log-deviation of consumption and the (gross) effective savings rate from the steady state, respectively. In a canonical New Keynesian model (Woodford, 2003), the effective savings rate $\hat{r}_{t+1}^S$ coincides with the policy rate set by the monetary authority. In our framework, however, this rate is a composite measure reflecting the returns on both short-term and long-term bonds, each weighted by its share in the aggregate portfolio. The (gross) effective savings rate is given by:

$$R_{t+1}^S = \sum_{f=1}^2 \lambda_t^f R_{t+1}^f, \quad (2)$$

where λ_t^f is the household's portfolio share of the f -maturity bond in period t , and R_{t+1}^f is the f -maturity bond (one-period) return. As stated, the simplified model considers two assets: short-term and long-term bonds, so $f = 1, 2$. Log-linearizing (2) yields:

$$\hat{r}_{t+1}^S = \sum_{f=1}^2 \frac{\lambda_t^f R^f}{R^S} (\hat{\lambda}_t^f + \hat{r}_{t+1}^f), \quad (3)$$

¹⁷Our full model relaxes this assumption and allows for inflation dynamics and multiple asset classes.

where λ^f , R^f , and R^S are the steady state levels of the portfolio share, the bond return, and the effective savings rate, respectively. Note that the effective savings rate changes when either the portfolio share $\hat{\lambda}_t^f$ or the bond return $\hat{r}_{t+1}^f$ changes. For instance, a contractionary monetary policy shock, corresponding to a higher $\hat{r}_{t+1}^1$, will increase the effective savings rate given $\hat{\lambda}_t^f$. At the same time, $\hat{\lambda}_t^1$ increases as households shift their portfolios toward short-term bonds. This substitution mutes the increase in the effective savings rate.¹⁸

The household's portfolio share of the short-term bond is given by:

$$\hat{\lambda}_t^1 = \kappa_B \mathbb{E}_t [\hat{r}_{t+1}^1 - \hat{r}_{t+1}^2], \quad (4)$$

where κ_B is the elasticity of the demand for the short-term bond with respect to the expected return spread between the short-term and long-term bonds. Although the full model later provides a microfoundation for this elasticity, we treat it as exogenous and finite here.¹⁹ $\kappa_B \rightarrow \infty$ corresponds to perfect substitution between the two bonds, rendering returns on the long-term bond irrelevant for the business cycle dynamics and leading to the expectations hypothesis.²⁰ For expositional simplicity, we assume that in that case the households weakly prefer to save into the short-maturity asset, i.e. $\lim_{\kappa_B \rightarrow \infty} \lambda^1 = 1$, which replicates the traditional single-asset model employed in the literature.

Since $\lambda_t^1 + \lambda_t^2 = 1$, we obtain

$$\hat{\lambda}_t^2 = -\frac{\lambda^1}{\lambda^2} \hat{\lambda}_t^1, \quad (5)$$

where λ^1 and λ^2 denote levels of the portfolio share of the short-term and long-term bonds at the steady state, respectively.

Finally, due to rigid prices, the economy is demand-determined. Thus, we abstract from the supply side of the economy and assume that the output is determined by the aggregate consumption demand determined by the Euler equation (1).

¹⁸We assume that $\lambda^1 R^1 < \lambda^2 R^2$, which implies that a larger share of short-term bonds lowers the effective savings rate given R_{t+1}^1 and R_{t+1}^2 . This compositional change dampens the increase in the effective savings rate.

¹⁹In addition, we estimate this parameter based on U.S. data.

²⁰Note that as $\kappa_B \rightarrow \infty$, $R^1 = R^2 = R^S$ at the steady state.

Conventional Monetary Policy The central bank follows a Taylor rule to set the short-term interest rate:

$$\hat{r}_{t+1}^1 = \phi_C \hat{c}_t + \varepsilon_t \quad (6)$$

where $\phi_C > 0$ is the policy response coefficient to the output gap, and ε_t is a policy shock. Under conventional policy, we assume that the long-term interest rate is fixed at the steady state level, i.e., $\hat{r}_{t+1}^2 = 0$.²¹ For simplicity, we further assume that ε_t is a i.i.d. random variable with $\mathbb{E}[\varepsilon_t] = 0$.

Yield-Curve Control Policy In addition to the short-term rate given by (6), the central bank follows a Taylor rule to set the expected long-term interest rate:

$$\mathbb{E}_t \hat{r}_{t+1}^2 = \phi_{C,2} \hat{c}_t \quad (7)$$

where $\phi_{C,2} > 0$ is the policy response coefficient to the output gap.²²

2.1 Monetary Policy Transmission

We first consider the transmission of monetary policy under conventional policy and yield-curve control policy. The short-term interest rate is given by (6), and the long-term interest rate is fixed at the steady state level, i.e., $\hat{r}_{t+1}^2 = 0$, under conventional policy and given by (7), under yield-curve control policy. The effective savings rate is given by (3), and the portfolio share of the short-term bond is given by (4). For illustrative purposes, we assume that the only shock that drives the business cycle is the monetary policy shock ε_t .

Conventional Policy Merging (3) and (5) with $\hat{r}_{t+1}^2 = 0$, we obtain:

$$\hat{r}_{t+1}^S = \underbrace{\frac{\lambda^1 R^1}{R^S}}_{\equiv \eta_1 > 0} \hat{r}_{t+1}^1 - \underbrace{\frac{\lambda^1 (R^2 - R^1)}{R^S}}_{\equiv \eta_2 > 0} \hat{\lambda}_t^1 \quad (8)$$

²¹Our full model relaxes this assumption—the long-term interest rates are determined by the household's endogenous portfolio share of the long-term bonds and government bond issuance. In those cases, the long-term rates will be affected by the policy rate change and the portfolio rebalancing channel, with an incomplete pass-through.

²²Note that the long-term rate $\hat{r}_{t+1}^2$ is intrinsically uncertain, as future bond prices cannot be known with certainty from today's perspective. In the full model, rather than assuming the central bank directly selects $\mathbb{E}_t \hat{r}_{t+1}^2$, we posit that it chooses long-term bond yields using Taylor rules as current yields are observable.

where $\eta_1 < 1$ due to our assumption that $\lambda^1 R^1 < \lambda^2 R^2$, and $\eta_2 > 0$. This equation shows an important feature of our model: when $\hat{r}_{t+1}^1$ increases, its direct effect on the effective savings rate $\hat{r}_{t+1}^S$ is mitigated by a factor of $\eta_1 < 1$. In addition, households will substitute their portfolio towards the short-term bond, raising $\hat{\lambda}_t^1$. The endogenous change in portfolio shares further mitigates the increase in the effective savings rate $\hat{r}_{t+1}^S$ given $R^2 > R^1$ at the steady state.

The equilibrium is characterized by merging the Euler equation (1), the effective savings rate (8), the portfolio share of the short-term bond (4), and the short-term interest rate (6), which results in:²³

$$\hat{c}_t^{\text{CP}} = -\frac{1}{1 + (\eta_1 - \eta_2 \kappa_B) \phi_C} (\eta_1 - \eta_2 \kappa_B) \varepsilon_t \quad (9)$$

where the denominator is assumed to be positive, i.e., $1 + (\eta_1 - \eta_2 \kappa_B) \phi_C > 0$. We categorize different cases of the strength of substitution between the two bonds as follows:

Case 1 $\kappa_B \rightarrow \infty$: Perfect substitution between the two bonds, leading to the *expectations hypothesis*. In this case, $R^1 = R^2 = R^S$ at the steady state, so $\eta_2 = 0$; under the perfect-substitution normalization $\lim_{\kappa_B \rightarrow \infty} \lambda^1 = 1$ stated above, $\eta_1 = 1$. Thus, the equilibrium consumption is given by:

$$\hat{c}_t^{\text{CP}} = -\frac{1}{1 + \phi_C} \varepsilon_t, \quad (10)$$

where stronger monetary policy response to the output gap ϕ_C leads to more stabilizing effect on the business cycle. This corresponds to a standard New Keynesian model (Woodford, 2003).

Case 2 $\kappa_B \rightarrow 0$: No substitution between the two bonds. In this case, $\hat{\lambda}_t^1 = 0$ and $\hat{r}_{t+1}^S = \eta_1 \hat{r}_{t+1}^1$. The equilibrium consumption is given by:

$$\hat{c}_t^{\text{CP}} = -\frac{1}{1 + \eta_1 \phi_C} \eta_1 \varepsilon_t. \quad (11)$$

Note that compared with (10), equation (11) shows that both the monetary policy shock ε_t and the monetary policy responsiveness $\phi_C > 0$ are reduced by a factor

²³This is the only rational expectation equilibrium that is consistent with the assumptions of the model. Because the policy shock is iid and the simplified model has no endogenous state variable, the equilibrium policy function depends only on the current shock, implying $\mathbb{E}_t[\hat{c}_{t+1}] = 0$.

of $\eta_1 < 1$. This occurs because in this case, a shock to the policy rate $\hat{r}_{t+1}^1$ is not fully transmitted to the effective savings rate $\hat{r}_{t+1}^S$, with the degree of pass-through determined by the fixed weighted-return share $\eta_1 < 1$.

Case 3 $\kappa_B > 0$ and $\eta_2 \kappa_B < \eta_1$: Substitution between the two bonds, which further mitigates the pass-through from the policy rate to the effective savings rate $\hat{r}_{t+1}^S$ —both the monetary policy shock ε_t and the monetary policy responsiveness $\phi_C > 0$ are reduced by a smaller factor of $\eta_1 - \kappa_B \eta_2 < 1$. In this case, $\hat{r}_{t+1}^S$ still moves positively with the policy rate $\hat{r}_{t+1}^1$. Consumption $\hat{c}_t^{\text{CP}}$ decreases under contractionary monetary policy shocks $\varepsilon_t > 0$, but less than in the above two cases.²⁴

Case 4 $\kappa_B > 0$ and $\eta_2 \kappa_B > \eta_1$: Substitution between the two bonds is now strong enough that $\hat{r}_{t+1}^S$ moves inversely with the policy rate $\hat{r}_{t+1}^1$. Contractionary monetary policy shocks $\varepsilon_t > 0$ reduces the effective savings rate $\hat{r}_{t+1}^S$ and raises consumption $\hat{c}_t^{\text{CP}}$

As a result, monetary policy becomes less effective in stabilizing the business cycle—remember, the effective monetary responsiveness ϕ_C is dampened by a factor of $\eta_1 - \eta_2 \kappa_B$ —and, in certain cases, can even produce outcomes that run counter to those predicted by the standard New Keynesian model when the degree of substitution between the two bonds is high enough.²⁵

Yield-Curve Control Policy Under YCC policy, the equilibrium is characterized by the Euler equation (1), the general effective savings rate (3), the portfolio share of the short-term bond (4), the short-term interest rate (6), and the long-term interest rate (7). This results in:

$$\hat{c}_t^{\text{YCC}} = - \frac{1}{1 + \underbrace{(\eta_1 \phi_C + (1 - \eta_1) \phi_{C,2})}_{\text{Policy Response}} - \eta_2 \kappa_B \underbrace{(\phi_C - \phi_{C,2})}_{\text{Cross-Substitution}}} (\eta_1 - \eta_2 \kappa_B) \varepsilon_t, \quad (12)$$

²⁴This is because the first effect—where the monetary policy shock ε_t is dampened by the smaller factor $\eta_1 - \kappa_B \eta_2 < 1$ —outweighs the second effect, in which the policy responsiveness $\phi_C > 0$ is also mitigated by the same factor. Note that the coefficient

$$\frac{\eta_1 - \eta_2 \kappa_B}{1 + (\eta_1 - \eta_2 \kappa_B) \phi_C} > 0$$

is less than $\frac{\eta_1}{1 + \eta_1 \phi_C}$ for **Case 2** and $\frac{1}{1 + \phi_C}$ for **Case 1** when $\eta_2 \kappa_B < \eta_1$.

²⁵A higher κ_B reduces $R^2 - R^1$ at the steady state, reducing η_2 . Under our calibration in the main text, $\eta_2 \kappa_B$ still increases with κ_B .

where in the policy-response term, the short-rate and long-rate responses enter the effective savings rate as a weighted average, with weights η_1 and $1 - \eta_1$, respectively.

Comparing equations (12) and (9) elucidates the key advantage of YCC policy: it allows the central bank to influence both short- and long-term interest rates simultaneously. This effectively limits substitution across the two bonds— $\eta_2 \kappa_B \phi_C$ term in the denominator of (9) is replaced by $\eta_2 \kappa_B (\phi_C - \phi_{C,2})$ in the denominator of (12), raising the effect of monetary policy in stabilizing the business cycle. In contrast, the policy shock ε_t is dampened by the same factor $\eta_1 - \eta_2 \kappa_B$ as under conventional policy.

In addition, YCC policy effectively raises the overall responsiveness of monetary policy to the output gap— $(\eta_1 \phi_C + (1 - \eta_1) \phi_{C,2})$ term in the denominator of (12) is greater than $\eta_1 \phi_C$ in the denominator of (9). This is because the central bank can now shift the entire yield curve in response to economic conditions.

Overall, these two channels make YCC policy more effective in stabilizing the business cycle than conventional policy when the degree of substitution between the two bonds is finite— $|\hat{c}_t^{\text{YCC}}| < |\hat{c}_t^{\text{CP}}|$. While our simplified model is defined using a monetary policy shock, the stabilizing effect of YCC policy remains robust across various types of shocks. Under all shock scenarios, YCC proves more effective than conventional policy in stabilizing the business cycle. And this result continues to hold even when we consider a zero lower bound (ZLB) on the short-term interest rate.

In the following sections, we provide a more formal model specification and empirical analysis to support this conclusion.

3 Model

As in Section 2, the model is built around a representative household framework. To provide a tractable foundation, we consider a large household with internal frictions, allowing us to isolate the effects of yield-curve interventions in a clean and controlled setting.

3.1 Representative Household

The representative household maximizes its expected discounted utility:

$$\max_{\{C_{t+j}, N_{t+j}\}} \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[\log(C_{t+j}) - \frac{\eta}{\eta + 1} \left(\frac{N_{t+j}}{\bar{N}_{t+j}} \right)^{1 + \frac{1}{\eta}} \right], \quad (13)$$

where C_t denotes final good consumption and $N_t = \left(\int_0^1 N_t(\nu)^{\frac{\eta+1}{\eta}} d\nu \right)^{\frac{\eta}{\eta+1}}$ is the aggregate labor index with $N_t(\nu)$ representing labor supplied in intermediate industry ν . The parameter η is the Frisch elasticity of labor supply, and $\bar{N}_t$ is the balanced-growth population, growing at a constant gross rate GN .

Each period, the household invests in f -period zero-coupon government bonds (with $f = 1, \dots, F$) and provides loans to firms.²⁶ The budget constraint is:

$$C_t + \frac{L_t}{P_t} + \frac{\sum_{f=1}^F B_t^{H,f}}{P_t} = \frac{\sum_{f=0}^{F-1} R_t^f B_{t-1}^{H,f+1}}{P_t} + \frac{R_t^K L_{t-1}}{P_t} + \int_0^1 \frac{W_t(\nu) N_t(\nu)}{P_t} d\nu + \frac{\Lambda_t}{P_t}, \quad (14)$$

where L_t denotes one-period loans to firms, yielding a return R_t^K ; $W_t(\nu)$ is the wage in industry ν ; and Λ_t aggregates transfers from lump-sum taxation and profits from the central bank and firms.

$B_t^{H,f}$ denotes nominal investment in f -maturity government bonds that pay one dollar at time $t+f$, with Q_t^f denoting the bond price (by definition, $Q_t^0 = 1$ at maturity). Households cannot issue risk-free bonds. The one-period return on an f -period bond is given by $R_t^f = \frac{Q_t^f}{Q_{t-1}^{f+1}}$, capturing the one-period holding returns. The gross yield of a zero-coupon bond with f -maturity is defined as $YD_t^f := (Q_t^f)^{-1/f}$.²⁷

3.1.1 Inelastic Portfolio Demands

The representative household determines optimal consumption, employment, and savings S_t , which are allocated between government bonds $B_t^H = \sum_{f=1}^F B_t^{H,f}$ and firm loans L_t , so that $S_t = B_t^H + L_t$.

Following the literature (Koijen and Yogo, 2019; Gabaix and Koijen, 2022), we assume that households' portfolio shares are determined by downward-sloping demand functions with finite own- and cross-price elasticities. We microfound these portfolio shares through heterogeneous information within the household. In particular, after choosing total savings S_t , the representative household delegates portfolio decisions to a continuum of families $m \in [0, 1]$. Each family chooses between government bonds and private loans; if a family selects bonds, a continuum of members $n \in [0, 1]$ within that family chooses the bond

²⁶Alternatively, this may be interpreted as households purchasing one-period corporate bonds. Banks and financial intermediaries are abstracted from our framework; in the absence of intermediation frictions, both representations yield equivalent results.

²⁷It follows that bond returns can alternatively be expressed as $R_t^f = \frac{(YD_t^f)^{-f}}{(YD_{t-1}^{f+1})^{-(f+1)}}$.

maturity. Consumption is fully insured within the household as in [Del Negro et al. \(2017\)](#). Because there is no common signal about future asset returns, savings are allocated symmetrically across families and members before idiosyncratic information is realized, and each member then allocates assigned savings to the asset with the highest perceived expected discounted return.

Bond Family For bond-investing families, member n selects the maturity f that maximizes $\mathbb{E}_{m,n,t}[Q_{t,t+1}R_{t+1}^{f-1}]$. Under the rational expectations, all members would choose the same maturity, thereby equalizing expected discounted returns across maturities and reproducing the expectations hypothesis in the log-linearized model. To generate a downward-sloping demand curve, we instead assume that individual expectations satisfy

$$\mathbb{E}_{m,n,t}[Q_{t,t+1}R_{t+1}^{f-1}] = z_{n,t}^f \mathbb{E}_t[Q_{t,t+1}R_{t+1}^{f-1}], \quad \forall f = 1, \dots, F,$$

where $z_{n,t}^f$ follows an i.i.d. Fréchet distribution with scale parameter z_t^f and shape parameter κ_B . Aggregating across members implies that the household's portfolio share of the f -maturity bond in total government bond holdings, $\lambda_t^{HB,f} \equiv \frac{B_t^{H,f}}{B_t^H}$, is given by:

$$\lambda_t^{HB,f} = \left(\frac{z_t^f \mathbb{E}_t[Q_{t,t+1}R_{t+1}^{f-1}]}{\Phi_t^B} \right)^{\kappa_B}, \quad (15)$$

where $\Phi_t^B \equiv \left[\sum_{j=1}^F (z_t^j \mathbb{E}_t[Q_{t,t+1}R_{t+1}^{j-1}])^{\kappa_B} \right]^{\frac{1}{\kappa_B}}$. Here, κ_B is the elasticity of portfolio demand for the f -maturity bond with respect to its expected discounted return.²⁸ We assume $\kappa_B > 0$ throughout the paper.

Additionally, we introduce z_t^f as a shock to the portfolio preference for the f -maturity bond. For example, an increase in z_t^f raises the household's f -maturity share of government bond holdings, given returns and other macroeconomic states. In equilibrium, this shock will raise the price of the f -maturity bond Q_t^f and induce households to reallocate their portfolios toward other maturities and private loans, thus pushing down their returns.²⁹ This shock, combined with the household's endogenous portfolio “rebalancing”, is an important

²⁸In our framework, $\mathbb{E}_t[Q_{t,t+1}R_{t+1}^{f-1}]$ does not equal 1 due to the inelasticity of demand in (15), consistent with the literature.

²⁹Based on our calibration and the estimated value of κ_B , decreases in returns of other assets are smaller than the decrease in the original return R_{t+1}^f . This produces a phenomenon known as *localization* of demand shocks, as documented in [Vayanos and Vila \(2021\)](#) and [Droste et al. \(2021\)](#).

source of variation that generates an upward-sloping yield curve both at and away from the steady state.³⁰

Given equation (15), the aggregate return on the household's bond portfolio is:

$$R_{t+1}^{HB} = \sum_{f=0}^{F-1} \lambda_t^{HB,f+1} R_{t+1}^f. \quad (16)$$

Bond vs. Loans Family At the upper nest, family m compares the aggregate bond return R_{t+1}^{HB} with the loan return R_{t+1}^K . Under benchmark rational expectations, all families would make the same choice, equalizing $\mathbb{E}_t[Q_{t,t+1} R_{t+1}^{HB}]$ and $\mathbb{E}_t[Q_{t,t+1} R_{t+1}^K]$. We instead assume that family-specific expectations for loans satisfy

$$\mathbb{E}_{m,t} [Q_{t,t+1} R_{t+1}^K] = z_{m,t}^K \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K],$$

where $z_{m,t}^K$ is a Fréchet shock with scale parameter z_t^K and shape parameter κ_S . Aggregating across families yields the household's portfolio share of total savings allocated to loans:

$$\lambda_t^K = \left(\frac{z_t^K \mathbb{E}_t[Q_{t,t+1} R_{t+1}^K]}{\Phi_t^S} \right)^{\kappa_S}, \quad (17)$$

where $\Phi_t^S = [(\mathbb{E}_t[Q_{t,t+1} R_{t+1}^{HB}])^{\kappa_S} + (z_t^K \mathbb{E}_t[Q_{t,t+1} R_{t+1}^K])^{\kappa_S}]^{\frac{1}{\kappa_S}}$. Here, κ_S is the elasticity of the portfolio demand for private loans with respect to their expected discounted return.³¹

We assume $\kappa_S > 0$ throughout the paper. As the bond return R_{t+1}^{HB} depends on the bond portfolio $\{\lambda_t^{HB,f}\}_{f=1}^F$, given by (15), our portfolio demand follows a nested constant elasticity of substitution (CES) structure.

Additionally, we introduce z_t^K as a shock to the portfolio preference for private loans. Similar to the case for $\{z_t^f\}$, this shock induces households to rebalance their portfolios, which affects the equilibrium returns of bonds with different maturities and private loans.³²

Based on (15) and (17), the aggregate bond holdings by maturity f are:

$$B_t^{H,f} = (1 - \lambda_t^K) \lambda_t^{HB,f} S_t, \quad \forall f = 1, \dots, F,$$

³⁰In the context of the preferred-habitat literature (Vayanos and Vila, 2021), this shock can be interpreted as a shock to the preferred-habitat demand for the f -maturity bond.

³¹Again, $\mathbb{E}_t[Q_{t,t+1} R_{t+1}^K]$ deviates from 1 in our framework, due to the assumed inelasticity of demand in (17), consistent with the literature.

³²Equation (17) indicates that the household allocates a larger share of their portfolio to loans when loan returns R_{t+1}^K rise, and reduce this share when the aggregate bond return R_{t+1}^{HB} increases.

and the aggregate return on household savings is:

$$R_t^S = (1 - \lambda_{t-1}^K) R_t^{HB} + \lambda_{t-1}^K R_t^K. \quad (18)$$

Therefore, R_t^S is a weighted average of returns from bonds (across maturities) and loans,³³ representing the effective savings rate of households. The household's budget constraint (14) can be rewritten as:

$$C_t + \frac{S_t}{P_t} = \frac{R_t^S S_{t-1}}{P_t} + \int_0^1 \frac{W_t(\nu) N_t(\nu)}{P_t} d\nu + \frac{\Lambda_t}{P_t}. \quad (19)$$

Despite the richer asset structure and market segmentation, the representative household's problem ultimately resembles that of a conventional New Keynesian model. This structural representation offers two advantages. First, the estimated demand elasticities—see Online Appendix B—summarize market segmentation across maturities. Second, it nests the benchmark New Keynesian model as a special case.

Our microfoundation is related to a literature that derives smooth aggregate choice from discrete individual decisions under limited information. Matějka and McKay (2015) show that rational inattention in a discrete-choice environment yields generalized multinomial-logit choice probabilities. Woodford (2009, 2011, 2014a,b) develop closely related treatments in which information-processing constraints generate discrete adjustment, inattentive valuation, and stochastic choice. Our framework is similar in spirit, though not identical in formulation: each household member makes a discrete portfolio choice by selecting the asset or maturity with the highest perceived expected return, while aggregation across heterogeneously informed decision makers produces smooth portfolio shares and finite own- and cross-elasticities in a multi-asset term-structure environment.

Expectations Hypothesis Note that as $\kappa_B \rightarrow \infty$, $\kappa_S \rightarrow \infty$, and $z_t^f = z_t^K = 1$ for all f, t , our framework converges to a conventional New Keynesian model represented by:

$$\mathbb{E}_t[Q_{t,t+1} R_{t+1}^{f-1}] = \mathbb{E}_t[Q_{t,t+1} R_{t+1}^K] = 1, \quad \forall f. \quad (20)$$

³³While our analysis assumes a one-period duration for private loans, extending the model to incorporate a term structure in the loan market is feasible. However, given our focus on unconventional monetary policy via central bank purchases of government bonds, we restrict our discussion to the term structure for bonds to maintain model tractability.

In its log-linearized form, equation (20) becomes:

$$\mathbb{E}_t[\hat{r}_{t+1}^{f-1}] = \mathbb{E}_t[\hat{r}_{t+1}^K] = \hat{r}_{t+1}^0, \quad \forall f, \quad (21)$$

which renders all returns other than the policy rate $\hat{r}_{t+1}^0$ redundant. Equation (21) corresponds to the classic expectations hypothesis. Thus, our model specification with inelastic financial markets nests the conventional expectations hypothesis result as a special case.

3.1.2 Optimality Conditions

The solution to the household's problem (13), subject to the budget constraint (19), yields the following equilibrium conditions:

$$\left(\frac{N_t(\nu)}{\bar{N}_t} \right)^{\frac{1}{\eta}} = \left(\frac{C_t}{\bar{N}_t} \right)^{-1} \frac{W_t(\nu)}{P_t}, \quad (22)$$

$$1 = \beta \mathbb{E}_t \left[\frac{R_{t+1}^S C_t}{C_{t+1} \Pi_{t+1}} \right], \quad (23)$$

where $\Pi_{t+1} = P_{t+1}/P_t$ is the gross inflation rate. In (23), the effective savings rate R_{t+1}^S serves as a composite interest rate for intertemporal consumption decisions, reflecting the returns on bonds of various maturities and loans, each weighted by its share in the aggregate portfolio. As a result, the endogenous portfolio rebalancing channel becomes central to the business cycle dynamics under forward-looking unconventional monetary policies.

Remarks Our findings on imperfect substitutability across assets align with bond-in-the-utility models that feature finite elasticities (Alpanda and Kabaca, 2020) and with models incorporating portfolio adjustment costs (Costa, 2019). However, our approach retains the standard aggregate dynamics—see equations (22) and (23)—which improves tractability.³⁴

In contrast to the preferred-habitat literature, our framework does not depend on financial arbitrageurs to generate an upward-sloping yield curve. Instead, the financial market is segmented, with the extent of segmentation directly governed by the demand elasticities κ_B and κ_S . Household portfolios are also exposed to asset-specific and maturity-specific preference shocks $z_{t,f=1}^{fF}$ and z_t^K , which, together with central bank and government balance sheet policies, produce equilibrium variation in term premia across the yield curve.

³⁴Harrison (2024) likewise introduces a portfolio friction that generates a gap between short- and long-term bond returns, enabling the central bank's quantitative easing policy to be effective.

3.2 Capital Producer

A representative capital firm produces capital K_t and rents it to intermediate goods producers at price P_t^K . Capital is accumulated by investing final goods, depreciates at rate δ , and new investment I_t is implemented with a one-period lag. Hence, capital evolves according to

$$K_t = (1 - \delta)K_{t-1} + I_{t-1}.$$

The capital firm's profits are given by

$$\Lambda_t^K = P_t^K K_t - P_t I_t, \quad (24)$$

where P_t denotes the final good price index. Profit maximization with respect to I_t yields the following standard first-order condition:

$$1 = \mathbb{E}_t \left[Q_{t,t+1} \Pi_{t+1} \left((1 - \delta) + \frac{P_{t+1}^K}{P_{t+1}} \right) \right], \quad (25)$$

where $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$ is the gross inflation rate.

3.3 Firms

There is a continuum $\nu \in [0, 1]$ of intermediate varieties, each produced by a monopolist using capital and labor according to

$$Y_t(\nu) = \left(\frac{K_t(\nu)}{\alpha} \right)^\alpha \left(\frac{A_t N_t(\nu)}{1 - \alpha} \right)^{1-\alpha}, \quad (26)$$

where $A_t = \exp(u_t^A)$ denotes aggregate labor-augmenting productivity with $u_t^A = \mu + u_{t-1}^A + \varepsilon_t^A$, $\varepsilon_t^A \sim N(0, \sigma_A^2)$, and the gross growth rate is $GA_t \equiv A_t/A_{t-1} = \exp(\mu + \varepsilon_t^A)$.

A representative competitive firm aggregates intermediate varieties into a final good using a Dixit–Stiglitz aggregator:

$$Y_t = \left[\int_0^1 Y_t(\nu)^{\frac{\epsilon-1}{\epsilon}} d\nu \right]^{\frac{\epsilon}{\epsilon-1}},$$

with $\epsilon > 1$ as the elasticity of substitution. The demand for intermediate good ν is

$$Y_t(\nu) = \left(\frac{P_t(\nu)}{P_t} \right)^{-\epsilon} Y_t, \quad (27)$$

and the aggregate price index is defined as

$$P_t = \left[\int_0^1 P_t(\nu)^{1-\epsilon} d\nu \right]^{\frac{1}{1-\epsilon}}. \quad (28)$$

Intermediate producers face sticky prices à la **Calvo (1983)** and reset their prices each quarter with probability $1-\theta$. Firms that adjust set the optimal price P_t^* , and the equilibrium aggregate price index satisfies $P_t^{1-\epsilon} = (1-\theta)(P_t^*)^{1-\epsilon} + \theta P_{t-1}^{1-\epsilon}$.

Working-Capital Constraint and Loan Market Each intermediate firm rents capital from the capital producer at price P_t^K . To finance production, firm ν must borrow a fraction γ_L of its subsidized revenue, $(1+\zeta^F)P_t(\nu)Y_t(\nu)$, where ζ^F denotes the production subsidy. A firm borrowing $L_t(\nu)$ in period t repays $R_{t+1}^K L_t(\nu)$ in period $t+1$, where the gross loan rate R_{t+1}^K is contracted at time t . This loan requirement captures a financial friction faced by intermediate firms, which can be interpreted as a working-capital constraint.

In equilibrium, intermediate firm loan demand equals the household loan supply, so the loan market clears.

Firm ν maximizes its discounted profit stream:

$$\max \sum_{j=0}^{\infty} \mathbb{E}_t \left[Q_{t,t+j} \left((1+\zeta^F)P_{t+j}(\nu)Y_{t+j}(\nu) - W_{t+j}(\nu)N_{t+j}(\nu) - P_{t+j}^K K_{t+j}(\nu) - R_{t+j}^K L_{t+j-1}(\nu) + L_{t+j}(\nu) \right) \right], \quad (29)$$

where $Q_{t,t+j} = \beta^j \left(\frac{P_{t+j}}{P_t} \frac{C_{t+j}}{C_t} \right)^{-1}$ is the stochastic discount factor between periods t and $t+j$. At period $t+j$, firm ν repays $R_{t+j}^K L_{t+j-1}(\nu)$ on the loan taken in period $t+j-1$.

Minimizing production costs with respect to labor and capital yields the input demand functions:

$$N_t(\nu) = (1-\alpha) \frac{Y_t(\nu)}{A_t} \left(\frac{\frac{P_t^K}{P_t}}{\frac{W_t(\nu)}{P_t A_t}} \right)^{\alpha}, \quad \frac{K_t(\nu)}{A_t} = \alpha \frac{Y_t(\nu)}{A_t} \left(\frac{\frac{P_t^K}{P_t}}{\frac{W_t(\nu)}{P_t A_t}} \right)^{-(1-\alpha)}. \quad (30)$$

3.4 Bond Market

The bond market equilibrium condition is given by

$$B_t^{H,f} + B_t^{G,f} + B_t^{CB,f} = 0, \quad \forall f = 1, \dots, F, \quad (31)$$

where $B_t^{H,f}$ represents households' holdings of bonds with maturity f bonds, while $B_t^{G,f}$ and $B_t^{CB,f}$ denote the bond positions of the government³⁵ and the central bank, respectively. We assume that only the government and the central bank are able to issue risk-free claims and therefore may take on negative bond positions.³⁶

Allowing the government to issue bonds—and maintain negative positions—is crucial. Our assumptions regarding technology growth GA_t and population growth GN ensure that, in steady state, government debt remains positive yet stable, avoiding explosive dynamics. As a result, the government continues to supply risk-free debt even under cyclical fluctuations.

Defining $\lambda_t^{G,f}$ and $\lambda_t^{CB,f}$ as portfolio shares of f -maturity bonds held by the government and the central bank, respectively, we can rewrite (31) as

$$\lambda_t^{HB,f} B_t^H + \lambda_t^{G,f} B_t^G + \lambda_t^{CB,f} B_t^{CB} = 0, \quad \forall f = 1, \dots, F. \quad (32)$$

3.5 Government

The government's budget constraint is given by

$$G_t + \zeta^F Y_t + \frac{B_t^G}{P_t} = T_t + \frac{R_t^G B_{t-1}^G}{P_t}, \quad \text{with } B_t^G = \sum_{f=1}^F B_t^{G,f}, \quad R_t^G = \sum_{f=0}^{F-1} \lambda_{t-1}^{G,f+1} R_t^f. \quad (33)$$

B_t^G denotes the government's nominal bond position, G_t represents real government spending, T_t captures tax revenues, and R_t^G is the overall return on the government's bond portfolio. Here, $\lambda_t^{G,f}$ indicates the share of government debt allocated to bonds of maturity f . By assumption, the government's holdings satisfy $B_t^{G,f} = \lambda_t^{G,f} B_t^G$ for all $f = 1, \dots, F$,

³⁵By assumption, the government issues bonds across maturities, so $B_t^{G,f} \leq 0$ for all f .

³⁶In practice, a negative position of the central bank can be interpreted as issuing interest-bearing excess reserves—for this, see Frost (1971), Güntner (2015), Mattingly and Abou-Zaid (2015), Primus (2017), and Ennis (2018).

with both $\lambda_t^{G,f}$ and B_t^G treated as exogenous.³⁷

Rearranging (33), the budget constraint can be written as

$$\frac{B_t^G}{P_t} = \frac{R_t^G B_{t-1}^G}{P_t} - [\zeta_t^G + \zeta_t^F - \zeta_t^T] Y_t, \quad (34)$$

where $\zeta_t^G = \frac{G_t}{Y_t}$ and $\zeta_t^T = \frac{T_t}{Y_t}$ denote government spending and taxation as shares of GDP, respectively, and are exogenous in our framework.

3.6 Central Bank

The profits earned on the bonds held by the central bank are given by

$$\Lambda_t^{CB} = R_t^{CB} B_{t-1}^{CB} - B_t^{CB}, \text{ with } B_t^{CB} = \sum_{f=1}^F B_t^{CB,f}, \quad R_t^{CB} = \sum_{f=0}^{F-1} \lambda_{t-1}^{CB,f+1} R_t^f, \quad (35)$$

where B_t^{CB} denotes the central bank's total nominal bond holdings, and R_t^{CB} is the aggregate bond return on its portfolio $\{B_{t-1}^{G,f}\}_{f=1}^F$. The share of bonds allocated to maturity f is given by $\lambda_t^{CB,f}$. Formally,

$$B_t^{CB,f} = \lambda_t^{CB,f} \cdot B_t^{CB}, \quad \forall f = 1, \dots, F. \quad (36)$$

Both B_t^{CB} and $\lambda_t^{CB,f}$ depend on the monetary policy rules, which will be described shortly. The profits earned on central bank balance sheets Λ_t^{CB} are transferred as a lump-sum payment to the household, forming part of the total transfer Λ_t in (14).

3.7 Monetary Policy

The above equation (36) introduces F additional conditions, so that the central bank's monetary policy now has F degrees of freedom that must be specified to achieve a determinate nominal equilibrium.³⁸ The monetary authority implements policy by one of the following three approaches:

³⁷We abstract from the government's optimal maturity structure problem and assume its gross bond positions and portfolios across maturities are exogenous, focusing on the central bank's monetary policy. For this issue, see e.g., Bhandari et al. (2017); Aparisi de Lannoy et al. (2022).

³⁸The central bank selects its bond portfolio across maturities and its total debt position. In conventional New Keynesian models, this portfolio problem is typically abstracted away since the explicit term structure is absent, as discussed in Section 3.1.1.

Policy 1 Setting a rule for each f -maturity bond's holdings $B_t^{CB,f}$, allowing bond prices (and yields) to adjust accordingly.

Policy 2 Setting a rule for each f -maturity bond's yield YD_t^f —or equivalently, its price Q_t^f , and then adjusting the purchase amounts $B_t^{CB,f}$ to achieve the target yield.

Policy 3 Employing a combination of the two approaches across different maturities.

In **Policy 1**, the central bank directly controls the expansion of its bond holdings, resembling traditional rules on money supply growth.³⁹ **Policy 2** exemplifies yield-curve control (YCC) policies, as implemented by the Bank of Japan in 2016.⁴⁰ **Policy 3** combines elements of both and includes the traditional short-term rate target of conventional monetary policy.

While unconventional policies could be implemented according to any of the three cases above, we assume that their defining characteristic is an intent to influence returns along the entire yield curve (in contrast to conventional policies, which focus on short-term rates). Consequently, we choose a YCC policy rule as the representative unconventional policy within our framework. In what follows, we formally characterize the equations describing both conventional and YCC policy rules.

Conventional Policy Conventional monetary policy targets the short-term interest rate—in **Policy 3**, whereby the central bank sets a rule for the one-period bond yield YD_t^1 while leaving longer-term bonds unchanged. We assume the central bank maintains normalized positions for long-term bonds (i.e., adjusted for technology, population, and price growth) as follows:

$$R_{t+1}^0 \equiv YD_t^1 = \max \left\{ YD_t^{1*}, 1 \right\}, \quad (37a)$$

$$\frac{YD_t^{1*}}{\overline{YD}^1} = \left(\frac{YD_{t-1}^{1*}}{\overline{YD}^1} \right)^{\rho_1} \left(\frac{YD_{t-2}^{1*}}{\overline{YD}^1} \right)^{\rho_2} \left[\left(\frac{\Pi_t}{\overline{\Pi}} \right)^{\gamma_\pi} \left(\frac{Y_t}{\overline{Y}_t} \right)^{\gamma_y} \exp \left(\tilde{\epsilon}_t^{YD^1} \right) \right]^{1-(\rho_1+\rho_2)}, \quad (37b)$$

$$\frac{B_t^{CB,f}}{A_t \bar{N}_t P_t} = \frac{\overline{B_t^{CB,f}}}{\overline{ANP}}, \quad \forall f = 2, \dots, F, \quad (37c)$$

³⁹This approach reflects the idea that bond purchases increase money supply. It also specifies the portfolio composition across maturities, even though we do not explicitly incorporate money as a separate variable.

⁴⁰On September 21, 2016, the Bank of Japan set a short-term rate target of -0.1% and capped its 10-year government bond yield near zero. For the U.S. case, see e.g., [Humpage \(2016\)](#) regarding the Fed's yield-curve control during World War II.

where YD_t^{1*} follows a standard Taylor rule that targets deviations in inflation and output, with $\tilde{\varepsilon}_t^{YD^1}$ representing a monetary policy shock. When YD_t^{1*} falls below 1, the ZLB binds, so that $R_{t+1}^0 \equiv YD_t^1 = 1$ as specified in (37a).

Yield-Curve Control Policy Yield-curve control targets the entire yield curve by applying a Taylor rule to each bond maturity, including the short-term rate rule in (37). For each maturity $f \geq 2$, the first line defines the notional YCC target YD_t^{f*} , while the second line combines that target with the conventional-policy yield $YD_t^{CP,f}$. This formulation nests pure YCC and conventional policy as special cases.

$$\frac{YD_t^{f*}}{\overline{YD}^f} = \left(\frac{YD_{t-1}^{f*}}{\overline{YD}^f} \right)^{\rho_1} \left(\frac{YD_{t-2}^{f*}}{\overline{YD}^f} \right)^{\rho_2} \left[\left(\frac{\Pi_t}{\overline{\Pi}} \right)^{\gamma_\pi^f} \left(\frac{Y_t}{\overline{Y}_t} \right)^{\gamma_y^f} \exp \left(\tilde{\varepsilon}_t^{YD^f} \right) \right]^{1-(\rho_1+\rho_2)}, \quad (38a)$$

$$YD_t^{YCC,f} = \overline{YD}^f \left(\frac{YD_t^{CP,f}}{\overline{YD}^f} \right)^{\gamma_{CP}^f} \left[\frac{YD_t^{f*}}{\overline{YD}^f} \right]^{1-\gamma_{CP}^f}, \quad (38b)$$

where γ_π^f and γ_y^f capture the responsiveness to inflation and output deviations for maturities $f = 1, \dots, F$, and $\tilde{\varepsilon}_t^{YD^f}$ is a monetary policy shock affecting the yield on an f -maturity bond. The shock processes, including the factor representation of $\{\tilde{\varepsilon}_t^{YD^f}\}_{f=1}^F$, are specified in Appendix I. The term $YD_t^{CP,f}$ denotes the yield that would prevail under conventional monetary policy—see equation (37). $\gamma_{CP}^f \in [0, 1]$ governs the influence of the conventional target within the yield-curve control framework. When $\gamma_{CP}^f = 1$, the rule reverts to conventional policy; when $\gamma_{CP}^f = 0$, the yield for the f -maturity bond is determined solely by the YCC rule, that is:

$$YD_t^{YCC,f} = \overline{YD}^f \left(\frac{YD_{t-1}^{f*}}{\overline{YD}^f} \right)^{\rho_1} \left(\frac{YD_{t-2}^{f*}}{\overline{YD}^f} \right)^{\rho_2} \left[\left(\frac{\Pi_t}{\overline{\Pi}} \right)^{\gamma_\pi^f} \left(\frac{Y_t}{\overline{Y}_t} \right)^{\gamma_y^f} \exp \left(\tilde{\varepsilon}_t^{YD^f} \right) \right]^{1-(\rho_1+\rho_2)}.$$

In this pure case, the central bank ignores balance sheet exposure concerns for bonds with maturity f —see equation (37c). Intermediate values $0 < \gamma_{CP}^f < 1$ allow the central bank to balance yield-curve intervention with considerations of balance sheet composition and size. For simplicity, in our subsequent analysis we set $\gamma_{CP}^f = 0$ for all f , representing a pure yield-curve control policy.

3.8 Market Clearing

Using the bond market equilibrium condition (31), we obtain the total transfers to households—comprising profits from firms, the central bank, and capital producers, net of government taxes as

$$\Lambda_t = P_t Y_t - P_t G_t - P_t I_t - \int_0^1 W_t(\nu) N_t(\nu) d\nu - R_t^K L_{t-1} + L_t + B_t^H - R_t^{HB} B_{t-1}^H. \quad (39)$$

Combining (39) with the household's budget constraint (14) yields the standard aggregate market-clearing condition:

$$C_t + G_t + I_t = Y_t. \quad (40)$$

3.9 Aggregation

Aggregating the labor demand—equation (30)—across intermediate firms yields:

$$\frac{N_t}{\bar{N}_t} = (1 - \alpha)^{\left(\frac{\eta}{\eta+\alpha}\right)} \left(\frac{C_t}{A_t \bar{N}_t}\right)^{-\alpha\left(\frac{\eta}{\eta+\alpha}\right)} \left(\frac{Y_t}{A_t \bar{N}_t}\right)^{\left(\frac{\eta}{\eta+\alpha}\right)} \left(\frac{P_t^K}{P_t}\right)^{\alpha\left(\frac{\eta}{\eta+\alpha}\right)} \Delta_t^{\frac{\eta}{\eta+1}}, \quad (41)$$

where Δ_t is a measure of price dispersion, recursively defined as:

$$\Delta_t = (1 - \theta) \left(\frac{P_t^*}{P_t}\right)^{-\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)} + \theta \Pi_t^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)} \Delta_{t-1}. \quad (42)$$

Note from equation (41) that, for a given (normalized) level of consumption and output, the required labor N_t increases with price dispersion Δ_t —a proxy for the inefficiency induced by nominal rigidities. Moreover, a higher $\frac{P_t^K}{P_t}$ raises the rental cost of capital, prompting firms to substitute capital with labor, thereby raising N_t . This substitution channel is also captured in the following aggregate capital equilibrium condition.⁴¹

$$\frac{K_t}{A_{t-1} \bar{N}_{t-1}} = \alpha(1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot G A_t \cdot G N \cdot \left(\frac{C_t}{A_t \bar{N}_t}\right)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} \left(\frac{Y_t}{A_t \bar{N}_t}\right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t}\right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha}\right)} \Delta_t. \quad (43)$$

⁴¹We normalize K_t by $A_{t-1} \bar{N}_{t-1}$, as K_t is determined in period $t-1$, while N_t , C_t , and Y_t are normalized by $A_t \bar{N}_t$, and P_t^K is normalized by P_t .

Consequently, aggregate capital K_t increases when consumption, output, or price dispersion increases and/or when the rental price of capital decreases. These two equations highlight the role of firms' substitution between capital and labor during production.

The household satisfies the Euler equation (23), with effective savings rate R_{t+1}^S determined by (18), λ_t^K given by (17), and $\lambda_t^{HB,f}$ specified in (15). The equilibrium condition for the household's allocation between loans and bonds is:

$$\frac{L_t}{B_t^H} = \frac{\gamma_L(1 + \zeta^F)P_t Y_t}{B_t^H} = \frac{\lambda_t^K}{1 - \lambda_t^K}, \quad (44)$$

where B_t^H and L_t represent the aggregate household bond and loan holdings, respectively.

3.9.1 Conventional Policy

Under conventional monetary policy—equation (37)—the central bank does not adjust its normalized holdings of long-term bonds. Consequently, for $f > 1$ the ratio $\frac{B_t^{CB,f}}{A_t N_t P_t}$ remains constant. In this setting, the household's bond portfolio share, $\lambda_t^{HB,f}$, must satisfy:

$$\lambda_t^{HB,f} = -\frac{\frac{B_t^{G,f}}{A_t N_t P_t} + \frac{\overline{B^{CB,f}}}{\overline{ANP}}}{\frac{B_t^H}{A_t N_t P_t}}, \quad \forall f > 1. \quad (45)$$

3.9.2 Yield-Curve Control Policy

In the yield-curve control case—equation (38)—monetary policy affects households' bond portfolio across maturities $\{\lambda_t^{HB,f}\}_{f=1}^F$ and the effective bond return R_t^{HB} via the following relations:

$$\lambda_t^{HB,f} = \left(\frac{z_t^f \mathbb{E}_t [Q_{t,t+1} R_{t+1}^{f-1}]}{\Phi_t^B} \right)^{\kappa_B}, \quad R_{t+1}^{f-1} = \frac{(YD_{t+1}^{f-1})^{-(f-1)}}{(YD_t^f)^{-f}}.$$

Changes in the household's bond portfolio composition affect the effective bond rate R_t^{HB} from (16), the loan rate R_t^K from (17), and the effective savings rate R_t^S from (18). These adjustments, in turn, influence consumption via the Euler equation (23) and other aggregate outcomes as captured by equations (40), (41), (43), and (44).

Figure III.2 in Appendix III provides a graphical illustration of the model. Appendix I provides shock processes for portfolio preference shocks $\{z_t^f\}_{f=1}^F$ and z_t^K , government

spending ratio ζ_t^G and the revenue ratio ζ_t^T , government bond shares $\{\lambda_t^{G,f}\}_{f=1}^F$, and monetary policy shocks $\{\tilde{\varepsilon}_t^{YDf}\}_{f=1}^F$.

We present all other equilibrium conditions and derivations in Online Appendix A.

4 Steady-State (Long-Run) Analysis

4.1 Steady-State Relations

We assume that the central bank's total bond holdings, B^{CB} , equal a constant fraction ζ^{CB} of the total government bond issuance, B^G ; that is, $B^{CB} = \zeta^{CB} B^G$.⁴² Given the portfolio shares $\{\lambda^{CB,f}\}_{f=1}^F$, the steady-state relation in the bond market—equation (32)—becomes:

$$\lambda^{HB,f} = \frac{\lambda^{G,f} + \lambda^{CB,f} \zeta^{CB}}{1 + \zeta^{CB}}. \quad (46)$$

Thus, the household's steady-state bond portfolio shares $\{\lambda^{HB,f}\}_{f=1}^F$ are determined by the exogenous parameters $\{\lambda^{G,f}, \lambda^{CB,f}\}_{f=1}^F$ and ζ^{CB} .

At the steady state, the government's budget constraint can be expressed as:

$$\frac{B^G}{ANP} = - \left(1 - \frac{R^G}{\Pi \cdot GA \cdot GN} \right)^{-1} [\zeta^G + \zeta^F - \zeta^T] \frac{Y}{AN}. \quad (47)$$

Given the normalized output level $\frac{Y}{AN}$ and a positive primary deficit ratio $\zeta^G + \zeta^F - \zeta^T > 0$, an increase in the interest rate on government debt, R^G , results in a larger volume of bond issuance $|B^G|$ (recall that $B^G < 0$) and a higher debt-to-output ratio, as the government must pay more in interest on its debt.⁴³

The remaining steady-state relationships and the procedures for characterizing these conditions are detailed in Online Appendix A.1.3.

⁴²Since $B^G < 0$ and $B^{CB} > 0$ at the steady state, it follows that $\zeta^{CB} < 0$.

⁴³To ensure a positive primary deficit through non-explosive bond issuance at the steady state, we require that $R^G < \Pi \cdot GA \cdot GN$, a condition that holds under our calibration. This condition is consistent with recent literature on debt sustainability, such as Blanchard (2019).

4.2 Results

4.2.1 Calibration

We use publicly available data on (i) Treasury yields, (ii) the Federal Reserve’s holdings of Treasury bonds (December 2002–June 2007), and (iii) the U.S. Treasury’s outstanding bonds (January 1990–January 2007) to calibrate the model’s term structure parameters. We set $F = 120$ to represent maturities up to 30 years (i.e., 120 quarters).⁴⁴ Parameter values are summarized in Table II.2 in Appendix II, with standard parameters calibrated to widely accepted values in the literature.

Household demand for maturity- f bonds, $\lambda_t^{HB,f}$, is driven by the maturity-specific preference shock z_t^f and the demand elasticity κ_B ; similarly, loan demand λ_t^K is driven by z_t^K and κ_S . We calibrate the slope of the yield curve based on $\{z^f\}_{f=1}^F$, and the level of the yield curve using z^K , with fixed values for the demand elasticities κ_B and κ_S . Specifically, we set $\kappa_B = 10$, estimated from macro data using our bond portfolio equation (15) in Online Appendix B,⁴⁵ and $\kappa_S = 6$, based on [Kekre and Lenel \(2023\)](#). With these values, $\{z^f\}$ is selected to match the relative yields across maturities, while z^K is calibrated to reproduce the steady-state return on the household bond portfolio, R^{HB} . The detailed calibration procedure is provided in Online Appendix B.1. We also set $\gamma_L = 3$, which constitutes an upper bound for private debt-to-GDP ratios in advanced economies. Under our calibration, the untargeted steady-state value of R^K is 8.12%, close to the average Moody’s Seasoned BAA Corporate Bond Yield of 7.88% during 1990–2007.

Figure 1 illustrates the distribution of bond holdings across maturities for households, the government, and the central bank, as well as the implied steady-state yield curve. The calibrated values of z^K and $\{z^f\}_{f=1}^F$ are reported in Table II.1 and Figure II.1 in Appendix II. Notably, $z^1 = 1$ is relatively large compared to z^f for $f \geq 2$, reflecting the fact that short-term yields have historically been lower than longer-term yields. This gap likely reflects a safety and/or liquidity premium on short-term bonds, a well-documented feature in the literature—see, for example, [Krishnamurthy and Vissing-Jorgensen \(2012\)](#) and [Caballero and Farhi \(2017\)](#).

⁴⁴When yield data are missing, we interpolate to generate a smooth yield curve. Data source: <https://fiscaldata.treasury.gov/datasets/monthly-statement-public-debt>

⁴⁵Note that our κ_B estimate is in the range of the elasticity estimates in [Jansen et al. \(2024\)](#), who find that elasticity estimates range roughly from 3.8 to 436.6 across different granular investors.

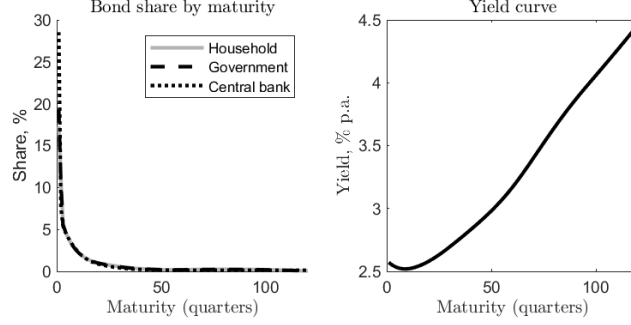


Figure 1: Steady-state bond shares of different entities, with the yield curve

Lastly, our calibration of Taylor coefficients, γ_π^1 , $\gamma_\pi^{f \geq 2}$, γ_y , $\gamma_y^{f \geq 2}$, satisfies model determinacy both in conventional policy and yield-curve control policy regimes, where increasing monetary responsiveness to output and inflation generically guarantees that our model yields unique equilibrium. We characterize the determinacy conditions in Online Appendix G.

4.2.2 Supply and Demand for Bonds

We now examine how changes in the government's debt structure—captured by its Treasury issuance shares $\{\lambda^{G,f}\}_{f=1}^F$ —affect the yield curve at the steady state. Figure 2 shows alternative debt issuance arrangements (left panel) and the corresponding yield curve shifts (right panel); dashed and dotted lines indicate higher long-term issuance compared to the benchmark (solid line).⁴⁶ The model generates a positive correlation between yields and the relative supply $\lambda^{G,f}$; higher issuance at a given maturity increases its yield. This effect not only targets that maturity but also influences equilibrium returns in both Treasury and private loan markets via household portfolio rebalancing, thereby affecting the government's overall bond issuance in general equilibrium.

⁴⁶For illustration purposes, the shifts in portfolio shares are arbitrary.

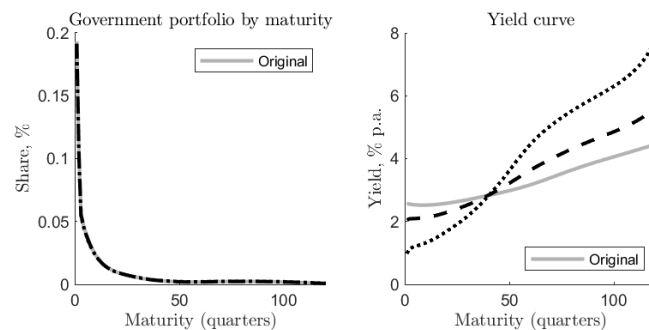


Figure 2: Government's bond issuance portfolio and yield curve

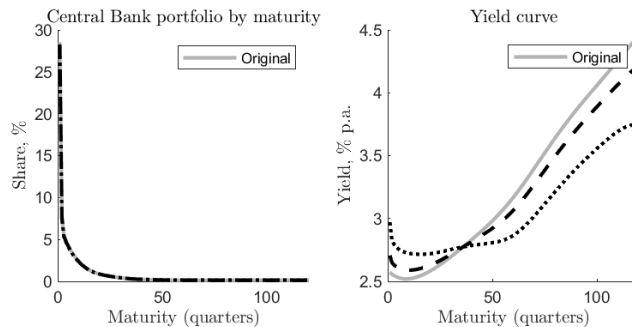


Figure 3: Variations in central bank's bond portfolio across maturities

Figure 3 illustrates an alternative scenario where the central bank adjusts its bond portfolio composition. Here, the dashed and dotted lines represent higher long-term bond purchases relative to the benchmark (solid line). Note that an increased relative purchase of a given maturity is associated with a lower yield, which is consistent with evidence that central bank bond purchases act as an additional demand shock in segmented markets under

general equilibrium.⁴⁷

4.2.3 Comparative Statics with Deficit Ratio

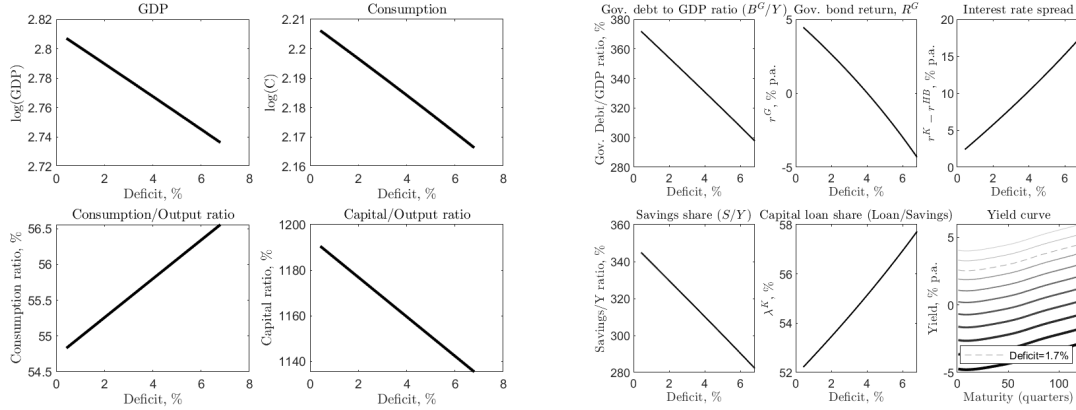


Figure 4: Variations in deficit ratio $\zeta^F + \zeta^G - \zeta^T$

Figure 4 shows comparative statics for changes in the deficit ratio $\zeta^F + \zeta^G - \zeta^T$. A higher deficit ratio can be sustained only if (i) the government issues more bonds, (ii) the effective bond rate R^G decreases, or (iii) output declines, thereby lowering nominal deficit expenditure.

Examining the first case, if the government issues more debt for a given output level, the supply effect discussed in Section 4.2.2 raises R^G , which in turn forces even more debt issuance to cover higher interest costs—a process that spirals indefinitely. In contrast, the second and third cases operate jointly: a higher deficit ratio reduces output, consumption, and capital, which lowers the nominal deficit and government bond issuance, thereby depressing R^G . Meanwhile, the loan rate R^K remains relatively stable, leading to an increased credit spread $r^K - r^{HB}$. Notably, our finding that the debt-to-GDP ratio $\frac{B^G}{Y}$ declines and the entire yield curve shifts downward in response to an increased deficit ratio aligns with previous literature.⁴⁸ Households in turn shift their portfolios towards loans, increasing the capital loan share λ^K .

⁴⁷Krishnamurthy and Vissing-Jorgensen (2011) show that QE2, which concentrated on Treasury markets, had a larger effect on Treasuries and Agency bonds than on mortgage-backed securities and corporate bonds. D’Amico and King (2013) similarly find evidence of both stock and flow effects of QEs on Treasury yields, consistent with imperfect substitutability in the Treasury market.

⁴⁸Laubach (2009) estimates that a 1% point increase in the projected debt-to-GDP ratio raises long-term interest rates by approximately 3–4 basis points.

Additional comparative statics results are provided in Online Appendix E.

5 Short-Run Analysis

5.1 Log-linearization

We now present the log-linearized solution of the model. Lower-case letters denote normalized variables, e.g., $k_t \equiv \frac{K_t}{A_{t-1}N_{t-1}}$, $y_t \equiv \frac{Y_t}{A_tN_t}$, $c_t \equiv \frac{C_t}{A_tN_t}$, $n_t \equiv \frac{N_t}{N_t}$, $p_t^K \equiv \frac{P_t^K}{P_t}$, while hats denote log-deviations from the steady state. Given our system's complexity, we highlight a few key equilibrium equations here; complete derivations are provided in Online Appendix A.

Linearizing the Euler equation (23) yields the dynamic IS equation featuring the effective savings rate $\hat{r}_{t+1}^S$.⁴⁹

$$\hat{c}_t = \mathbb{E}_t [\hat{c}_{t+1} - (\hat{r}_{t+1}^S - \hat{\pi}_{t+1})], \quad (48)$$

where $\hat{r}_t^S$ is derived from (18) as

$$\hat{r}_t^S = \frac{\lambda^K (R^K - R^{HB})}{R^S} \hat{\lambda}_{t-1}^K + \frac{(1 - \lambda^K) R^{HB}}{R^S} \hat{r}_t^{HB} + \frac{\lambda^K R^K}{R^S} \hat{r}_t^K. \quad (49)$$

Note that $\hat{r}_t^S$ depends on the effective bond rate $\hat{r}_t^{HB}$, the loan rate $\hat{r}_t^K$, and the loan share $\hat{\lambda}_{t-1}^K$, capturing the portfolio rebalancing between bonds and loans. The effective bond rate $\hat{r}_t^{HB}$ is obtained by linearizing (16):

$$\hat{r}_t^{HB} = \sum_{f=1}^F \frac{\lambda^{HB,f} (Y D^{f-1})^{-(f-1)}}{R^{HB} (Y D^f)^{-f}} \left[\hat{\lambda}_{t-1}^{HB,f} - (f-1) \cdot \hat{y} d_t^{f-1} + f \cdot \hat{y} d_{t-1}^f \right]. \quad (50)$$

It depends on past yields $\{\hat{y} d_{t-1}^f\}_{f=1}^F$, current yields $\{\hat{y} d_t^{f-1}\}_{f=1}^F$, and household bond portfolio shares $\{\hat{\lambda}_{t-1}^{HB,f}\}_{f=1}^F$, capturing the impact of holding-period returns and portfolio composition on overall bond returns.

To study portfolio reallocation, we log-linearize the household's optimal bond portfolio equation (15) and express the bond shares as functions of past and current yields:

$$\hat{\lambda}_{t-1}^{HB,f} = \kappa_B \mathbb{E}_{t-1} \left[\hat{z}_{t-1}^f - \hat{\pi}_t + \hat{c}_{t-1} - \hat{c}_t - (f-1) \cdot \hat{y} d_t^{f-1} + f \cdot \hat{y} d_{t-1}^f - \hat{\phi}_{t-1}^B \right], \quad (51)$$

⁴⁹Note the similarity between this equation and the one in the simple model—see equation (1).

where $\hat{\phi}_t^B$ depends on $\{\hat{y}d_t^{f-1}, \hat{y}d_{t-1}^f\}_{f=1}^F$ and other aggregate variables. Substituting equation (51) into (50) represents the household's effective bond rate as a function of the entire yield curve.

Finally, linearizing the household's optimal portfolio choice between bonds and loans (17) yields:

$$\hat{\lambda}_t^K = \kappa_S (1 - \lambda^K) (\hat{z}_t^K + \mathbb{E}_t [\hat{r}_{t+1}^K - \hat{r}_{t+1}^{HB}]) . \quad (52)$$

Here, increases in $\hat{z}_t^K$ and in the expected spread $\mathbb{E}_t[\hat{r}_{t+1}^K - \hat{r}_{t+1}^{HB}]$ raise the loan share $\hat{\lambda}_t^K$. Since $\hat{r}_{t+1}^K$ directly enters the effective savings rate $\hat{r}_{t+1}^S$ (48), it influences consumption via the intertemporal substitution channel. Changes in $\hat{r}_{t+1}^K$ also affect λ_t^K , altering loan issuance L_t , output (44), and, consequently, aggregate labor (41) and capital (43) accumulation.

5.2 Results

5.2.1 Impulse-Response without the ZLB

We first analyze impulse responses to various shocks in an economy that remains above the ZLB. The shocks considered are z_t^1 (household preference shock for short-term bonds), z_t^K (household preference shock for loans), and ε_t^T (tax shock). Graphs for additional shocks appear in Online Appendix F.

Short-term Bond Preference Shock, z_t^1 : Figure 5a indicates that a positive shock in z_t^1 increases households' demand for shortest-maturity bonds, which carry the lowest returns along the steady-state yield curve. Under conventional monetary policy (dashed lines), the shock has expansionary effects: households rebalance their portfolios toward lower-yield short-term bonds, reducing the aggregate bond return. This lowers the effective savings rate and boosts aggregate consumption demand. In response, they further adjust their portfolios by allocating more savings toward loans.⁵⁰ Thus, the central bank raises the short-term rate (i.e., policy rate) to mitigate the expansionary effects of the shock.⁵¹

While conventional policy would typically induce an expansion, YCC achieves near-perfect stabilization as it mitigates portfolio reallocation along the yield curve.

⁵⁰Note that a higher loan share lowers the loan rate as seen in Figure 5a, boosting capital accumulation of firms and output through supply-side channels.

⁵¹This increase in the short-term policy rate, in equilibrium, induces households to reallocate more savings toward shorter-term bonds, thereby amplifying the portfolio reallocation toward the shortest-maturity bonds.

Loan Preference Shock, z_t^K : Figure 5b illustrates that a positive z_t^K shock causes households to allocate more savings toward loans. This portfolio reallocation increases aggregate capital and output via supply-side channels, which in turn boosts inflation as aggregate consumption demand increases as well. As a result, the central bank raises the policy rate. As in the previous case, YCC dampens these fluctuations more effectively than conventional policy.⁵²

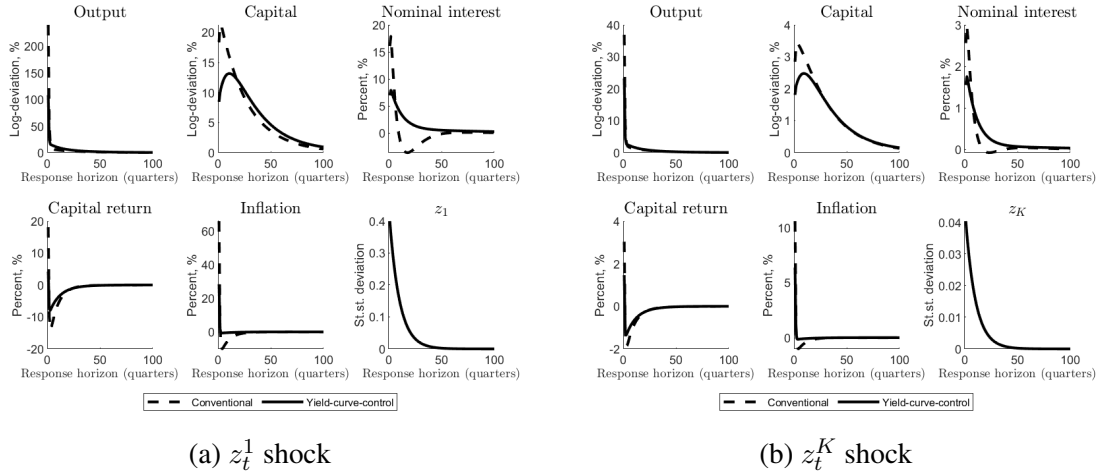


Figure 5: Impulse response to z_t^1 and z_t^K without ZLB

Tax Shock, ε_t^T : Figure 6 presents the impulse responses to a positive tax shock, ε_t^T . Under conventional policy, higher tax revenues allow the government to reduce bond issuance. This decline leads to a lower aggregate bond return and triggers a substantial reallocation of household portfolios toward loans. Consequently, the effective savings rate rises, suppressing aggregate consumption demand. At the same time, borrowing costs of firms decrease, reducing inflation⁵³ and further weakening aggregate demand. Taken together, these effects generate a pronounced recession.

More broadly, Figure 6 highlights *non-Ricardian* features of our model, as tax shocks directly influence the business cycle through their effect on government bond issuance.

In contrast, the YCC policy achieves better stabilization because the entire yield curve shifts downward in response to the shock.

⁵²In this case, YCC policy shifts the entire yield curve upward, diluting the impact of the shock on aggregate capital and output

⁵³This lowers wages as firms substitute capital for labor. The decline in wages further suppresses aggregate demand.

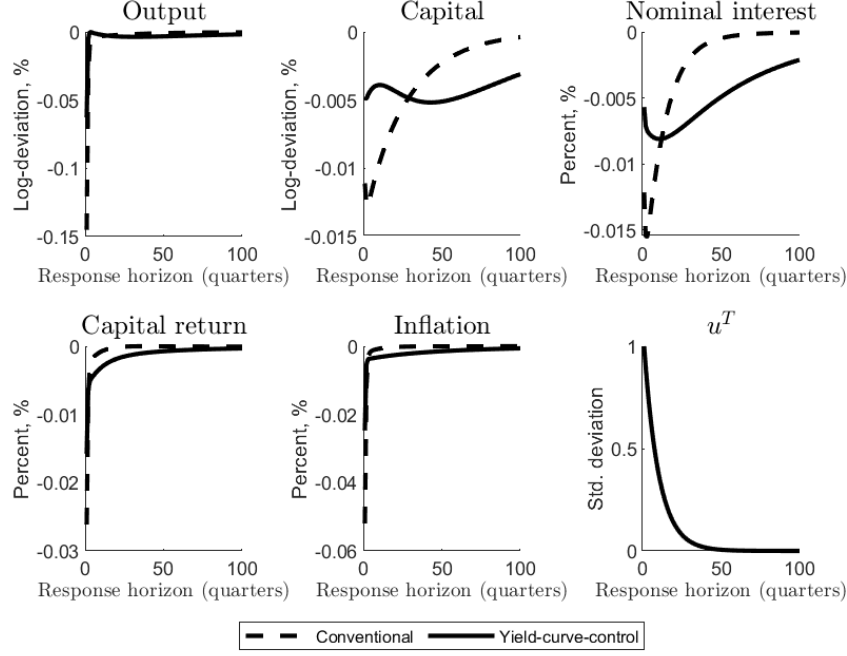


Figure 6: Impulse-response to ε_t^T shock without ZLB

These results highlight that while conventional policy may induce significant business cycle fluctuations following shocks, a yield-curve control regime can more effectively stabilize the economy by actively managing the entire term structure.

5.2.2 Impulse-Response at the ZLB

This section presents impulse responses to various shocks when the ZLB is binding. To illustrate independent ZLB episodes, we calibrate shocks to extremely large—though highly unlikely—magnitudes.

Short-term Bond Preference Shock, z_t^1 : Figure 7a presents that a negative z_t^1 shock—decreasing household demand for shortest-term bonds—lowers the short-term rate under conventional policy (dashed lines) and generates a recession with deflation and output drop by 3–4% per standard deviation. In contrast, YCC (solid lines) mitigates the recession.

However, this intervention also extends the ZLB episode, as the downward pressure on long-term yields feeds back into the household’s portfolio reallocation. Households further tilt their portfolios toward loans, tightening the zero lower bound on the short-term rate as the loan rate declines. This, in turn, puts downward pressure on inflation, as firms substitute

labor with relatively cheaper capital, further lengthening the ZLB episode.

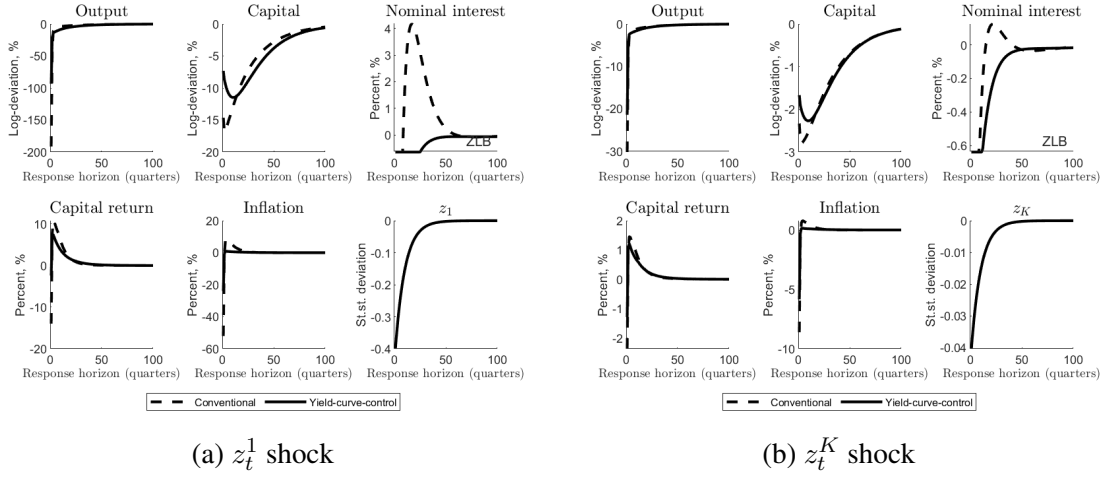


Figure 7: Impulse response to z_t^1 and z_t^K with ZLB

Loan Preference Shock, z_t^K : Figure 7b depicts that a sizable negative z_t^K shock induces households to reduce loan investments and reallocate their portfolios toward bonds. Consequently, bond rates decline and the policy rate remains at the ZLB, while output, capital, inflation, and loan rates all fall. As with the z_t^1 shock, YCC achieves better stabilization but extends the ZLB duration.

Tax Shock, ε_t^T : Figure 8 illustrates that, under conventional policy, a positive tax shock leads to reduced bond issuance. Similar to Figure 6, the resulting declines in bond returns, along with weaker aggregate consumption, interact with endogenous portfolio adjustments by households and firms' input substitution to intensify the downturn—lowering output, capital, inflation, and returns to capital. Under YCC, the central bank compresses the entire yield curve, thereby offsetting part of the contractionary effect. However, as in earlier cases, this policy also prolongs the ZLB episode.

In summary, while yield-curve control policies effectively stabilize the economy by offsetting shocks across the term structure, they tend to extend the duration of ZLB episodes compared with conventional policy, highlighting a trade-off in the stabilization benefits of unconventional monetary measures.

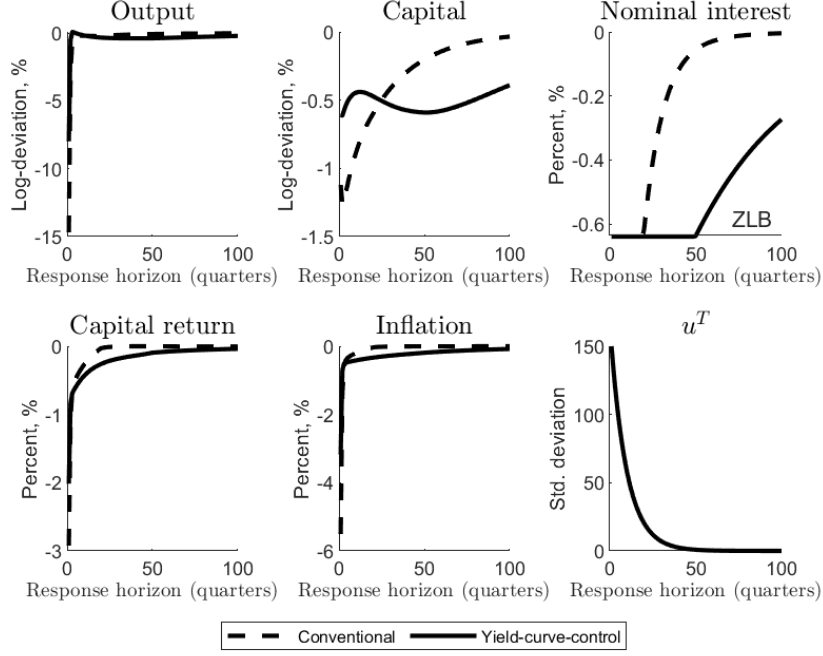


Figure 8: Impulse-response to ε_t^T shock with ZLB

5.2.3 Policy Comparison

For policy comparison, we follow the literature (Woodford, 2003; Coibion et al., 2012) in calculating a second-order approximation of household utility.

Proposition 1 *A second-order approximation of the expected per-period household welfare is given by*

$$\mathbb{E}U_t - \bar{U}^F = \Omega_n \text{Var}(\hat{n}_t) + \Omega_\pi \text{Var}(\hat{\pi}_t) + t.i.p. + h.o.t.,$$

where Ω_n and Ω_π are coefficients that capture the disutility arising from fluctuations in labor and inflation, respectively; $\bar{U}^F$ denotes the efficient (i.e., flexible-price) steady-state⁵⁴ utility around which the approximation is centered; and the acronyms *t.i.p.* and *h.o.t.* refer to terms independent of policy and to higher-order terms, respectively.

Proof. See Online Appendix C. ■

⁵⁴Due to assumed trend inflation at the steady state ($\Pi > 1$), our steady state is not efficient; see Coibion et al. (2012).

Based on the welfare criterion provided in Proposition 1, we compare three monetary policy regimes: (i) conventional policy, represented by (37); (ii) yield-curve control (YCC) policy, represented by (38); and (iii) a *mixed* policy in which the central bank adopts YCC only when the policy rate is constrained by the ZLB.⁵⁵ We evaluate these regimes using ex-ante per-period welfare, mean and median ZLB duration, and ZLB frequency.⁵⁶

	Conventional	Yield-Curve-Control	Mixed Policy
Mean ZLB duration	4.6254 quarters	7.2579 quarters	6.6116 quarters
Median ZLB duration	3 quarters	4 quarters	3 quarters
ZLB frequency	16.2121%	16.202%	23.0404%
Welfare	−1.4229%	−1.0863%	−1.3546%

Table 1: Policy Comparisons

Table 1 shows that, compared with conventional policy, YCC and mixed policy improve welfare by 0.34 and 0.07 percentage points, respectively. However, both policies extend average ZLB spells relative to conventional policy: the mean ZLB duration rises from 4.63 quarters under conventional policy to 7.26 quarters under YCC and 6.61 quarters under mixed policy.

Under mixed policy, stabilization during ZLB episodes improves relative to conventional policy because YCC is activated whenever the short-term rate binds. However, the same household portfolio rebalancing channel can also prolong ZLB spells, and in the simulation mixed policy raises the unconditional ZLB frequency. YCC differs because it adjusts the entire yield curve even outside ZLB periods. By lowering long-term rates in response to adverse shocks, it supports aggregate demand before the short-term rate reaches the ZLB, which tends to reduce ZLB entry. Conditional on reaching the ZLB, however, YCC places additional downward pressure on short-term and loan rates, which lengthens the ZLB spell. These two forces largely offset in the unconditional ZLB frequency, which remains close to the conventional-policy value, but the average ZLB duration is longer under YCC.⁵⁷

⁵⁵Upon exiting the ZLB, the central bank immediately resets its holdings of $f \geq 2$ -maturity bonds to their steady-state levels. This approach differs from Karadi and Nakov (2021), who argue that a gradual exit from QE is optimal given that banks face reduced recapitalization incentives without additional QE.

⁵⁶The reported welfare measure in Table 1 includes terms independent of policy (*t.i.p.*).

⁵⁷Note that the statistics in Table 1 are unconditional, reflecting the potential impact of a range of shocks; specific shocks may produce different patterns across regimes.

6 Conclusion

This paper develops a New Keynesian model that incorporates the term structure of financial markets, as well as the size and maturity composition of government and central bank balance sheets, as active forces shaping the business cycle. The model underscores the importance of market segmentation—captured through households’ finite-elasticity portfolio demand across asset classes and maturities—and of endogenous portfolio reallocation in understanding unconventional monetary policy.

We show that government debt issuance and central bank purchases of bonds with different maturities are the primary determinants of the yield curve’s level and slope. Furthermore, the issuance of risk-free government bonds can stimulate the economy when conventional monetary policy is constrained by the ZLB, aligning with the literature on safe-asset shortages. We compare three policy regimes: (i) conventional policy, (ii) yield-curve control (YCC), where the central bank actively manipulates the entire yield curve, and (iii) a mixed regime, where YCC is employed only when the ZLB binds. Our results suggest that YCC interventions offer greater stabilization than conventional policy, both during normal periods and at the ZLB, though they also tend to extend the duration of ZLB episodes. This occurs because easing long-term rates induces households to shift their portfolios toward shorter maturities, further depressing short-term rates. In this sense, unconventional policies become *addictive*: the central bank relies on them for economic stabilization, yet their use exacerbates conditions that make conventional policy ineffective.

Our model offers a foundation for future research on quantitative tightening in high-inflation settings, particularly in the aftermath of the COVID-19 crisis and under geopolitical uncertainty. In addition, it enables analysis of the political economy of the expansion of central bank balance sheets. Finally, extending the model to an international macroeconomic context would be a valuable direction, allowing for a reexamination of issues such as global imbalances (e.g., [Caballero et al. \(2008, 2021\)](#)) and global monetary cycles (e.g., [Miranda-Agrippino and Rey \(2021\)](#)) while incorporating endogenous movements in the term structure.

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I Shock Processes

We assume that the household portfolio preference shocks $\{z_t^f\}_{f=1}^F$ and z_t^K follow $AR(1)$ processes, given by $z_t^f = \rho_z z_{t-1}^f + \varepsilon_t^{z,f}$ with $\text{Var}(\varepsilon_t^{z,f}) = \sigma_z^2$, and $z_t^K = \rho_z^K z_{t-1}^K + \varepsilon_t^{z,K}$ with $\text{Var}(\varepsilon_t^{z,K}) = (\sigma_z^K)^2$.

For the government spending ratio ζ_t^G and the revenue ratio ζ_t^T , we assume the following shock processes:

$$\zeta_t^G = \frac{1}{1 + a^G \exp(-u_t^G)}, \quad \zeta_t^T = \frac{1}{1 + a^T \exp(-u_t^T)}, \quad (\text{I.1})$$

where a^G and a^T are constants, and u_t^G and u_t^T follow $AR(1)$ processes $u_t^G = \rho_G u_{t-1}^G + \varepsilon_t^G$ and $u_t^T = \rho_T u_{t-1}^T + \varepsilon_t^T$, with ε_t^G and ε_t^T being i.i.d. shocks.

As the government's bond shares across maturities, $\{\lambda_t^{G,f}\}_{f=1}^F$, are exogenously given, we specify their processes as follows:

$$\lambda_t^{G,1} = \frac{1}{1 + \sum_{l=2}^F a^{B,l} \exp(\tilde{u}_t^{B,l})}, \quad \lambda_t^{G,f} = \frac{a^{B,f} \exp(\tilde{u}_t^{B,f})}{1 + \sum_{l=2}^F a^{B,l} \exp(\tilde{u}_t^{B,l})}, \quad \forall f > 1, \quad (\text{I.2})$$

where $a^{B,f}$ for all $f > 1$ are constants. When F is large, we reduce the number of independent shocks to $J \leq F$ by assuming

$$\tilde{u}_t^{B,f} = \sum_{j=2}^J \tau_{fj}^B u_t^{B,j}, \quad (\text{I.3})$$

where $u_t^{B,j}$ follows an independent $AR(1)$ process, and τ_{fj}^B is a constant for all j, f .

Similarly, we reduce the state space of shocks in the yield-curve control regime by representing the monetary policy shocks $\{\tilde{\varepsilon}_t^{YDf}\}_{f=1}^F$ for different maturities as linear combinations of several factors:

$$\tilde{\varepsilon}_t^{YDf} = \sum_{l=1}^L \tau_{f,l}^{YD} \varepsilon_t^{YDl}, \quad (\text{I.4})$$

where $\tau_{f,l}^{YD}$ (for all l, f) are constants, ε_t^{YDl} are i.i.d. shocks, and $L \leq F$.

II Calibration

Steady-state values		
$\frac{C}{AN}$	9.0112	Normalized consumption
$\frac{Y}{AN}$	16.3315	Normalized output
$\frac{K}{AN}$	192.6312	Normalized capital
$\frac{C}{Y}$	0.5518	Consumption per GDP
$\frac{K}{Y}$	11.7951	Capital per GDP
$\frac{P^K}{P}$	0.0335	Normalized rental price of capital
$\lambda^{HB,f}$	See Figure 1	Household's bond portfolio
λ^K	0.5322	Household's loan share out of total savings
R^K	1.0203	Household's loan rates (quarterly)
YD^f	See Figure 1	Equilibrium yield curve

Table II.1: Steady-state values derived from parameter calibration in Table II.2

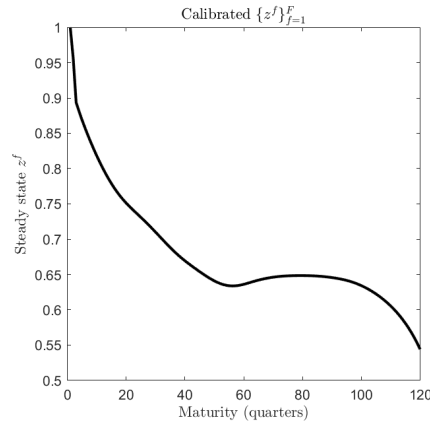


Figure II.1: Calibrated scale parameters of the Fréchet distribution, $\{z^f\}_{f=1}^F$

Households		
(β, η)	(0.998, 1)	Discount factor and Frisch labor elasticity
GN	1.00275	Gross population growth factor
Intermediate good firms		
μ	0.00375	Technology growth rate (log)
γ_L	3	Loan-to-GDP ratio
α	0.4	Capital income share
ϵ	10	Elasticity of substitution between differentiated goods
θ	0.45	Calvo price stickiness parameter
σ_A	0.0036	Standard deviation of technology shock
δ	0.025	Capital depreciation rate
Term structure		
$\{z_t^f\}_{f=1}^F$	See Figure II.1	Bond-maturity preference scale parameters (Appendix B.1)
z^K	1.00893	Capital-preference scale parameter (Appendix B.1)
κ_B	10	Bond maturity shape (volatility) parameter (Appendix B.2)
κ_S	6	Capital shape parameter: Kekre and Lenel (2023)
(ρ_z, ρ_z^K)	(0.9, 0.9)	Autoregressive coefficient: z_t^f and z_t^K
(σ_z, σ_z^K)	4×10^{-9}	Standard deviation for both z_t^f and z_t^K
Government		
ζ^F	0.11111	Government's optimal subsidy to firms
ζ^G	0.07889	Government expenditure per GDP (net subsidy ζ^F)
a^G	11.67606	Government expenditure coefficient
$\zeta^F + \zeta^G - \zeta^T$	0.017	Government deficit per GDP
ζ^T	0.173	Government tax revenue per GDP
(ρ_G, ρ_T)	(0.9, 0.9)	Autoregressive coefficient: expenditure and tax shocks
(σ_G, σ_T)	0.00147	Standard deviation for both expenditure and tax shocks
Central bank		
ζ^{CB}	-0.18	Central bank's balance sheet per issued government bond
$\bar{\pi}$	0.00575	Trend inflation (steady-state inflation), quarterly rate
γ_π^1	2.5	Taylor coefficient of YD_t^1 : responsiveness to inflation
$\gamma_\pi^{f \geq 2}$	0.5	Taylor coefficient of $YD_t^{f \geq 2}$: responsiveness to inflation
γ_y	0.15	Taylor coefficient: responsiveness to output
$\gamma_y^{f \geq 2}$	0	Taylor coefficient of $YD_t^{f \geq 2}$: responsiveness to output
(ρ_1, ρ_2)	(1.05, -0.13)	Autocorrelation in monetary policy: Coibion et al. (2012)
$\sigma_{\varepsilon^{YD^1}}$	0.00096	Standard deviation: monetary policy shock (for YD_t^1)
$\sigma_{\varepsilon^{YD^f}}, f \geq 2$	4×10^{-9}	Standard deviation: monetary policy shock (for $YD_t^{f \geq 2}$)
τ^{YD}	$I_{F \times F}$	State reduction matrix (for $YD_t^{f \geq 2}$)

Table II.2: Parameter values

III Additional Figures

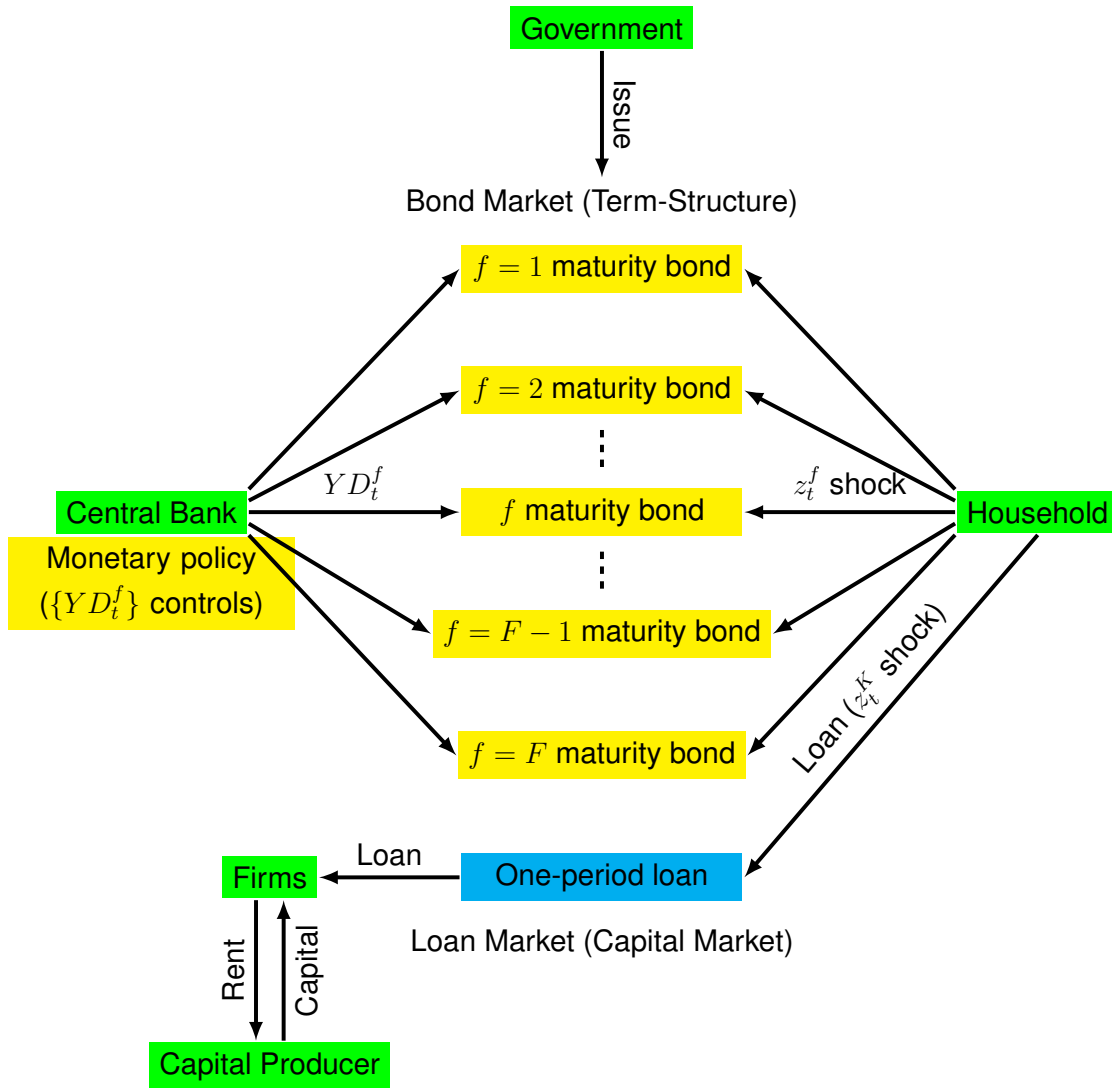


Figure III.2: Markets, agents, and mechanisms. Households allocate wealth between bonds and one-period loans to intermediate goods producers, the government issues bonds across maturities, and the central bank targets either the short rate alone or the full yield curve.

A Derivation and Proofs

A.1 Detailed Derivations in Section 3

As in (29), an intermediate firm ν resetting its price at period t maximizes ¹

$$\max \sum_{j=0}^{\infty} \mathbb{E}_t \left[\theta^j Q_{t,t+j} \cdot \left[\left[1 - \gamma_L \cdot \left(\tilde{R}_{t+j+1}^K - 1 \right) \right] \cdot (1 + \zeta^F) \cdot P_{t+j}(\nu) Y_{t+j}(\nu) \right. \right. \\ \left. \left. - W_{t+j}(\nu) N_{t+j}(\nu) - P_{t+j}^K K_{t+j}(\nu) \right] \right], \quad (\text{A.1})$$

where $Q_{t,t+j} = \beta^j \left(\frac{P_{t+j}}{P_t} \frac{C_{t+j}}{C_t} \right)^{-1}$ is the stochastic discount factor between periods t and $t+j$ and ζ^F is a production subsidy. Also, we define $\tilde{R}_{t+j+1}^K \equiv \mathbb{E}_{t+j}[Q_{t+j,t+j+1} R_{t+j+1}^K]$. Solving for the optimal resetting price at period t , P_t^* , we obtain

$$\frac{P_t^*}{P_t} = \frac{\mathbb{E}_t \left[\sum_{j=0}^{\infty} \theta^j Q_{t,t+j} \left(\frac{P_{t+j}}{P_t} \right)^{\epsilon+1} Y_{t+j} \left(\frac{(1 + \zeta^F)^{-1} \epsilon}{\epsilon - 1} \right) \left(\frac{MC_{t+j|t}(\nu)}{P_{t+j}} \right) \right]}{\mathbb{E}_t \left[\sum_{j=0}^{\infty} \theta^j Q_{t,t+j} \left(\frac{P_{t+j}}{P_t} \right)^{\epsilon} Y_{t+j} \left[1 - \gamma_L \cdot \left(\tilde{R}_{t+j+1}^K - 1 \right) \right] \right]}, \quad (\text{A.2})$$

where subindex $t+j|t$ represents the value of the variable conditional on the firm having reset its price last time at period t , and $MC_{t+j|t}(\nu)/P_t$ is the real marginal cost of production, defined as

$$\frac{MC_{t+j|t}(\nu)}{P_{t+j}} = \left(\frac{P_{t+j}^K}{P_{t+j}} \right)^{\alpha} \left(\frac{W_{t+j|t}(\nu)}{P_{t+j} A_{t+j}} \right)^{1-\alpha}. \quad (\text{A.3})$$

The expression for the one-period ahead stochastic discount factor can be expressed as

$$Q_{t,t+1} = \frac{\beta}{\Pi_{t+1} G A_{t+1} G N} \frac{\left(\frac{C_t}{A_t N_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} N_{t+1}} \right)} \quad (\text{A.4})$$

where Π_{t+1} , $G A_{t+1}$, and $G N$ stand the gross growth rates of inflation, TFP and population, respectively.

¹The factor θ^j is the probability that the price chosen at period t remains in effect through period $t+j$, so the resetting-price problem conditions on the path along which the firm does not reoptimize.

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Combining the expression for $Q_{t,t+1}$ in (A.4) with the household's Euler equation (23), we obtain

$$\mathbb{E}_t [Q_{t,t+1} R_{t+1}^S] = 1. \quad (\text{A.5})$$

A.1.1 Detailed Derivation in Section 3.9

Using equations (22), (26), (27) and (A.3), we can express firm-specific marginal costs as a function of the aggregate variables as in

$$\frac{MC_{t+j|t}(\nu)}{P_{t+j}} = (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{C_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} \left(\frac{Y_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{P_{t+j}^K}{P_{t+j}} \right)^{\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} \left(\frac{P_t^*}{P_{t+j}} \right)^{-\left(\frac{\epsilon(1-\alpha)}{\eta+\alpha} \right)}.$$

Similarly, we integrate the firm-level capital and labor demand—equation (30), with the wage and firm output substituted out via (22) and (27)—across the continuum of firms. We obtain the following capital and labor aggregation conditions.

$$\frac{K_t}{A_{t-1}\bar{N}_{t-1}} = \alpha(1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot GA_t \cdot GN \cdot \left(\frac{C_t}{A_t\bar{N}_t} \right)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} \left(\frac{Y_t}{A_t\bar{N}_t} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t} \right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha} \right)} \Delta_t, \quad (\text{A.6})$$

$$\frac{N_t}{\bar{N}_t} = (1 - \alpha)^{\left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{C_t}{A_t\bar{N}_t} \right)^{-\alpha \left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{Y_t}{A_t\bar{N}_t} \right)^{\left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{P_t^K}{P_t} \right)^{\alpha \left(\frac{\eta}{\eta+\alpha} \right)} \Delta_t^{\frac{\eta}{\eta+1}}, \quad (\text{A.7})$$

where Δ_t is a measure of price-dispersion that can be recursively defined as

$$\Delta_t = (1 - \theta) \left(\frac{P_t^*}{P_t} \right)^{-\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} + \theta \Pi_t^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} \Delta_{t-1}. \quad (\text{A.8})$$

Because N_t is the aggregate labor index of (13), the labor-side integration $\int_0^1 N_t(\nu)^{(\eta+1)/\eta} d\nu$ leading to (A.7) rescales the firm-level price-dispersion factor exactly onto the same Δ_t that appears on the capital side (A.6), raised to $\eta/(\eta+1)$.

Plugging the real marginal cost and the expressions for Q_{t+j} into the optimal resetting

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price equation (A.2), we obtain

$$\begin{aligned} & \left(\frac{P_t^*}{P_t} \right)^{1+\epsilon \left(\frac{1-\alpha}{\eta+\alpha} \right)} \\ &= \frac{\mathbb{E}_t \left[\sum_{j=0}^{\infty} (\theta\beta)^j (1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}} \frac{(1+\varsigma_F)^{-1}\epsilon}{\epsilon-1} \left(\frac{C_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right)^{-\alpha \frac{\eta+1}{\eta+\alpha}} \left(\frac{Y_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_{t+j}}{P_t} \right)^{\epsilon \frac{\eta+1}{\eta+\alpha}} \left(\frac{P_{t+j}^K}{P_{t+j}} \right)^{\alpha \frac{\eta+1}{\eta+\alpha}} \right]}{\mathbb{E}_t \left[\sum_{j=0}^{\infty} (\theta\beta)^j \left(\frac{P_{t+j}}{P_t} \right)^{\epsilon-1} \left(\frac{C_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right)^{-1} \left(\frac{Y_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right) \left[1 - \gamma_L \cdot (\tilde{R}_{t+j+1}^K - 1) \right] \right]}. \end{aligned} \quad (\text{A.9})$$

We can simplify this expression as

$$\frac{P_t^*}{P_t} = \left(\frac{F_t}{H_t} \right)^{\frac{1}{1+\epsilon \left(\frac{1-\alpha}{\eta+\alpha} \right)}}, \quad (\text{A.10})$$

where F_t and H_t are recursively written as

$$\begin{aligned} F_t &= (1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{(1+\varsigma_F)^{-1}\epsilon}{\epsilon-1} \right) \left(\frac{C_t}{A_t\bar{N}_t} \right)^{-\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} \left(\frac{Y_t}{A_t\bar{N}_t} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t} \right)^{\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} + \theta\beta \mathbb{E}_t \left[\Pi_{t+1}^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} F_{t+1} \right], \\ H_t &= \left(\frac{C_t}{A_t\bar{N}_t} \right)^{-1} \frac{Y_t}{A_t\bar{N}_t} \left[1 - \gamma_L \cdot (\tilde{R}_{t+1}^K - 1) \right] + \theta\beta \mathbb{E}_t \left[\Pi_{t+1}^{\epsilon-1} H_{t+1} \right]. \end{aligned} \quad (\text{A.11})$$

Using (A.11), we obtain the following equilibrium price-resetting condition:

$$\frac{F_t}{H_t} = \left(\frac{1-\theta}{1-\theta\Pi_{t+1}^{\epsilon-1}} \right)^{\left(\frac{1}{\epsilon-1} \right) [1+\epsilon \left(\frac{1-\alpha}{\eta+\alpha} \right)]}. \quad (\text{A.12})$$

Combining the market clearing condition (40), the definition of the government spending share (34), and the capital accumulation equation (24), we obtain the following expression:

$$\frac{C_t}{A_t\bar{N}_t} = (1-\varsigma_t^G) \left(\frac{Y_t}{A_t\bar{N}_t} \right) + \left(\frac{1-\delta}{GA_t \cdot GN} \right) \left(\frac{K_t}{A_{t-1}\bar{N}_{t-1}} \right) - \left(\frac{K_{t+1}}{A_t\bar{N}_t} \right). \quad (\text{A.13})$$

We now rewrite equation (23) as

$$1 = \beta \cdot \mathbb{E}_t \left[\frac{R_{t+1}^S}{\Pi_{t+1} GA_{t+1} GN} \cdot \frac{\left(\frac{C_t}{A_t\bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1}\bar{N}_{t+1}} \right)} \right]. \quad (\text{A.14})$$

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Since R_{t+1}^S depends on bonds return R_{t+1}^{HB} and loans return R_{t+1}^K while shares of savings that flow into bonds $(1 - \lambda_t^K)$ and loans (λ_t^K) are endogenous, we start from analyzing R_{t+1}^{HB} . We can rewrite the aggregate return indices as functions of the bond yields $\{YD_t^f\}_{f=1}^F$ as

$$R_t^j = \sum_{f=0}^{F-1} \lambda_{t-1}^{j,f+1} \frac{(YD_t^f)^{-f}}{(YD_{t-1}^{f+1})^{-(f+1)}}, \quad j \in \{HB, G, CB\},$$

and also the household's bond portfolio share as

$$\lambda_t^{HB,f} = \left(\frac{\mathbb{E}_t \left[\frac{\beta \cdot z_t^f}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \cdot \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right)} \cdot \frac{(YD_{t+1}^{f-1})^{-(f-1)}}{(YD_t^f)^{-f}} \right]}{\Phi_t^B} \right)^{\kappa_B}, \quad \forall f,$$

$$\Phi_t^B = \left[\sum_{j=1}^F \mathbb{E}_t \left[\frac{\beta \cdot z_t^j}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \cdot \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right)} \cdot \frac{(YD_{t+1}^{j-1})^{-(j-1)}}{(YD_t^j)^{-j}} \right] \right]^{\kappa_B \frac{1}{\kappa_B}}.$$

Now we find the equilibrium condition for the bond shares of the agents. Using the bond market equilibrium condition (31), we obtain

$$\lambda_t^{HB,f} = \frac{B_t^{G,f} + B_t^{CB,f}}{B_t^G + B_t^{CB}} = \frac{\lambda_t^{G,f} B_t^G + \lambda_t^{CB,f} B_t^{CB}}{B_t^G + B_t^{CB}}. \quad (\text{A.15})$$

We can rearrange the previous expression as

$$\lambda_t^{CB,f} = \lambda_t^{HB,f} + \left(\lambda_t^{HB,f} - \lambda_t^{G,f} \right) \cdot \frac{B_t^G}{B_t^{CB}}. \quad (\text{A.16})$$

Summing from $f = 2$ to F , and using $\sum_{f=2}^F \lambda_t^{j,f} = 1 - \lambda_t^{j,1}$, $j \in \{HB, G, CB\}$ we obtain

$$\sum_{f=2}^F \lambda_t^{CB,f} = 1 - \lambda_t^{HB,1} + \left(\lambda_t^{G,1} - \lambda_t^{HB,1} \right) \cdot \frac{B_t^G}{B_t^{CB}}. \quad (\text{A.17})$$

Plugging (A.17) into (A.16) and after some rearrangements, we obtain

$$\lambda_t^{CB,f} = \frac{\lambda_t^{HB,f} \left(\lambda_t^{CB,1} - \lambda_t^{G,1} \right) - \lambda_t^{G,f} \left(\lambda_t^{CB,1} - \lambda_t^{HB,1} \right)}{\lambda_t^{HB,1} - \lambda_t^{G,1}}, \quad f > 1. \quad (\text{A.18})$$

Now, we can obtain an expression for the central bank's bond holdings using (A.17) as

$$B_t^{CB} = \left(\frac{\lambda_t^{HB,1} - \lambda_t^{G,1}}{\lambda_t^{CB,1} - \lambda_t^{HB,1}} \right) \cdot B_t^G. \quad (\text{A.19})$$

Combining (31) and (A.19), we obtain

$$\frac{B_t^H}{A_t \bar{N}_t} = - \left(\frac{\lambda_t^{CB,1} - \lambda_t^{G,1}}{\lambda_t^{CB,1} - \lambda_t^{HB,1}} \right) \cdot \frac{B_t^G}{A_t \bar{N}_t}. \quad (\text{A.20})$$

Combining $L_t = \lambda_t^K S_t$ and $B_t^H = (1 - \lambda_t^K) S_t$ with $L_t = \gamma_L \cdot (1 + \zeta^F) \cdot P_t Y_t$, we obtain

$$\frac{B_t^H}{A_t \bar{N}_t P_t} = \gamma_L \cdot (1 + \zeta^F) \cdot \left(\frac{1 - \lambda_t^K}{\lambda_t^K} \right) \left(\frac{Y_t}{A_t \bar{N}_t} \right). \quad (\text{A.21})$$

Combining equations (A.20) and (A.21), we obtain:

$$- \left(\frac{\lambda_t^{CB,1} - \lambda_t^{G,1}}{\lambda_t^{CB,1} - \lambda_t^{HB,1}} \right) \cdot \frac{B_t^G}{A_t \bar{N}_t P_t} = \gamma_L \cdot (1 + \zeta^F) \cdot \left(\frac{1 - \lambda_t^K}{\lambda_t^K} \right) \left(\frac{Y_t}{A_t \bar{N}_t} \right). \quad (\text{A.22})$$

A.1.2 Conventional Policy in Section 3.9.1

Using the bond market equilibrium (31) with $\sum_{f=2}^F \lambda_t^{HB,f} = 1 - \lambda_t^{HB,1}$, we get

$$B_t^H = - \frac{\sum_{i=2}^F \left(B_t^{G,i} + B_t^{CB,i} \right)}{1 - \lambda_t^{HB,1}}. \quad (\text{A.23})$$

Combining (A.23) with (31) and (37c), we obtain the equilibrium set of equations,

$$\frac{\lambda_t^{HB,f}}{1 - \lambda_t^{HB,1}} = \frac{\frac{B_t^{G,f}}{A_t \bar{N}_t P_t} + \frac{\bar{B}^{CB,f}}{A \bar{N} P}}{\sum_{i=2}^F \left(\frac{B_t^{G,i}}{A_t \bar{N}_t P_t} + \frac{\bar{B}^{CB,i}}{A \bar{N} P} \right)}, \quad \forall f > 1. \quad (\text{A.24})$$

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Combining (A.21), (A.23), and (37c) yields the following equilibrium equation:

$$-\frac{\sum_{i=2}^F \left(\frac{B_t^{G,i}}{A_t \bar{N}_t P_t} + \frac{\overline{B^{CB,i}}}{A \bar{N} P} \right)}{1 - \lambda_t^{HB,1}} = \gamma_L \cdot (1 + \zeta^F) \cdot \left(\frac{1 - \lambda_t^K}{\lambda_t^K} \right) \left(\frac{Y_t}{A_t \bar{N}_t} \right), \quad (\text{A.25})$$

where normalized bond positions of the central bank are exogenously given. Finally, combining (A.24) and (A.25), we finally obtain

$$-\left(\frac{B_t^{G,f}}{A_t \bar{N}_t P_t} + \frac{\overline{B^{CB,f}}}{A \bar{N} P} \right) \cdot \left(\lambda_t^{HB,f} \right)^{-1} = \gamma_L \cdot (1 + \zeta^F) \cdot \left(\frac{1 - \lambda_t^K}{\lambda_t^K} \right) \left(\frac{Y_t}{A_t \bar{N}_t} \right), \quad \forall f > 1. \quad (\text{A.26})$$

A.1.3 Steady-State Derivations in Section 4.1

At the steady state, the central bank decides the level of bond holdings of maturities $B^{CB,f}$ that it wants to hold. It can be calibrated to match the data of the central bank's balance sheet. Given $\{\lambda^{CB,f}\}$ and the size of its portfolio B^{CB} , which is ζ^{CB} fraction of total government bond issuance satisfying $B^{CB} = \zeta^{CB} \cdot B^G$, we obtain the steady state households' bond shares as in the main-text relation (46):

$$\lambda^{HB,f} = \frac{\lambda^{G,f} + \lambda^{CB,f} \cdot \zeta^{CB}}{1 + \zeta^{CB}}.$$

Throughout this section we adopt the remaining-maturity-at-realization indexing for bond returns of the main text. Concretely, R^{f-1} is the steady-state value of R_{t+1}^{f-1} in (16) of the main text—the per-period return realized one period after holding an f -maturity bond. Equivalently, $R^{f-1} = (YD^f)^f / (YD^{f-1})^{f-1}$ at steady state, with $YD^0 = 1$.

From the definition of R^{HB} we have

$$\sum_{f=1}^F \lambda^{HB,f} \cdot \left(\frac{R^{f-1}}{R^{HB}} \right) = 1,$$

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which together with equation (15) can be rearranged as:

$$\lambda^{HB,f} = \left(\frac{z^f \cdot \frac{R^{f-1}}{R^{HB}}}{\tilde{\Phi}^B} \right)^{\kappa_B}, \quad \forall f, \quad \text{with } \tilde{\Phi}^B = \left[\sum_{j=1}^F \left[z^j \cdot \frac{R^{j-1}}{R^{HB}} \right]^{\kappa_B} \right]^{\frac{1}{\kappa_B}}. \quad (\text{A.27})$$

The above (46) and (A.27) jointly determine the steady state yields and household shares. Unfortunately, there is no analytical expression for them and we have to solve for the steady state values numerically. How we proceed, relying on simple iterations:

Step 1 Assume some initial guess for $\left\{ \frac{R^{f-1,guess}}{R^{HB}} \right\}_{f=1}^F$.

Step 2 Construct $\tilde{\Phi}^{B,old}$ using previous guess with $\tilde{\Phi}^B$ in equation (A.27).

Step 3 Update estimates on $\left\{ \frac{R^{f-1}}{R^{HB}} \right\}_{f=1}^F$ in two steps. First, for each $f > 1$, use $\tilde{\Phi}^{B,old}$ to compute

$$\frac{R^{f-1,new}}{R^{HB}} = (\lambda^{HB,f})^{\frac{1}{\kappa_B}} (z^f)^{-1} \tilde{\Phi}^{B,old}, \quad f > 1.$$

Then, using the just-updated $\left\{ \frac{R^{f-1,new}}{R^{HB}} \right\}_{f=2}^F$, compute

$$\frac{R^{0,new}}{R^{HB}} = \frac{1 - \sum_{f=2}^F \lambda^{HB,f} \left(\frac{R^{f-1,new}}{R^{HB}} \right)}{\lambda^{HB,1}}.$$

Step 4 Construct new shares $\lambda^{HB,f,new}$ by plugging $\left\{ \frac{R^{f-1,new}}{R^{HB}} \right\}_{f=1}^F$ into (A.27). Compute the discrepancy between these shares and the true ones found in (46). If the error is big, set $\frac{R^{f-1,guess}}{R^{HB}} = \frac{R^{f-1,new}}{R^{HB}}$ and repeat from Step 2 until convergence.

Using (18) and (23), we obtain

$$R^{HB} = \frac{\beta^{-1} \Pi \cdot GA \cdot GN}{1 - \lambda^K} - \frac{\lambda^K}{1 - \lambda^K} R^K. \quad (\text{A.28})$$

Combining the steady-state expression for R^S derived from equation (18) together with

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(A.28), we obtain

$$R^S = \beta^{-1} \Pi \cdot GA \cdot GN. \quad (\text{A.29})$$

From the Euler equation (A.5) and (A.29), we obtain an expression for the steady-state stochastic discount factor as

$$Q = \frac{1}{\beta^{-1} \Pi \cdot GA \cdot GN}. \quad (\text{A.30})$$

Therefore, by equation (A.30), the steady-state one-period ahead expected discounted return on capital, defined as $\tilde{R}_{t+1}^K \equiv \mathbb{E}_t[Q_{t,t+1} R_{t+1}^K]$, is given by

$$\tilde{R}^K = \frac{R^K}{\beta^{-1} \Pi \cdot GA \cdot GN}. \quad (\text{A.31})$$

We can rewrite R^G as

$$R^G = \chi^G \cdot R^{HB}, \quad \chi^G = \sum_{f=1}^F \lambda^{G,f} \cdot \left(\frac{R^{f-1}}{R^{HB}} \right), \quad (\text{A.32})$$

and from (A.28), it becomes

$$R^G = \chi^G \cdot \left[\frac{\beta^{-1} \Pi \cdot GA \cdot GN}{1 - \lambda^K} - \frac{\lambda^K}{1 - \lambda^K} R^K \right]. \quad (\text{A.33})$$

Using (17), we obtain an expression for the steady-state share of loans as

$$\lambda^K = \frac{\left(z^K \cdot \frac{R^K}{R^{HB}} \right)^{\kappa_S}}{1 + \left(z^K \cdot \frac{R^K}{R^{HB}} \right)^{\kappa_S}}. \quad (\text{A.34})$$

Further combining (A.28) and (A.34), we obtain

$$\frac{\lambda^K}{1 - \lambda^K} \equiv \left(z^K \cdot \frac{R^K}{R^{HB}} \right)^{\kappa_S} = \frac{\beta^{-1} \cdot \Pi \cdot GA \cdot GN - R^{HB}}{R^K - \beta^{-1} \cdot \Pi \cdot GA \cdot GN} \quad (\text{A.35})$$

The equilibrium government bonds are obtained from its budget constraint (34) and

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written as in the main-text relation (47):

$$\frac{B^G}{A\bar{N}P} = - \left(1 - \frac{R^G}{\Pi \cdot GA \cdot GN} \right)^{-1} [\zeta^G + \zeta^F - \zeta^T] \left(\frac{Y}{A\bar{N}} \right).$$

The government is a net borrower in the model, so $B^G < 0$ at the steady-state. Also, we would like to match the data in which the government runs primary deficit, i.e., $\zeta^G + \zeta^F - \zeta^T > 0$. The only way to achieve that is by having $R^G < \Pi \cdot GA \cdot GN$. Combining $B^{CB} = \zeta^{CB} \cdot B^G$, $B_t^H = -(B_t^G + B_t^{CB})$, (A.21), (A.32), and (A.35) yields

$$\gamma_L = \left(\frac{1 + \zeta^{CB}}{1 + \zeta^F} \right) \cdot [\zeta^G + \zeta^F - \zeta^T] \cdot \left(1 - \frac{\chi^G}{\Pi \cdot GA \cdot GN} \cdot R^{HB} \right)^{-1} \cdot \left[\frac{\beta^{-1} \cdot \Pi \cdot GA \cdot GN - R^{HB}}{R^K - \beta^{-1} \cdot \Pi \cdot GA \cdot GN} \right]. \quad (\text{A.36})$$

(A.35) and (A.36) form a nonlinear system of equations on the unknown steady-states of R^K and R^{HB} . After then, we can then simply back out bond returns as

$$R^{f-1} = R^{HB} \cdot \left(\frac{R^{f-1}}{R^{HB}} \right), \quad f = 1, \dots, F.$$

Now that we have characterized the bond returns, we can recursively obtain the bond yields using

$$YD^f = \left[R^{f-1} \cdot (YD^{f-1})^{f-1} \right]^{\frac{1}{f}},$$

where $YD^0 = 1$. The price dispersion is given by

$$\Delta = \left[\frac{1 - \theta}{1 - \theta \Pi^{\epsilon(\frac{\eta+1}{\eta+\alpha})}} \right] \left(\frac{1 - \theta \Pi^{\epsilon-1}}{1 - \theta} \right)^{\left(\frac{\epsilon}{\epsilon-1} \right) \left(\frac{\eta+1}{\eta+\alpha} \right)}. \quad (\text{A.37})$$

From the capital producer's optimization, we obtain an expression for P^K

$$\frac{P^K}{P} = \beta^{-1} \cdot GA \cdot GN - (1 - \delta). \quad (\text{A.38})$$

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The steady state representation of firms' pricing (A.11) can be written as

$$F = \xi^F \cdot \left(\frac{C}{A\bar{N}} \right)^{-\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} \left(\frac{Y}{A\bar{N}} \right)^{\frac{\eta+1}{\eta+\alpha}}, \quad (\text{A.39})$$

$$H = \xi^H \cdot \left(\frac{C}{A\bar{N}} \right)^{-1} \left(\frac{Y}{A\bar{N}} \right), \quad (\text{A.40})$$

$$\text{with } \xi^F = (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left[1 - \theta\beta\Pi^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} \right]^{-1} \left(\frac{(1 + \zeta^F)^{-1}\epsilon}{\epsilon - 1} \right) \left(\frac{P^K}{P} \right)^{\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)},$$

$$\xi^H = \left[1 - \gamma_L \cdot \left(\frac{R^K}{R^S} - 1 \right) \right] \cdot [1 - \theta\beta \cdot \Pi^{\epsilon-1}]^{-1}.$$

Using (A.39), (A.40) and (A.12), we obtain

$$\left(\frac{C}{A\bar{N}} \right)^{\left(\frac{(1-\alpha)\eta}{\eta+\alpha} \right)} = \xi^Y \cdot \left(\frac{Y}{A\bar{N}} \right)^{-\left(\frac{1-\alpha}{\eta+\alpha} \right)}, \quad \text{with } \xi^Y = \left(\frac{\xi^H}{\xi^F} \right) \left(\frac{1 - \theta}{1 - \theta\Pi^{\epsilon-1}} \right)^{\left(\frac{1}{\epsilon-1} \right) \left[1 + \epsilon \left(\frac{1-\alpha}{\eta+\alpha} \right) \right]}. \quad (\text{A.41})$$

Combining (A.41) and (A.6), we obtain

$$\frac{K}{A\bar{N}} = \xi^K \cdot \left(\frac{Y}{A\bar{N}} \right), \quad \text{with } \xi^K = \alpha \cdot (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot GA \cdot GN \cdot \left(\frac{P^K}{P} \right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha} \right)} \Delta \cdot \xi^Y. \quad (\text{A.42})$$

Henceforth, $C/A\bar{N}$ and $Y/A\bar{N}$ stand for the steady-states of $C_t/(A_t\bar{N}_t)$ and $Y_t/(A_t\bar{N}_t)$, while $K/A\bar{N}$ stands for $K_t/(A_{t-1}\bar{N}_{t-1})$, matching the timing convention in (A.6).

Plugging (A.42) into the aggregate resource constraint, we obtain

$$\frac{C}{A\bar{N}} = \xi^C \cdot \left(\frac{Y}{A\bar{N}} \right), \quad \text{with } \xi^C = (1 - \zeta^G) - \xi^K \cdot \left[1 - \left(\frac{1 - \delta}{GA \cdot GN} \right) \right]. \quad (\text{A.43})$$

Combining (A.41) and (A.43), we obtain

$$\frac{Y}{A\bar{N}} = (\xi^Y)^{\left(\frac{1}{1-\alpha} \right) \left(\frac{\eta+\alpha}{\eta+1} \right)} (\xi^C)^{-\left(\frac{\eta}{\eta+1} \right)}. \quad (\text{A.44})$$

A.1.4 Log-linearization

We start by log-linearizing the equations that are common to the conventional policy model and the yield-curve-control model, then derive the ones that are different. Log-linearize the

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gross labor-productivity growth rate, defined as $GA_t \equiv \exp(\mu + \varepsilon_t^A)$, and the spending and revenue ratio expressions (I.1):

$$\hat{g}a_t = \hat{\varepsilon}_t^A, \quad \hat{\zeta}_t^G = \frac{a^G}{1 + a^G} \cdot \hat{u}_t^G, \quad \hat{\zeta}_t^T = \frac{a^T}{1 + a^T} \cdot \hat{u}_t^T. \quad (\text{A.45})$$

Equations (A.13) and (A.14), with the help of (A.45), can be linearized as

$$\hat{c}_t = \left[(1 - \zeta^G) \cdot \frac{Y}{C} \right] \left[\hat{y}_t - \frac{1}{1 + a^G} \cdot \hat{u}_t^G \right] + \left[\frac{1 - \delta}{GA \cdot GN} \frac{K}{C} \right] (\hat{k}_t - \hat{\varepsilon}_t^A) - \frac{K}{C} \hat{k}_{t+1}, \quad (\text{A.46})$$

$$\hat{c}_t = \mathbb{E}_t [\hat{c}_{t+1} - (\hat{r}_{t+1}^S - \hat{\pi}_{t+1})], \quad (\text{A.47})$$

where we use (A.45) to substitute for $\hat{\zeta}_t^G$ and $\hat{g}a_t$. The term $\mathbb{E}_t \hat{g}a_{t+1}$ drops out of the Euler equation because technology growth is i.i.d. Plugging (A.46) into (A.47), we obtain the following dynamic IS equation for output $\hat{y}_t$.

$$\begin{aligned} \hat{y}_t = \mathbb{E}_t \left[\hat{y}_{t+1} - \left[\frac{(1 - \zeta^G)^{-1}(1 - \delta)}{GA \cdot GN} \cdot \frac{K}{Y} \right] (\hat{k}_t - \hat{\varepsilon}_t^A) + (1 - \zeta^G)^{-1} \left[1 + \frac{1 - \delta}{GA \cdot GN} \right] \frac{K}{Y} \hat{k}_{t+1} \right. \\ \left. - (1 - \zeta^G)^{-1} \frac{K}{Y} \hat{k}_{t+2} - (1 - \zeta^G)^{-1} \frac{C}{Y} (\hat{r}_{t+1}^S - \hat{\pi}_{t+1}) + \frac{1 - \rho_G}{1 + a^G} \cdot \hat{u}_t^G \right]. \end{aligned} \quad (\text{A.48})$$

where the coefficient $(1 - \rho_G)/(1 + a^G)$ comes from substituting the AR(1) process for u_{t+1}^G into the expression.

Substituting the expression for the stochastic discount factor (A.4) into the household's bond portfolio conditions (15) and log-linearizing the resulting expression yields

$$\hat{\lambda}_t^{HB,f} = \kappa^B \mathbb{E}_t \left[\hat{z}_t^f - \hat{\pi}_{t+1} + \hat{c}_t - \hat{c}_{t+1} - (f - 1) \hat{y} \hat{d}_{t+1}^{f-1} + f \hat{y} \hat{d}_t^f - \hat{\phi}_t^B \right], \quad (\text{A.49})$$

where

$$\begin{aligned} \hat{\phi}_t^B = \mathbb{E}_t (-\hat{\pi}_{t+1} + \hat{c}_t - \hat{c}_{t+1}) + \sum_{j=1}^F \left[\frac{\beta z^j (YD^{j-1})^{-(j-1)}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^j)^{-j}} \right]^{\kappa^B} \hat{z}_t^j \\ + \sum_{j=1}^F j \left[\frac{\beta z^j (YD^{j-1})^{-(j-1)}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^j)^{-j}} \right]^{\kappa^B} \hat{y} \hat{d}_t^j - \sum_{j=0}^{F-1} j \left[\frac{\beta z^{j+1} (YD^j)^{-j}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^{j+1})^{-(j+1)}} \right]^{\kappa^B} \mathbb{E}_t (\hat{y} \hat{d}_{t+1}^j). \end{aligned} \quad (\text{A.50})$$

These expressions use $\mathbb{E}_t \hat{g}a_{t+1} = 0$, following from the i.i.d. assumption of the technology-growth shock. The final sum is indexed by $j = 0, \dots, F - 1$ because it collects the future-

yield terms $\hat{y}d_{t+1}^j$ associated with $j + 1$ -period bonds.

Combining (A.49) and (A.50), we obtain the following expression for $\hat{\lambda}_t^{HB,f}$:

$$\hat{\lambda}_t^{HB,f} = \sum_{j=1}^F \Psi_{fj}^1 \hat{z}_t^j + \sum_{j=1}^F \Psi_{fj}^2 \hat{y}d_t^j + \sum_{j=1}^F \Psi_{fj}^3 \mathbb{E}_t \left[\hat{y}d_{t+1}^j \right],$$

where

$$\Psi_{fj}^1 = \begin{cases} \left[1 - \left[\frac{\beta \cdot z^j (YD^{j-1})^{-(j-1)}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^j)^{-j}} \right]^{\kappa^B} \right] \cdot \kappa^B & , \text{ if } f = j, \\ - \left[\frac{\beta \cdot z^j (YD^{j-1})^{-(j-1)}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^j)^{-j}} \right]^{\kappa^B} \cdot \kappa^B & , \text{ if } f \neq j, \end{cases}$$

$$\Psi_{fj}^2 = j \cdot \Psi_{fj}^1,$$

$$\Psi_{fj}^3 = \begin{cases} -j \cdot \left[1 - \left[\frac{\beta \cdot z^{j+1} (YD^j)^{-j}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^{j+1})^{-(j+1)}} \right]^{\kappa^B} \right] \cdot \kappa^B & , \text{ if } j = f - 1, \\ j \cdot \left[\frac{\beta \cdot z^{j+1} (YD^j)^{-j}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^{j+1})^{-(j+1)}} \right]^{\kappa^B} \cdot \kappa^B & , \text{ if } j \neq f - 1, \\ 0 & , \text{ if } j = F. \end{cases}$$

We can put the system of F equations in matrix format as

$$\overrightarrow{\hat{\lambda}_t^{HB}} = \Psi^1 \cdot \overrightarrow{\hat{z}_t} + \Psi^2 \cdot \overrightarrow{\hat{y}d_t} + \Psi^3 \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}} \right], \quad (\text{A.51})$$

where $\{\Psi^1, \Psi^2, \Psi^3\}$ are $F \times F$ matrices containing elements of $\{\Psi_{fj}^1, \Psi_{fj}^2, \Psi_{fj}^3\}$, with f representing rows and j columns. Linearizing equations (I.2) and (I.3) yields

$$\overrightarrow{\hat{\lambda}_t^G} = \Xi \cdot \overrightarrow{\hat{u}_t^B}, \quad \overrightarrow{\hat{u}_t^B} = \mathcal{T}^B \cdot \overrightarrow{\hat{u}_t^B},$$

where $\overrightarrow{\hat{u}_t^B}$ is the vector of transformed maturity-specific shocks, while $\overrightarrow{\hat{u}_t^B}$ is the reduced

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shock vector. $\tilde{\Xi}$ is a matrix whose elements $\tilde{\Xi}_{fj}$ (f -rows, j -columns) are

$$\tilde{\Xi}_{fj} = \begin{cases} 0 & , \text{ if } f = 1 \text{ \& } j = f, \\ 1 - \lambda^{G,f} & , \text{ if } f \geq 2 \text{ \& } j = f, \\ -\lambda^{G,j} & , \text{ if } j \neq f, \end{cases}$$

and similarly $\mathcal{T}^B$ is a matrix containing elements τ_{fj}^B from (I.3). By defining $\Xi = \tilde{\Xi} \cdot \mathcal{T}^B$, we can combine the previous two equations to obtain

$$\vec{\hat{\lambda}}_t^G = \Xi \cdot \vec{\hat{u}}_t^B. \quad (\text{A.52})$$

Let $\tilde{B}^{G,f} = -B^{G,f}$ denote the positive amount of government debt, so hats below are log deviations of $\tilde{B}_t^{G,f}$. Combining this definition with (A.52) gives

$$\vec{\hat{b}}_t^G = \Xi \cdot \vec{\hat{u}}_t^B + \vec{1}_{F \times 1} \cdot \hat{b}_t^G, \quad (\text{A.53})$$

where $\vec{1}_{F \times 1}$ is a unit vector of size F . Log-linearizing the household's stochastic discount factor (A.4) yields:

$$\hat{q}_{t,t+1} = \hat{c}_t - \hat{c}_{t+1} - \hat{\pi}_{t+1} - \hat{g}a_{t+1}. \quad (\text{A.54})$$

Log-linearizing Φ_t^S in the household's portfolio between loans and bonds (17), we obtain

$$\hat{\phi}_t^S = \mathbb{E}_t[\hat{q}_{t,t+1}] + \frac{(R^{HB})^{\kappa^S}}{(R^{HB})^{\kappa^S} + (z^K R^K)^{\kappa^S}} \mathbb{E}_t[\hat{r}_{t+1}^{HB}] + \frac{(z^K R^K)^{\kappa^S}}{(R^{HB})^{\kappa^S} + (z^K R^K)^{\kappa^S}} (\hat{z}_t^K + \mathbb{E}_t[\hat{r}_{t+1}^K]). \quad (\text{A.55})$$

Log-linearizing the household's portfolio decision between loans and bonds (17) and making use of the previous expression (A.55), we obtain the main-text relation (52):

$$\begin{aligned} \hat{\lambda}_t^K &= \kappa^S \cdot \left[\frac{(R^{HB})^{\kappa^S}}{(R^{HB})^{\kappa^S} + (z^K R^K)^{\kappa^S}} \right] (\hat{z}_t^K + \mathbb{E}_t[\hat{r}_{t+1}^K - \hat{r}_{t+1}^{HB}]) \\ &= \kappa^S (1 - \lambda^K) (\hat{z}_t^K + \mathbb{E}_t[\hat{r}_{t+1}^K - \hat{r}_{t+1}^{HB}]). \end{aligned}$$

where the second equality uses the steady-state portfolio share in (A.34).

By linearizing the formula for the effective savings rate of the household (i.e., (18)), we

obtain

$$\hat{r}_t^S = \frac{\lambda^K (R^K - R^{HB})}{R^S} \hat{\lambda}_{t-1}^K + \frac{(1 - \lambda^K) R^{HB}}{R^S} \hat{r}_t^{HB} + \frac{\lambda^K R^K}{R^S} \hat{r}_t^K. \quad (\text{A.56})$$

Log-linearizing the effective bond rates of households, government, and central bank,

$$\hat{r}_t^j = \sum_{f=1}^F \frac{\lambda^{j,f} (YD^{f-1})^{-(f-1)}}{R^j (YD^f)^{-f}} \cdot \left[\hat{\lambda}_{t-1}^{j,f} - (f-1) \hat{y}d_t^{f-1} + f \hat{y}d_{t-1}^f \right], \quad j \in \{HB, G, CB\},$$

with which we can express these equations on matrix format as

$$\hat{r}_t^j = \Psi^{j,4} \cdot \overrightarrow{\hat{\lambda}_{t-1}^j} - \Psi^{j,5} \cdot \overrightarrow{\hat{y}d_t} + \Psi^{j,6} \cdot \overrightarrow{\hat{y}d_{t-1}}, \quad j \in \{HB, G, CB\}, \quad (\text{A.57})$$

where $\{\Psi^{j,4}, \Psi^{j,5}, \Psi^{j,6}\}$ are $1 \times F$ -sized matrices whose elements are defined as:

$$\begin{aligned} \Psi_{1f}^{j,4} &= \frac{\lambda^{j,f} (YD^{f-1})^{-(f-1)}}{R^j (YD^f)^{-f}}, \quad \Psi_{1f}^{j,6} = \frac{\lambda^{j,f} (YD^{f-1})^{-(f-1)}}{R^j (YD^f)^{-f}} f. \\ \Psi_{1f}^{j,5} &= \begin{cases} \frac{\lambda^{j,f+1} (YD^f)^{-f}}{R^j (YD^{f+1})^{-(f+1)}} f & , \text{ if } f < F, \quad j \in \{HB, G, CB\}, \\ 0 & , \text{ if } f = F, \end{cases} \end{aligned}$$

where the vector $\overrightarrow{\hat{y}d_t}$ in equation (A.57) is indexed by current remaining maturity, so the current yield term for an $f + 1$ -period bond enters position f in $\Psi^{j,5}$.

By plugging (A.52) into $\hat{r}^G$ in (A.57), we obtain

$$\hat{r}_t^G = \Psi^{G,4} \cdot \overrightarrow{\Xi} \cdot \overrightarrow{\hat{u}_{t-1}^B} - \Psi^{G,5} \cdot \overrightarrow{\hat{y}d_t} + \Psi^{G,6} \cdot \overrightarrow{\hat{y}d_{t-1}}. \quad (\text{A.58})$$

By plugging (A.51) into $\hat{r}^{HB}$ in (A.57), we obtain

$$\hat{r}_t^{HB} = \Psi^{HB,4} \Psi^1 \cdot \overrightarrow{\hat{z}_{t-1}} + [\Psi^{HB,4} \Psi^2 + \Psi^{HB,6}] \cdot \overrightarrow{\hat{y}d_{t-1}} + \Psi^{HB,4} \Psi^3 \cdot \mathbb{E}_{t-1} \left[\overrightarrow{\hat{y}d_t} \right] - \Psi^{HB,5} \cdot \overrightarrow{\hat{y}d_t}. \quad (\text{A.59})$$

Taking the expectation operator $\mathbb{E}_t$ on the previous equation (A.59), we obtain

$$\mathbb{E}_t [\hat{r}_{t+1}^{HB}] = \Psi^{HB,4} \Psi^1 \cdot \overrightarrow{\hat{z}_t} + [\Psi^{HB,4} \Psi^2 + \Psi^{HB,6}] \cdot \overrightarrow{\hat{y}d_t} + [\Psi^{HB,4} \Psi^3 - \Psi^{HB,5}] \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}} \right]. \quad (\text{A.60})$$

Plugging equations (52) and (A.60) into the $t + 1$ expression of equation (A.56) and apply-

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ing the expectations operator $\mathbb{E}_t$, we obtain the expected effective savings rate as:

$$\mathbb{E}_t [\hat{r}_{t+1}^S] = \Psi^7 \vec{\hat{z}}_t + \Psi^8 \vec{\hat{y}d}_t + \Psi^9 \mathbb{E}_t [\vec{\hat{y}d}_{t+1}] + \Psi^{10} \hat{r}_{t+1}^K + \Psi^{11} \hat{z}_t^K, \quad (\text{A.61})$$

where

$$\begin{aligned} \Psi^7 &= \Psi^{HB,4} \Psi^1 \left[\frac{(1 + \kappa^S \lambda^K)(1 - \lambda^K) R^{HB} - \kappa^S (1 - \lambda^K) \lambda^K R^K}{R^S} \right], \\ \Psi^8 &= [\Psi^{HB,4} \Psi^2 + \Psi^{HB,6}] \left[\frac{(1 + \kappa^S \lambda^K)(1 - \lambda^K) R^{HB} - \kappa^S (1 - \lambda^K) \lambda^K R^K}{R^S} \right], \\ \Psi^9 &= [\Psi^{HB,4} \Psi^3 - \Psi^{HB,5}] \left[\frac{(1 + \kappa^S \lambda^K)(1 - \lambda^K) R^{HB} - \kappa^S (1 - \lambda^K) \lambda^K R^K}{R^S} \right], \\ \Psi^{10} &= \frac{[1 + \kappa^S (1 - \lambda^K)] \lambda^K R^K - \kappa^S (1 - \lambda^K) \lambda^K R^{HB}}{R^S}, \quad \Psi^{11} = \frac{\kappa^S \lambda^K (1 - \lambda^K) (R^K - R^{HB})}{R^S} \end{aligned}$$

Plugging back the expression of the household's expected bonds rate (i.e., (A.60)) into her portfolio decision between loans and bonds (i.e., (52)), we obtain

$$\hat{\lambda}_t^K = \kappa^S (1 - \lambda^K) (\hat{z}_t^K + \hat{r}_{t+1}^K) - \Psi^{12} \cdot \vec{\hat{z}}_t - \Psi^{13} \cdot \vec{\hat{y}d}_t - \Psi^{14} \cdot \mathbb{E}_t [\vec{\hat{y}d}_{t+1}], \quad (\text{A.62})$$

where $\Psi^{12} = \kappa^S (1 - \lambda^K) \Psi^{HB,4} \Psi^1$, $\Psi^{13} = \kappa^S (1 - \lambda^K) [\Psi^{HB,4} \Psi^2 + \Psi^{HB,6}]$, and $\Psi^{14} = \kappa^S (1 - \lambda^K) [\Psi^{HB,4} \Psi^3 - \Psi^{HB,5}]$. If we linearize the capital aggregation (A.6) and use (A.45), we obtain

$$\hat{k}_t = \hat{\varepsilon}_t^A + \left(\frac{\eta + 1}{\eta + \alpha} \right) \hat{y}_t - \left(\frac{\eta(1 - \alpha)}{\eta + \alpha} \right) \cdot [\hat{p}_t^K - \hat{c}_t]. \quad (\text{A.63})$$

Combining (A.46) and the above (A.63), we obtain

$$\begin{aligned} \hat{p}_t^K &= \left[(1 - \zeta^G) \frac{Y}{C} + \frac{\eta + 1}{\eta(1 - \alpha)} \right] \cdot \hat{y}_t - \left[(1 - \zeta^G) \cdot \frac{Y}{C} \right] \left(\frac{1}{1 + a^G} \right) \cdot \hat{u}_t^G \\ &\quad + \left[\frac{1 - \delta}{GA \cdot GN} \cdot \frac{K}{C} - \frac{\eta + \alpha}{\eta(1 - \alpha)} \right] \cdot [\hat{k}_t - \hat{\varepsilon}_t^A] - \frac{K}{C} \cdot \hat{k}_{t+1}. \end{aligned} \quad (\text{A.64})$$

If we linearize the supply block (A.11) and (A.12), we obtain

$$\begin{aligned} \hat{f}_t &= \left[1 - \theta \beta \Pi^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} \right] \left(\frac{\eta + 1}{\eta + \alpha} \right) [\hat{y}_t + \alpha \mathbb{E}_t [\hat{q}_{t,t+1} + \hat{p}_t^K - \hat{c}_t]] \\ &\quad + \theta \beta \Pi^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} \mathbb{E}_t \left[\epsilon \left(\frac{\eta + 1}{\eta + \alpha} \right) \hat{\pi}_{t+1} + \hat{f}_{t+1} \right], \end{aligned} \quad (\text{A.65})$$

$$\begin{aligned} \hat{h}_t &= [1 - \theta \beta \Pi^{\epsilon-1}] \left[\hat{y}_t - \hat{c}_t - \left(\frac{\gamma_L \cdot R^K}{R^S - \gamma_L \cdot (R^K - R^S)} \right) \cdot [\hat{r}_{t+1}^K + \mathbb{E}_t [\hat{q}_{t,t+1}]] \right] \\ &\quad + \theta \beta \Pi^{\epsilon-1} \mathbb{E}_t [(\epsilon - 1) \hat{\pi}_{t+1} + \hat{h}_{t+1}], \end{aligned} \quad (\text{A.66})$$

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leading to

$$\hat{f}_t - \hat{h}_t = \left[1 + \epsilon \left(\frac{1 - \alpha}{\eta + \alpha} \right) \right] \left(\frac{\theta \Pi^{\epsilon-1}}{1 - \theta \Pi^{\epsilon-1}} \right) \hat{\pi}_t. \quad (\text{A.67})$$

Combining (A.47), (A.54), and (A.63) with equations (A.66) and (A.65), we obtain:

$$\begin{aligned} \hat{f}_t = & -\Psi^{16} \cdot \vec{\hat{z}}_t - \Psi^{17} \cdot \hat{z}_t^K - \Psi^{18} \cdot [\hat{k}_t - \hat{\epsilon}_t^A] + \Psi^{19} \cdot \hat{y}_t - \Psi^{20} \cdot \vec{\hat{y}d}_t - \Psi^{21} \cdot \hat{r}_{t+1}^K \\ & - \Psi^{22} \cdot \mathbb{E}_t [\vec{\hat{y}d}_{t+1}] + \Psi^{23} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{24} \cdot \mathbb{E}_t [\hat{f}_{t+1}]. \end{aligned} \quad (\text{A.68})$$

$$\begin{aligned} \hat{h}_t = & \Psi^{26} \cdot \vec{\hat{z}}_t + \Psi^{27} \cdot \hat{z}_t^K - \Psi^{28} \cdot [\hat{k}_t - \hat{\epsilon}_t^A] + \Psi^{29} \cdot \hat{u}_t^G + \Psi^{30} \cdot \hat{y}_t + \Psi^{31} \cdot \vec{\hat{y}d}_t \\ & - \Psi^{32} \cdot \hat{r}_{t+1}^K + \Psi^{33} \cdot \hat{k}_{t+1} + \Psi^{34} \cdot \mathbb{E}_t [\vec{\hat{y}d}_{t+1}] + \Psi^{35} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{36} \cdot \mathbb{E}_t [\hat{h}_{t+1}]. \end{aligned} \quad (\text{A.69})$$

where

$$\begin{aligned} \Psi^{15} &= \left[1 - \theta \beta \Pi^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} \right] \left(\frac{\eta+1}{\eta+\alpha} \right), & \Psi^{26} &= \Psi^{25} \Psi^7, \\ \Psi^{16} &= \alpha \Psi^{15} \Psi^7, & \Psi^{27} &= \Psi^{25} \Psi^{11}, \\ \Psi^{17} &= \alpha \Psi^{15} \Psi^{11}, & \Psi^{28} &= [1 - \theta \beta \Pi^{\epsilon-1}] \left[\frac{1-\delta}{GA \cdot GN} \cdot \frac{K}{C} \right], \\ \Psi^{18} &= \Psi^{15} \left(\frac{\alpha}{1-\alpha} \right) \left(\frac{\eta+\alpha}{\eta} \right), & \Psi^{29} &= [1 - \theta \beta \Pi^{\epsilon-1}] (1 - \zeta^G) \frac{Y}{C} \left(\frac{1}{1+a^G} \right), \\ \Psi^{19} &= \Psi^{15} \left[1 + \left(\frac{\alpha}{1-\alpha} \right) \left(\frac{\eta+1}{\eta} \right) \right], & \Psi^{30} &= [1 - \theta \beta \Pi^{\epsilon-1}] \left[1 - (1 - \zeta^G) \frac{Y}{C} \right], \\ \Psi^{20} &= \alpha \Psi^{15} \Psi^8, & \Psi^{31} &= \Psi^{25} \Psi^8, \\ \Psi^{21} &= \alpha \Psi^{15} \Psi^{10}, & \Psi^{32} &= \Psi^{25} \cdot (1 - \Psi^{10}), \\ \Psi^{22} &= \alpha \Psi^{15} \Psi^9, & \Psi^{33} &= [1 - \theta \beta \Pi^{\epsilon-1}] \frac{K}{C}, \\ \Psi^{23} &= \theta \beta \Pi^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} \epsilon \left(\frac{\eta+1}{\eta+\alpha} \right), & \Psi^{34} &= \Psi^{25} \Psi^9, \\ \Psi^{24} &= \theta \beta \Pi^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)}, & \Psi^{35} &= \theta \beta \Pi^{\epsilon-1} (\epsilon - 1), \\ \Psi^{25} &= [1 - \theta \beta \Pi^{\epsilon-1}] \left(\frac{\gamma_L \cdot R^K}{R^S - \gamma_L \cdot (R^K - R^S)} \right), & \Psi^{36} &= \theta \beta \Pi^{\epsilon-1}. \end{aligned}$$

Linearizing the government's budget constraint (i.e., (34)) yields

$$\begin{aligned} \hat{b}_t^G &= \frac{R^G}{\Pi \cdot GA \cdot GN} [\hat{r}_t^G - \hat{\pi}_t - \hat{g}a_t + \hat{b}_{t-1}^G] \\ &- [\zeta^G + \zeta^F - \zeta^T] \left(\frac{Y}{B/P} \right) \left[\hat{y}_t + \left(\frac{\zeta^G}{\zeta^G + \zeta^F - \zeta^T} \right) \left(\frac{a^G}{1+a^G} \right) \hat{u}_t^G - \left(\frac{\zeta^T}{\zeta^G + \zeta^F - \zeta^T} \right) \left(\frac{a^T}{1+a^T} \right) \hat{u}_t^T \right]. \end{aligned} \quad (\text{A.70})$$

Here, steady-state ratios use the original government bond position B/P , while hatted debt variables use the positive-debt variable $\tilde{B}^G = -B^G$.

Using the steady state equilibrium condition (i.e., (47)) with (A.45) and (A.58), we can

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express the previous (A.70) as

$$\begin{aligned} \hat{b}_t^G = & \frac{R^G}{\Pi \cdot GA \cdot GN} \left[\Psi^{G,4} \cdot \Xi \cdot \overrightarrow{\hat{u}_{t-1}^B} - \Psi^{G,5} \cdot \overrightarrow{\hat{y}d_t} + \Psi^{G,6} \cdot \overrightarrow{\hat{y}d_{t-1}} - \hat{\pi}_t - \hat{\varepsilon}_t^A + \hat{b}_{t-1}^G \right] \\ & + \left(1 - \frac{R^G}{\Pi \cdot GA \cdot GN} \right) \left[\hat{y}_t + \left(\frac{\zeta^G}{\zeta^G + \zeta^F - \zeta^T} \right) \left(\frac{a^G}{1 + a^G} \right) \hat{u}_t^G - \left(\frac{\zeta^T}{\zeta^G + \zeta^F - \zeta^T} \right) \left(\frac{a^T}{1 + a^T} \right) \hat{u}_t^T \right], \end{aligned} \quad (\text{A.71})$$

where the coefficient on the fiscal-output term follows directly from substituting the steady-state debt-output relation in (47).

Linearizing the capital producer's optimization condition (25) yields

$$0 = \mathbb{E}_t \left[\hat{q}_{t,t+1} + \hat{\pi}_{t+1} + \left(\frac{P^K/P}{1 - \delta + P^K/P} \right) \hat{p}_{t+1}^K \right]. \quad (\text{A.72})$$

By plugging (A.47) and (A.54) into the previous (A.72) and rearranging, we get

$$\mathbb{E}_t [\hat{r}_{t+1}^S - \hat{\pi}_{t+1}] = \left(\frac{P^K/P}{1 - \delta + P^K/P} \right) \mathbb{E}_t [\hat{p}_{t+1}^K]. \quad (\text{A.73})$$

This rearrangement uses $\mathbb{E}_t \hat{g}a_{t+1} = 0$, because technology growth is i.i.d.

Plugging expressions on the effective savings rate (i.e., (A.61)) and the rental price of capital (i.e., (A.64)) into (A.73) we obtain

$$\begin{aligned} \hat{r}_{t+1}^K = & -\Psi^{37} \cdot \overrightarrow{\hat{z}_t} - \Psi^{38} \cdot \hat{z}_t^K - \Psi^{39} \cdot \overrightarrow{\hat{y}d_t} - \Psi^{40} \cdot \mathbb{E}_t [\overrightarrow{\hat{y}d_{t+1}}] + \Psi^{41} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] \\ & + \Psi^{42} \cdot \mathbb{E}_t [\hat{y}_{t+1}] + \Psi^{43} \cdot \hat{k}_{t+1} - \Psi^{44} \cdot \mathbb{E}_t [\hat{k}_{t+2}] - \Psi^{45} \cdot \hat{u}_t^G, \end{aligned} \quad (\text{A.74})$$

where we define

$$\begin{aligned} \Psi^{37} &= (\Psi^{10})^{-1} \Psi^7, & \Psi^{42} &= (\Psi^{10})^{-1} \left(\frac{\frac{P^K}{P}}{1 - \delta + \frac{P^K}{P}} \right) \left[(1 - \zeta^G) \frac{Y}{C} + \left(\frac{\eta + 1}{\eta(1 - \alpha)} \right) \right], \\ \Psi^{38} &= (\Psi^{10})^{-1} \Psi^{11}, \\ \Psi^{39} &= (\Psi^{10})^{-1} \Psi^8, & \Psi^{43} &= (\Psi^{10})^{-1} \left(\frac{\frac{P^K}{P}}{1 - \delta + \frac{P^K}{P}} \right) \left[\frac{1 - \delta}{GA \cdot GN} \frac{K}{C} - \left(\frac{\eta + \alpha}{\eta(1 - \alpha)} \right) \right], \\ \Psi^{40} &= (\Psi^{10})^{-1} \Psi^9, \\ \Psi^{41} &= (\Psi^{10})^{-1}, & \Psi^{44} &= (\Psi^{10})^{-1} \left(\frac{\frac{P^K}{P}}{1 - \delta + \frac{P^K}{P}} \right) \frac{K}{C}, \\ & & \Psi^{45} &= (\Psi^{10})^{-1} \left(\frac{\frac{P^K}{P}}{1 - \delta + \frac{P^K}{P}} \right) (1 - \zeta^G) \frac{Y}{C} \left(\frac{\rho^G}{1 + a^G} \right). \end{aligned}$$

The coefficient Ψ^{45} uses the AR(1) expectation of the government spending shock, so

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$$\mathbb{E}_t \hat{u}_{t+1}^G = \rho^G \hat{u}_t^G.$$

Finally, plugging the effective savings rate (A.61) into the Euler equation (A.48), we obtain

$$\begin{aligned} \hat{y}_t = \mathbb{E}_t \Big[& \hat{y}_{t+1} + \Psi^{46} \cdot \hat{\pi}_{t+1} - \Psi^{47} \cdot \vec{\hat{z}}_t - \Psi^{48} \cdot \hat{z}_t^K - \Psi^{49} \cdot \vec{\hat{y}d}_t - \Psi^{50} \cdot \mathbb{E}_t \left[\vec{\hat{y}d}_{t+1} \right] \\ & - \Psi^{51} \cdot \hat{r}_{t+1}^K - \Psi^{52} \cdot (\hat{k}_t - \hat{\varepsilon}_t^A) + \Psi^{53} \cdot \hat{k}_{t+1} - \Psi^{54} \cdot \hat{k}_{t+2} + \Psi^{55} \cdot \hat{u}_t^G \Big], \end{aligned} \quad (\text{A.75})$$

where we define

$$\begin{aligned} \Psi^{46} &= (1 - \zeta^G)^{-1} \frac{C}{Y}, & \Psi^{52} &= \frac{(1 - \zeta^G)^{-1} (1 - \delta) K}{GA \cdot GN Y}, \\ \Psi^{47} &= \Psi^{46} \Psi^7, & \Psi^{53} &= (1 - \zeta^G)^{-1} \left[1 + \frac{1 - \delta}{GA \cdot GN} \right] \frac{K}{Y}, \\ \Psi^{48} &= \Psi^{46} \Psi^{11}, & \Psi^{54} &= (1 - \zeta^G)^{-1} \frac{K}{Y}, \\ \Psi^{49} &= \Psi^{46} \Psi^8, & \Psi^{55} &= \frac{1 - \rho_G}{1 + a^G}. \\ \Psi^{50} &= \Psi^{46} \Psi^9, \\ \Psi^{51} &= \Psi^{46} \Psi^{10}, \end{aligned}$$

Linearizing the labor aggregation condition (A.7) yields

$$\hat{n}_t = -\alpha \left(\frac{\eta}{\eta + \alpha} \right) \hat{c}_t + \left(\frac{\eta}{\eta + \alpha} \right) \hat{y}_t + \alpha \left(\frac{\eta}{\eta + \alpha} \right) \cdot \hat{p}_t^K, \quad (\text{A.76})$$

where we rely on that price dispersion has no first-order effect around the symmetric steady state.

Plugging equations (A.46) and (A.64) into equation (A.76), we obtain

$$\hat{n}_t = \left(\frac{\eta}{\eta + \alpha} \right) \left[1 + \left(\frac{\alpha}{1 - \alpha} \right) \left(\frac{\eta + 1}{\eta} \right) \right] \cdot \hat{y}_t - \left(\frac{\alpha}{1 - \alpha} \right) \cdot [\hat{k}_t - \hat{\varepsilon}_t^A], \quad (\text{A.77})$$

where the fiscal-shock and future-capital terms present in (A.46) and (A.64) cancel algebraically after the substitution.

Log-linearization: the conventional policy specific derivations Linearizing the bond market equilibrium condition (i.e., (A.26)), we obtain

$$\hat{\lambda}_t^{HB,f} = \left(\frac{B^{G,f}}{B^{G,f} + B^{CB,f}} \right) \hat{b}_t^{G,f} - \hat{y}_t + \frac{1}{1 - \lambda^K} \cdot \hat{\lambda}_t^K, \quad f \geq 2. \quad (\text{A.78})$$

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As above, steady-state ratios use the original bond positions, while hatted government debt uses $\tilde{B}^{G,f} = -B^{G,f}$.

From $\lambda_t^{HB,1} = 1 - \sum_{f=2}^F \lambda_t^{HB,f}$ we obtain

$$\hat{\lambda}_t^{HB,1} = - \sum_{f=2}^F \frac{\lambda^{HB,f}}{\lambda^{HB,1}} \hat{\lambda}_t^{HB,f}. \quad (\text{A.79})$$

We can rearrange the previous expressions (i.e., (A.78) and (A.79)) in the matrix form as

$$\Theta^1 \cdot \overrightarrow{\hat{\lambda}_t^{HB}} = \Theta^2 \cdot \overrightarrow{\hat{b}_t^G} - \Theta^3 \cdot \hat{y}_t + \Theta^4 \cdot \hat{\lambda}_t^K, \quad (\text{A.80})$$

where $\{\Theta^1, \Theta^2\}$ are $F \times F$ -sized matrices with elements Θ_{jf}^1 and Θ_{jf}^2 (row j , column f) and $\{\Theta^3, \Theta^4\}$ are $F \times 1$ vectors with j -element Θ_{j1}^3 . We define their elements as

$$\begin{aligned} \Theta_{jf}^1 &= \begin{cases} 1 & , \text{ if } j = f, \\ \frac{\lambda^{HB,f}}{\lambda^{HB,1}} & , \text{ if } j = 1 \text{ \& } f > 1, \\ 0 & , \text{ otherwise,} \end{cases} & \Theta_{j1}^3 &= \begin{cases} 0 & , \text{ if } j = 1, \\ 1 & , \text{ otherwise,} \end{cases} \\ \Theta_{jf}^2 &= \begin{cases} \frac{B^{G,f}}{B^{G,f} + B^{CB,f}} & , \text{ if } j > 1 \text{ \& } j = f, \\ 0 & , \text{ otherwise,} \end{cases} & \Theta^4 &= \frac{1}{1 - \lambda^K} \cdot \Theta^3. \end{aligned}$$

By inverting Θ^1 in (A.80), we can rewrite (A.80) as

$$\overrightarrow{\hat{\lambda}_t^{HB}} = \Theta^5 \cdot \overrightarrow{\hat{b}_t^G} - \Theta^6 \cdot \hat{y}_t + \Theta^7 \cdot \hat{\lambda}_t^K, \quad (\text{A.81})$$

where we define $\Theta^5 = (\Theta^1)^{-1} \Theta^2$, $\Theta^6 = (\Theta^1)^{-1} \Theta^3$, $\Theta^7 = (\Theta^1)^{-1} \Theta^4$. By plugging the government's bond portfolio (A.53), and the household's loan share (A.62) into (A.81), we obtain

$$\overrightarrow{\hat{\lambda}_t^{HB}} = \Theta^8 \cdot \hat{b}_t^G - \Theta^6 \cdot \hat{y}_t + \Theta^9 \cdot (\hat{z}_t^K + \hat{r}_{t+1}^K) - \Theta^{10} \cdot \overrightarrow{\hat{y}_{d,t}} - \Theta^{11} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}_{d,t+1}} \right] - \Theta^{12} \cdot \hat{z}_t + \Theta^{13} \cdot \overrightarrow{\hat{u}_t^B}, \quad (\text{A.82})$$

where we define

$$\begin{aligned} \Theta^8 &= \Theta^5 \cdot \overrightarrow{1_{F \times 1}}, & \Theta^{10} &= \Theta^7 \cdot \Psi^{13}, & \Theta^{12} &= \Theta^7 \cdot \Psi^{12}, \\ \Theta^9 &= \Theta^7 \cdot \kappa^S (1 - \lambda^K), & \Theta^{11} &= \Theta^7 \cdot \Psi^{14}, & \Theta^{13} &= \Theta^5 \cdot \Xi. \end{aligned}$$

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Since Θ^5 is $F \times F$ and Ξ is $F \times J$, Θ^{13} is $F \times J$, conformable with the reduced bond-shock vector $\vec{\hat{u}}_t^B$.

By plugging the household's optimal portfolio (A.51) into equation (A.82), we obtain

$$\vec{\hat{y}}_t^d = \Theta^{14} \cdot \hat{b}_t^G - \Theta^{15} \cdot \hat{y}_t + \Theta^{16} \cdot \hat{r}_{t+1}^K - \Theta^{17} \cdot \mathbb{E}_t \left[\vec{\hat{y}}_{t+1}^d \right] - \Theta^{18} \cdot \vec{\hat{z}}_t + \Theta^{19} \cdot \hat{z}_t^K + \Theta^{20} \cdot \vec{\hat{u}}_t^B, \quad (\text{A.83})$$

where $\Theta^{14} = [\Theta^{10} + \Psi^2]^{-1} \Theta^8$, $\Theta^{15} = [\Theta^{10} + \Psi^2]^{-1} \Theta^6$, $\Theta^{16} = [\Theta^{10} + \Psi^2]^{-1} \Theta^9$, $\Theta^{17} = [\Theta^{10} + \Psi^2]^{-1} [\Theta^{11} + \Psi^3]$, $\Theta^{18} = [\Theta^{10} + \Psi^2]^{-1} [\Theta^{12} + \Psi^1]$, $\Theta^{19} = [\Theta^{10} + \Psi^2]^{-1} \Theta^9$, $\Theta^{20} = [\Theta^{10} + \Psi^2]^{-1} \Theta^{13}$.

Log-linearization: the Yield-Curve-Control (YCC) policy specific derivations Linearizing the Taylor rule for the f -maturity bond in (38b) gives two equations. First, the notional YCC target satisfies

$$\hat{y}_t^{*,f} = \rho_1 \hat{y}_{t-1}^{*,f} + \rho_2 \hat{y}_{t-2}^{*,f} + (1 - \rho_1 - \rho_2) \left[\gamma_\pi^f \hat{\pi}_t + \gamma_y^f \hat{y}_t + \hat{\varepsilon}_t^{YDf} \right], \quad f \geq 2. \quad (\text{A.84})$$

Second, the actual YCC yield combines the counterfactual conventional-policy yield with this notional target:

$$\hat{y}_t^{YCC,f} = \gamma_{CP}^f \hat{y}_t^{CP,f} + (1 - \gamma_{CP}^f) \hat{y}_t^{*,f}, \quad f \geq 2. \quad (\text{A.85})$$

Here, we define a $(F-1) \times (F-1)$ matrix Γ^{CP} with $\Gamma_{ff}^{CP} = \gamma_{CP}^{f+1}$ for $f = 1, \dots, F-1$ and $\Gamma_{ij}^{CP} = 0$ for $i \neq j$, the vectors $\vec{\gamma}_{\pi(f \geq 2)} = [\gamma_\pi^2, \dots, \gamma_\pi^F]'$ and $\vec{\gamma}_{y(f \geq 2)} = [\gamma_y^2, \dots, \gamma_y^F]'$, and from (I.4) the $(F-1) \times L$ matrix $\mathcal{T}_{(f \geq 2)}^{YD}$ with elements $\mathcal{T}_{(f \geq 2),f,l}^{YD} = \tau_{f+1,l}^{YD}$ (row f , column l). Let $\vec{\varepsilon}_t^{YD} = [\varepsilon_t^{YD,1}, \dots, \varepsilon_t^{YD,L}]'$. If we construct the policy-specific yield vectors

$$\vec{\hat{y}}_{(f \geq 2)}^{YCC} = [\hat{y}_t^{YCC,2}, \dots, \hat{y}_t^{YCC,F}]', \quad \vec{\hat{y}}_{(f \geq 2)}^{CP} = [\hat{y}_t^{CP,2}, \dots, \hat{y}_t^{CP,F}]',$$

and the notional YCC-target vector

$$\vec{\hat{y}}_{(f \geq 2)}^* = [\hat{y}_t^{*,2}, \dots, \hat{y}_t^{*,F}]',$$

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then the notional target rule can be written in vector form as

$$\begin{aligned} \overrightarrow{\hat{y}d_t^*}_{(f \geq 2)} &= \overrightarrow{\rho_1 \hat{y}d_{t-1}^*}_{(f \geq 2)} + \overrightarrow{\rho_2 \hat{y}d_{t-2}^*}_{(f \geq 2)} \\ &+ (1 - \rho_1 - \rho_2) \left[\overrightarrow{\gamma_{\pi(f \geq 2)}} \hat{\pi}_t + \overrightarrow{\gamma_{y(f \geq 2)}} \hat{y}_t + \mathcal{T}_{(f \geq 2)}^{YD} \overrightarrow{\varepsilon_t^{YD}} \right]. \end{aligned} \quad (\text{A.86})$$

Combining the notional target rule with the actual YCC-yield equation gives

$$\overrightarrow{\hat{y}d_t^{YCC}}_{(f \geq 2)} = \Gamma^{CP} \overrightarrow{\hat{y}d_t^{CP}}_{(f \geq 2)} + (I - \Gamma^{CP}) \overrightarrow{\hat{y}d_t^*}_{(f \geq 2)}, \quad (\text{A.87})$$

where I is the identity matrix of size $F - 1$. Since $\overrightarrow{\hat{y}d_t^{CP}}$ is the yield vector that prevails in the counterfactual scenario where the current yield is determined by conventional monetary policy (A.83), its dynamics follow

$$\begin{aligned} \overrightarrow{\hat{y}d_t^{CP}} &= \Theta^{14} \cdot \hat{b}_t^G - \Theta^{15} \cdot \hat{y}_t + \Theta^{16} \cdot \hat{r}_{t+1}^K - \Theta^{17} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}^{YCC}} \right] \\ &- \Theta^{18} \cdot \hat{z}_t + \Theta^{19} \cdot \hat{z}_t^K + \Theta^{20} \cdot \hat{u}_t^B. \end{aligned}$$

where coefficients Θ^i for $i = 14, \dots, 20$ are the same as in the conventional policy case, and the expectation is evaluated with the YCC yield vector, as it is the one that prevails in the YCC scenario. The full policy-specific yield vectors are

$$\overrightarrow{\hat{y}d_t^{YCC}} = \left[\hat{y}d_t^{YCC,1}, \overrightarrow{\hat{y}d_t^{YCC}}_{(f \geq 2)} \right]', \quad \overrightarrow{\hat{y}d_t^{CP}} = \left[\hat{y}d_t^{CP,1}, \overrightarrow{\hat{y}d_t^{CP}}_{(f \geq 2)} \right]'$$

The short-term YCC yield satisfies $\hat{y}d_t^{YCC,1} = \hat{y}d_t^{CP,1}$ because the short-maturity rate follows the conventional Taylor rule in (37).

Henceforth, each yield-dependent expression is evaluated with the yield vector relevant to that policy: YCC-related equations use $\overrightarrow{\hat{y}d_t^{YCC}}$, while the counterfactual conventional-policy yield equation uses $\overrightarrow{\hat{y}d_t^{CP}}$. Now that $\overrightarrow{\hat{y}d_t^{YCC}}$ governs households' intertemporal decisions, equation (A.75) becomes

$$\begin{aligned} \hat{y}_t &= \mathbb{E}_t \left[\hat{y}_{t+1} + \Psi^{46} \cdot \hat{\pi}_{t+1} - \Psi^{47} \cdot \hat{z}_t - \Psi^{48} \cdot \hat{z}_t^K - \Psi^{49} \cdot \overrightarrow{\hat{y}d_t^{YCC}} - \Psi^{50} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}^{YCC}} \right] \right. \\ &\quad \left. - \Psi^{51} \cdot \hat{r}_{t+1}^K - \Psi^{52} \cdot (\hat{k}_t - \hat{\varepsilon}_t^A) + \Psi^{53} \cdot \hat{k}_{t+1} - \Psi^{54} \cdot \hat{k}_{t+2} + \Psi^{55} \cdot \hat{u}_t^G \right]. \end{aligned}$$

B Calibration and Estimation Strategy

B.1 Calibrating $\{z^f\}_{f=1}^F$ and z^K at the Steady State

Calibration of $\{z^f\}_{f=1}^F$ This section explains the calibration of $\{z^f\}_{f=1}^F$ in matching the steady-state yield curve. Following the convention of Appendix A.1.3, R^{f-1} denotes the per-period return on an f -maturity bond at the steady state (i.e., the steady-state value of R_{t+1}^{f-1} in (16)). Based on the data on yields of bonds with different maturities, we calculate f -maturity bond's average holding returns $\{R^{f-1}\}_{f=1}^F$, which are our calibration target.

Step 1 Calculate steady-state bond shares $\{\lambda^{HB,f}\}_{f=1}^F$ using equation (46), given the data on $\{\lambda^{G,f}\}_{f=1}^F$, $\{\lambda^{CB,f}\}_{f=1}^F$, and ζ^{CB} .

Step 2 Aggregate the household's bond-portfolio return as $R^{HB} = \sum_{f=1}^F \lambda^{HB,f} \cdot R^{f-1}$ and compute the return ratios $\left\{\frac{R^{f-1}}{R^{HB}}\right\}_{f=1}^F$.

Step 3 Normalize $z^1 = 1$ and obtain initial guess for $\{z^{j,guess}\}$. Set $z^{j,old} = z^{j,guess}$ in the iteration below.

Step 4 Construct $\tilde{\Phi}^{B,old}$ based on the following formula, where the return ratios $\left\{\frac{R^{f-1}}{R^{HB}}\right\}_{f=1}^F$ across maturities are obtained from the data:

$$\tilde{\Phi}^{B,old} = \left[\sum_{f=1}^F \left[z^{f,old} \left(\frac{R^{f-1}}{R^{HB}} \right) \right]^{\kappa_B} \right]^{\frac{1}{\kappa_B}}.$$

Step 5 Back out new $z^{f,new}$, $f = 2, \dots, F$ estimates using:

$$z^{f,new} = (\lambda^{HB,f})^{\frac{1}{\kappa_B}} \left(\frac{R^{f-1}}{R^{HB}} \right)^{-1} \tilde{\Phi}^{B,old}.$$

Step 6 If $\{z^{f,new}\}_{f=2}^F$ differs from $\{z^{f,old}\}_{f=2}^F$ above a tolerance, set $z^{f,old} = z^{f,new}$ and return to Step 4.

Calibration of z^K We calibrate z^K such that the model's steady-state return on the household's bond portfolio, R^{HB} , matches with the observed data average. This alignment is

achieved by finding the steady-state value of R^K from (A.36),² and subsequently replacing the values of R^{HB} and R^K into equation (A.35), which enables us to recover the z^K value consistent with the model's moment.

B.2 Elasticity Estimation, κ_B

Combining the log-linear approximation of equation (15), we obtain

$$\hat{\lambda}_t^{HB,f} = \kappa_B \cdot (1 - \lambda^{HB,f}) \cdot E_t [\hat{r}_{t+1}^{f-1}] - \kappa_B \cdot \sum_{j \neq f} \lambda^{HB,j} \cdot E_t [\hat{r}_{t+1}^{j-1}] + \varepsilon_t^f,$$

where $\varepsilon_t^f \equiv \kappa_B \cdot \sum_{j \neq f} \lambda^{HB,j} \cdot [\hat{z}_t^f - \hat{z}_t^j]$ is a residual term containing the effects on the households' bond portfolio of shocks to different maturity preferences along the yield curve. Differentiation across maturities yields the following expression

$$\log(\lambda_t^{HB,f}) - \log(\lambda_t^{HB,l}) = \alpha^{fl} + \kappa_B \cdot E_t [r_{t+1}^{f-1} - r_{t+1}^{l-1}] + \varepsilon_t^{fl}, \quad (\text{B.1})$$

where α^{fl} denotes a constant, and $\varepsilon_t^{fl} = \varepsilon_t^f - \varepsilon_t^l$ represents a residual term embodying the discrepancy between the preference shocks for bond maturities f and l . The expected bond return spread, denoted as $E_t [r_{t+1}^{f-1} - r_{t+1}^{l-1}]$, poses challenges for empirical observation. Therefore, we turn to the following approximation centered on the bond yield spread:

$$\begin{aligned} E_t [r_{t+1}^{f-1} - r_{t+1}^{l-1}] &= yd_t^f - yd_t^l \\ &\quad - \underbrace{(f-1) \cdot E_t [yd_{t+1}^f - yd_t^f]}_{\text{add as control}} + \underbrace{(l-1) \cdot E_t [yd_{t+1}^l - yd_t^l]}_{\text{add as control}} \\ &\quad + \underbrace{(f-1) \cdot E_t [yd_{t+1}^f - yd_{t+1}^{f-1}]}_{\approx 0 \text{ (by assumption)}} - \underbrace{(l-1) \cdot E_t [yd_{t+1}^l - yd_{t+1}^{l-1}]}_{\approx 0 \text{ (by assumption)}}, \end{aligned}$$

where the spread $yd_t^f - yd_t^l$ in the initial line is directly observable from the data, the expected yield change in the subsequent line can be approximated by employing the realized changes as control variables, and the terms in the final line may be assumed to be close to zero over short time intervals. This last condition is the identifying assumption that allows the level spread $yd_t^f - yd_t^l$ to serve as the regressor: it requires the expected term-premium

²In the context of our present calibration, the derived R^K is situated within the range corresponding to the average corporate debt rate across varied ratings.

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differences between adjacent maturities to be small and, conditional on the controls, uncorrelated with the regressor. The final equation used for the empirical estimation of the elasticity parameter κ_B becomes:

$$\log \left(\lambda_{t+h}^{HB,f} \right) - \log \left(\lambda_{t+h}^{HB,l} \right) = \alpha_h^{fl} + \kappa_{B,h} \cdot \left[yd_t^f - yd_t^l \right] + \mathbf{x}_t' \beta_h^{\text{fl}} + \varepsilon_{t+h}^{fl}, \quad h \geq 0, \quad (\text{B.2})$$

where $\mathbf{x}_t$ denotes a vector of control variables accompanied by the corresponding coefficients β_h^{fl} , and the time subindex $h \geq 0$ accommodates a lagged effect from the yield spread to the bond portfolio composition which might occur in practice. Equation (B.2) is estimated across various horizons h utilizing Jordà local projection methods, thereby facilitating the determination of a plausible range for the elasticity parameter κ_B .

The unbiased estimation of κ_B in equation (B.2) requires the fluctuations in the bond yield spread $yd_t^f - yd_t^l$ to be uncorrelated with shocks in the relative preferences for each maturity, as captured by ε_{t+h}^{fl} . This implies that any other aggregate shocks, uncorrelated to the contemporaneous (and/or future) maturity preferences of households, could serve as potentially valid instruments for the yield spread. Following this rationale, we instrument the contemporaneous yield spread with its own lagged value, while at the same time, we incorporate the lags of the dependent variable as controls to eliminate any potential serial correlation of the preference shocks. The dependent variable is defined as the log-difference between the aggregate household portfolio shares in a group of long-maturity bonds and a group of short-maturity bonds, respectively. For the long-maturity group, we calculate the share of bonds with maturities ranging from 5 to 10 years within the households' portfolio, whereas for the short-maturity group, bonds with maturities above 15 days and up to 90 days are considered. For both groups, the aggregate household portfolio holdings are computed on a monthly basis by deducting the U.S. Treasury securities held by the Federal Reserve from the Government's outstanding Treasury amounts within the selected maturity ranges.³ The principal regressor employed is the spread between the market yields of the 7-year and the 1-month constant maturity U.S. Treasury securities, which lie within the maturity bands of the selected portfolio shares in the dependent variable. Additionally, we control for the one-month-ahead changes in the 7-year and the 1-month constant maturity yields, along with the first three lags of the dependent variable. The regressions are estimated across the sample period extending from 2003m3 to 2019m3.⁴

³The outstanding amounts of Government Treasuries are reported in the U.S. Treasury Monthly Statement of the Public Debt (MSPD).

⁴The lower bound of the sample period is dictated by the availability of maturity-disaggregated statistics

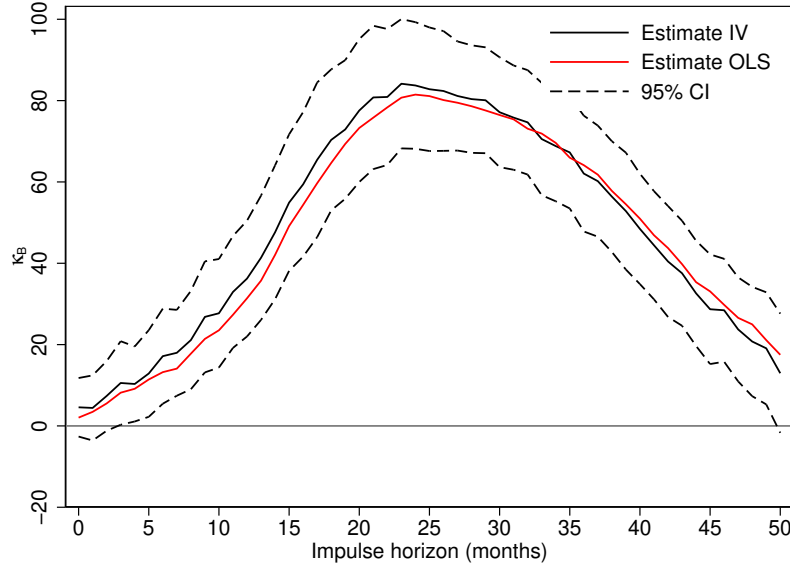


Figure B.1: Impulse-Response to a shock in the yield spread, $yd_t^f - yd_t^l$. The figure presents the coefficient estimates for the bond portfolio elasticity, κ_B , in equation (B.2), following the estimation methodology detailed in appendix B.2. The solid black line illustrates the estimate from the instrumental variables (IV) regression, with dashed lines indicating the 95 percent robust confidence intervals. The red line exhibits alternative OLS estimates. The sample period is from 2003m3 to 2019m3.

Figure B.1 delineates the IV and OLS estimates derived from Jordà local projections across a fifty-month horizon, accompanied by the 95% robust confidence bands pertinent to the primary IV regression. Both estimates largely align with the anticipated reaction of aggregate household portfolio shares, as posited by the model, in response to a shock to the yield spread. For calibration purposes, we select a value of $\kappa_B = 10$, which is consistent with the short-horizon response (roughly $h \leq 6$ months) shown in Figure B.1.

C Welfare

C.1 Deriving a second-order welfare approximation

In order to approximate welfare up to a second-order, due to the presence of trend inflation, we cannot discard $\hat{\Delta}_t$, which is the price dispersion term's log-deviation from its steady-

concerning the Federal Reserve's Treasury bond portfolio, commencing from 2002m12; the first three months are absorbed by the lag controls L^1 to L^3 on the dependent variable.

state value. For this issue, see [Coibion et al. \(2012\)](#).

Step 1: Following the literature, for any variable X , we define $\bar{X}$ as its steady-state value (with the positive trend inflation $\bar{\Pi} > 1$) and $\bar{X}^F$ as its flexible price (efficient) steady-state value. Also define (small) letter $\tilde{x}$ as log-deviation of X around $\bar{X}^F$, and $\hat{x}$ as log-deviation of X around $\bar{X}$.

We first characterize the constrained efficient (i.e., flexible-price) steady state. With the optimal production subsidy $\zeta^F = (\epsilon - 1)^{-1}$ that eliminates the monopolistic competition distortion, there is no distortion other than the firms' financing constraint in the flexible-price steady state economy anymore.⁵ In particular, each firm's price resetting condition (A.2) becomes

$$1 = \frac{P_t^*}{P_t} = \underbrace{\left(\frac{(1 + \zeta^F)^{-1} \epsilon}{\epsilon - 1} \right)}_{=1} \left[1 - \gamma_L (\tilde{R}_{t+1}^K - 1) \right]^{-1} \cdot \frac{MC_t}{P_t},$$

which can be simplified to

$$\frac{MC_t}{P_t} = 1 - \gamma_L (\tilde{R}_{t+1}^K - 1), \quad (\text{C.1})$$

where we use that all firms become identical, and thus $MC_t(\nu) = MC_t$ for all $\nu \in [0, 1]$. Therefore, the real marginal cost becomes the same for all firms. Plugging the real marginal cost (A.3) into the individual firm's labor demand (30) with $W_t(\nu) = W_t$ for $\forall \nu$, and defining $n_t = \frac{N_t}{\bar{N}_t}$ and $y_t = \frac{Y_t}{A_t \bar{N}_t}$, we obtain

$$n_t = (1 - \alpha) y_t \left(\frac{P_t^K}{P_t} \right)^\alpha \left(\frac{W_t}{P_t A_t} \right)^{-\alpha} = (1 - \alpha) \left[1 - \gamma_L (\tilde{R}_{t+1}^K - 1) \right] y_t \left(\frac{W_t}{P_t A_t} \right)^{-1}, \quad (\text{C.2})$$

which, with the household's intra-temporal consumption-labor decision (22), becomes:

$$\frac{n_t^{\frac{1}{c_t}}}{c_t^{-1}} = (1 - \alpha) \left[1 - \gamma_L (\tilde{R}_{t+1}^K - 1) \right] \frac{y_t}{n_t}, \quad (\text{C.3})$$

which is the efficiency condition that ensures that the household's marginal rate of substitu-

⁵The capital producing firm is competitive and thus our economy features no friction other than the firms' financing constraint if it were not nominal rigidity nor trend inflation. Still, the constraint on loan issuance does not affect firms' marginal decisions on labor and capital.

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tion matches with the marginal rate of technical substitution. Therefore, at the flexible-price steady state, the new constant Φ , which will turn out to enter in the per-period welfare later, can be calculated as

$$\Phi \equiv (\bar{n}^F)^{1+\frac{1}{\eta}} = (1-\alpha) \left[1 - \gamma_L \left(\frac{R^K}{\beta^{-1}\Pi \cdot GA \cdot GN} - 1 \right) \right] \frac{\bar{y}^F}{\bar{c}^F}, \quad (\text{C.4})$$

where the steady state expression for $\tilde{R}^K$ has been replaced using equation (A.31), and $\bar{n}^F$, $\bar{y}^F$, and $\bar{c}^F$ denote values of normalized labor, output, and consumption, respectively.

Step 2: With aggregation equations (A.6) and (A.7), we obtain

$$\left(\frac{N_t}{\bar{N}_t} \right)^{1-\alpha} \left(\frac{K_t}{A_{t-1}\bar{N}_{t-1}} \right)^\alpha = \alpha^\alpha (1-\alpha)^{1-\alpha} (GA_t \cdot GN)^\alpha \left(\frac{Y_t}{A_t \bar{N}_t} \right) \Delta_t^{(1-\alpha)\left[\frac{\eta}{\eta+1} + \frac{\alpha}{1-\alpha}\right]}, \quad (\text{C.5})$$

which is the aggregate production function with the price dispersion Δ_t . Plugging steady-state (with trend-inflation) capital (A.42) and output (A.44) into equation (C.5) yields

$$\frac{N}{\bar{N}} = \left[\alpha^\alpha (1-\alpha)^{1-\alpha} (GA \cdot GN)^\alpha (\xi^K)^{-\alpha} \right]^{\frac{1}{1-\alpha}} \Delta^{\frac{\eta+\alpha}{(\eta+1)(1-\alpha)}} (\xi^Y)^{\frac{1}{1-\alpha}\left(\frac{\eta+\alpha}{\eta+1}\right)} (\xi^C)^{-\frac{\eta}{\eta+1}} \quad (\text{C.6})$$

where ξ^K in (A.42), ξ^Y in (A.41), and ξ^C in (A.43) all depend on θ and the trend inflation Π . Therefore, we see that $\bar{n} \neq \bar{n}^F$ and define $\log X_n \equiv \log \bar{n} - \log \bar{n}^F$, which will turn out to be useful later when we calculate the household's first-order labor cost.

Step 3: Price dispersion with positive trend inflation

Delta method We use the following approximation throughout this section. For a random variable X with $\mathbb{E}[X] = \mu_X$,

$$\text{Var}(f(X)) = f'(\mu_X)^2 \cdot \text{Var}(X) + \text{h.o.t.} \quad (\text{C.7})$$

where h.o.t. denotes higher-order terms. In the remainder of this section, displays derived from Taylor or delta-method expansions omit such terms unless they are shown explicitly.

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Price dispersion We use lower-case p_t and $p_t(\nu)$ as logarithms of P_t and $P_t(\nu)$. By applying the above delta method to $P_t^{1-\epsilon} = \mathbb{E}_\nu (P_t(\nu)^{1-\epsilon})$, we obtain

$$p_t = \underbrace{\int_0^1 p_t(\nu) d\nu}_{\equiv \bar{p}_t} + \frac{1}{2} \left(\frac{1}{1-\epsilon} \right) \frac{\text{Var}_\nu (P_t(\nu)^{1-\epsilon})}{\mathbb{E}_\nu (P_t(\nu)^{1-\epsilon})^2} + \text{h.o.t.} \quad (\text{C.8})$$

where we define $\bar{p}_t \equiv \mathbb{E}_\nu (p_t(\nu))$. Applying the delta method to $\text{Var}_\nu (P_t(\nu)^{1-\epsilon})$, we have

$$\text{Var}_\nu (P_t(\nu)^{1-\epsilon}) = (1-\epsilon)^2 \cdot [\exp((1-\epsilon)\bar{p}_t)]^2 \cdot \text{Var}_\nu (p_t(\nu)), \quad (\text{C.9})$$

where we define $D_t \equiv \text{Var}_\nu (p_t(\nu))$. Applying the delta method to $\mathbb{E}_\nu (P_t(\nu)^{1-\epsilon})$, we obtain

$$\mathbb{E}_\nu (P_t(\nu)^{1-\epsilon}) = \exp((1-\epsilon)\bar{p}_t) \left[1 + \frac{(1-\epsilon)^2}{2} D_t \right]. \quad (\text{C.10})$$

Plugging (C.9) and (C.10) into (C.8), we obtain

$$p_t = \bar{p}_t + \frac{1-\epsilon}{2} \cdot \frac{D_t}{\left[1 + \frac{(1-\epsilon)^2}{2} D_t \right]^2}, \quad (\text{C.11})$$

which we linear-approximate around $D_t = \bar{D}$ and obtain

$$p_t - \bar{p}_t = \underbrace{\frac{1-\epsilon}{2} \cdot \frac{\bar{D}}{\left[1 + \frac{(1-\epsilon)^2}{2} \bar{D} \right]^2}}_{\equiv \Theta_1^p} + \underbrace{\frac{1-\epsilon}{2} \cdot \frac{1 - \frac{(1-\epsilon)^2}{2} \bar{D}}{\left[1 + \frac{(1-\epsilon)^2}{2} \bar{D} \right]^3}}_{\equiv \Theta_2^p} \cdot (D_t - \bar{D}) = \Theta_1^p + \Theta_2^p (D_t - \bar{D}). \quad (\text{C.12})$$

Now from our original definition of the price dispersion Δ_t (i.e., (42)), we take logarithm on both sides, linear-approximate around $\bar{D}$, and plug (C.12) into it to attain

$$\begin{aligned} \ln \Delta_t &= \ln \int_0^1 \left(\frac{P_t(\nu)}{P_t} \right)^{\frac{-\epsilon(\eta+1)}{\eta+\alpha}} d\nu \\ &= \frac{\epsilon(\eta+1)}{\eta+\alpha} (p_t - \bar{p}_t) + \ln \left(1 + \frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha} \right)^2 \bar{D} \right) + \frac{\frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha} \right)^2}{1 + \frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha} \right)^2 \bar{D}} (D_t - \bar{D}) \\ &= \Theta_1^\Delta + \Theta_2^\Delta \cdot (D_t - \bar{D}) + \text{h.o.t.}, \end{aligned}$$

where

$$\Theta_1^\Delta \equiv \frac{\epsilon(\eta+1)}{\eta+\alpha} \cdot \frac{1-\epsilon}{2} \cdot \frac{\bar{D}}{\left[1 + \frac{(1-\epsilon)^2}{2} \bar{D}\right]^2} + \ln \left(1 + \frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha}\right)^2 \bar{D}\right), \quad (\text{C.13})$$

$$\Theta_2^\Delta \equiv \frac{\epsilon(\eta+1)}{\eta+\alpha} \cdot \frac{1-\epsilon}{2} \cdot \frac{1 - \frac{(1-\epsilon)^2}{2} \bar{D}}{\left[1 + \frac{(1-\epsilon)^2}{2} \bar{D}\right]^3} + \frac{\frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha}\right)^2}{1 + \frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha}\right)^2 \bar{D}}. \quad (\text{C.14})$$

If we define b_t as the logarithm of the newly price-resetting firm's relative price P_t^*/P_t and $\bar{b}$ as its steady state value, we have $\bar{b} \neq 0$ due to the trend inflation. Combining (A.10) and (A.12) and linearizing, we obtain

$$b_t \equiv p_t^* - p_t = \bar{b} + \underbrace{\frac{\theta \Pi^{\epsilon-1}}{1 - \theta \Pi^{\epsilon-1}}}_{\equiv M} \hat{\pi}_t = \bar{b} + M \cdot \hat{\pi}_t, \quad \text{with } \bar{b} = \frac{1}{\epsilon-1} \ln \left(\frac{1-\theta}{1-\theta \Pi^{\epsilon-1}} \right). \quad (\text{C.15})$$

With $D_t = \text{Var}_\nu(p_t(\nu)) = \mathbb{E}_\nu((p_t(\nu) - p_t + p_t - \bar{p}_t)^2)$, we can write it as

$$D_t = \int_0^{1-\theta} (p_t^* - p_t)^2 d\nu + 2 \left(\int_0^{1-\theta} (p_t^* - p_t) d\nu \right) (p_t - \bar{p}_t) \quad (\text{C.16})$$

$$\begin{aligned} &+ (1-\theta)(p_t - \bar{p}_t)^2 + \int_{1-\theta}^1 (p_{t-1}(\nu) - \bar{p}_t)^2 d\nu \\ &= (1-\theta)(p_t^* - p_t)^2 + 2(1-\theta)(p_t^* - p_t)(p_t - \bar{p}_t) + (1-\theta)(p_t - \bar{p}_t)^2 + \theta D_{t-1} + \theta(\bar{p}_t - \bar{p}_{t-1})^2, \end{aligned} \quad (\text{C.17})$$

where we use

$$\int_{1-\theta}^1 (p_{t-1}(\nu) - \bar{p}_t)^2 d\nu = \theta D_{t-1} + \theta(\bar{p}_{t-1} - \bar{p}_t)^2. \quad (\text{C.18})$$

Conjecture Following Coibion et al. (2012), we conjecture the dynamics of D_t up to second order as⁶

$$D_t - \bar{D} = \kappa_D \hat{\pi}_t + Z_D (\hat{\pi}_t)^2 + F_D (D_{t-1} - \bar{D}) + G_D (D_{t-1} - \bar{D}) \hat{\pi}_t + H_D (D_{t-1} - \bar{D})^2. \quad (\text{C.19})$$

⁶Following Coibion et al. (2012), we treat κ_D as being of the same order as the shocks. Hence $\kappa_D \hat{\pi}_t$ is a second-order term, since $\hat{\pi}_t$ is a first-order deviation. This allows the first-order log-linearized model to omit price dispersion while retaining the effect of price dispersion in the second-order welfare expansion.

with

$$\begin{aligned}
 \bar{D} &= (\bar{b} + \Theta_1^p)^2 + \frac{\theta}{1-\theta}(\bar{\pi})^2 \quad (\text{i.e., steady state value of } D_t), \\
 \kappa_D &= [1 - 2(1-\theta)\Theta_2^p(\bar{b} + \Theta_1^p) + 2\theta\Theta_2^p\bar{\pi}]^{-1} [2(1-\theta)M(\bar{b} + \Theta_1^p) + 2\theta\bar{\pi}], \\
 Z_D &= [1 - 2(1-\theta)\Theta_2^p(\bar{b} + \Theta_1^p) + 2\theta\Theta_2^p\bar{\pi}]^{-1} [(1-\theta)M^2 + 2(1-\theta)M\Theta_2^p\kappa_D + (\Theta_2^p)^2(\kappa_D)^2 + \theta - 2\theta\Theta_2^p\kappa_D], \\
 F_D &= [1 - 2(1-\theta)\Theta_2^p(\bar{b} + \Theta_1^p) + 2\theta\Theta_2^p\bar{\pi}]^{-1} [\theta + 2\theta\Theta_2^p\bar{\pi}], \\
 G_D &= [1 - 2(1-\theta)\Theta_2^p(\bar{b} + \Theta_1^p) + 2\theta\Theta_2^p\bar{\pi}]^{-1} [2(1-\theta)M\Theta_2^pF_D + 2(\Theta_2^p)^2\kappa_DF_D - 2\theta\Theta_2^pF_D \\
 &\quad + 2\theta\Theta_2^p - 2\theta(\Theta_2^p)^2\kappa_D], \\
 H_D &= [1 - 2(1-\theta)\Theta_2^p(\bar{b} + \Theta_1^p) + 2\theta\Theta_2^p\bar{\pi}]^{-1} [(\Theta_2^p)^2(F_D)^2 + \theta(\Theta_2^p)^2 - 2\theta(\Theta_2^p)^2F_D].
 \end{aligned} \tag{C.20}$$

In the absence of trend inflation, we would have $\bar{\pi} = 0$ and $\bar{D} = 0$, implying that D_t is a second-order variable around zero and $\kappa_D = 0$. By contrast, when steady-state inflation is positive ($\bar{\pi} > 0$) and the price dispersion measure satisfies $\bar{D} > 0$, the linear response of price dispersion to inflation is generally nonzero, so κ_D is treated as a first-order coefficient. This does not make $\kappa_D\hat{\pi}_t$ a first-order object: because $\hat{\pi}_t$ is itself a first-order deviation, the product $\kappa_D\hat{\pi}_t$ remains second order and belongs in the welfare expansion.

Our goal is to derive (C.19) from firms' optimal pricing decisions and the impact of price dispersion on the aggregate price level. Substituting (C.12) and (C.15) into (C.16), and replacing $(D_t - \bar{D})$ with its conjectured form in (C.19) up to second order,⁷ and then matching coefficients yields the set of parameters reported in (C.20).

Consumption utility We can second-order approximate the utility of consumption as

$$\begin{aligned}
 u(c_t) &= \log c_t = u(\bar{c}^F) + u'_{\bar{c}^F} \cdot \bar{c}^F \cdot \underbrace{\left(\frac{c_t - \bar{c}^F}{\bar{c}^F} \right)}_{=\tilde{c}_t + \frac{1}{2}(\tilde{c}_t)^2} + \frac{1}{2}u''_{\bar{c}^F} \cdot (\bar{c}^F)^2 \cdot \underbrace{\left(\frac{c_t - \bar{c}^F}{\bar{c}^F} \right)^2}_{=(\tilde{c}_t)^2} + \text{h.o.t} \\
 &= u(\bar{c}^F) + \tilde{c}_t + \text{h.o.t.}
 \end{aligned} \tag{C.21}$$

⁷On the right-hand side, $(p_t - \bar{p}_t)^2$ appears and contains a second-order component, $(D_t - \bar{D})^2$, from (C.12). We use (C.19) to express this term in terms of $(\hat{\pi}_t)^2$, $(D_{t-1} - \bar{D})^2$, and $\hat{\pi}_t(D_{t-1} - \bar{D})$.

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Step 4: Labor aggregation and cost By applying the delta method—see (C.7)—to the labor aggregator, we can obtain⁸

$$\tilde{n}_t - \mathbb{E}_\nu(\tilde{n}_t(\nu)) = \underbrace{\frac{\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \bar{\nabla}}{\left[1 + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right)^2 \bar{\nabla} \right]^2}}_{\equiv \Theta_1^n} + \underbrace{\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \frac{1 - \frac{1}{2} \left(\frac{\eta+1}{\eta} \right)^2 \bar{\nabla}}{\left[1 + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right)^2 \bar{\nabla} \right]^3}}_{\equiv \Theta_2^n} \cdot (\nabla_t - \bar{\nabla}) \quad (\text{C.22})$$

where $\nabla_t \equiv \text{Var}_\nu(\log n_t(\nu))$. The second-order approximation to the firm ν -specific labor cost around the flexible-price steady state yields

$$\frac{\eta}{\eta+1} \left(\frac{N_t(\nu)}{\bar{N}_t} \right)^{\frac{\eta+1}{\eta}} = \frac{\eta}{\eta+1} (\bar{n}^F)^{\frac{\eta+1}{\eta}} + \Phi \left[\tilde{n}_t(\nu) + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \tilde{n}_t(\nu)^2 \right] + h.o.t \quad (\text{C.23})$$

where the constant Φ is from (C.4). Aggregating (C.23) over firms $\nu \in [0, 1]$ and plugging (C.22) results in

$$\begin{aligned} \frac{\eta}{\eta+1} \int_0^1 \left(\frac{N_t(\nu)}{\bar{N}_t} \right)^{\frac{\eta+1}{\eta}} d\nu - \frac{\eta}{\eta+1} (\bar{n}^F)^{\frac{\eta+1}{\eta}} &= \Phi \left[\mathbb{E}_\nu(\tilde{n}_t(\nu)) + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \int_0^1 \tilde{n}_t(\nu)^2 d\nu \right] \\ &= -\Phi \left(\Theta_1^n - \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_1^n)^2 \right) + \Phi \left[\left(1 - \left(\frac{\eta+1}{\eta} \right) \Theta_1^n \right) \bar{n}_t + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \bar{n}_t^2 \right. \\ &\quad \left. + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_2^n)^2 (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla})^2 - \frac{\eta+1}{\eta} \Theta_2^n \bar{n}_t (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla}) \right. \\ &\quad \left. + \left(\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (1 + 2\Theta_1^n \Theta_2^n - \Theta_2^n) (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla}) + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \bar{\nabla} \right] \right]. \end{aligned} \quad (\text{C.24})$$

Labor dispersion From individual firm's labor and capital demand (30) and the household's intra-marginal condition (22), we obtain

$$\tilde{k}_t(\nu) = \left(1 + \frac{1}{\eta} \right) \tilde{n}_t(\nu) + \text{aggregate}, \quad (\text{C.25})$$

where ‘aggregate’ stands for aggregate variables. Therefore, we obtain

$$\tilde{y}_t(\nu) = \left(1 + \frac{\alpha}{\eta} \right) \tilde{n}_t(\nu) + \text{aggregate}, \quad (\text{C.26})$$

⁸In the flexible-price steady-state, there is no heterogeneity among firms, i.e., $\bar{n}^F(\nu) = \bar{n}^F$ for $\forall \nu$.

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by plugging (C.25) into each firm's production function $\tilde{y}_t(\nu) = \alpha \tilde{k}_t(\nu) + (1 - \alpha) \tilde{n}_t(\nu)$. From the Dixit-Stiglitz good demand (27) and with (C.26), we can get

$$\text{Var}_\nu(\tilde{n}_t(\nu)) = \left(\frac{\epsilon}{1 + \frac{\alpha}{\eta}} \right)^2 \text{Var}_\nu(p_t(\nu)), \text{ with } \bar{\nabla} = \left(\frac{\epsilon}{1 + \frac{\alpha}{\eta}} \right)^2 \bar{D}. \quad (\text{C.27})$$

Step 5: Constructing a welfare function: Combining consumption utility (C.21) and labor disutility (C.24), we construct welfare as

$$\begin{aligned} \mathbb{E}U_t - \bar{U}^F = \mathbb{E} \left[\tilde{c}_t + \Phi \left(\Theta_1^n - \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_1^n)^2 \right) - \Phi \left\{ \left(1 - \left(\frac{\eta+1}{\eta} \right) \Theta_1^n \right) \tilde{n}_t + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \tilde{n}_t^2 \right. \right. \\ \left. \left. + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_2^n)^2 (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla})^2 - \frac{\eta+1}{\eta} \Theta_2^n \tilde{n}_t (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla}) \right. \right. \\ \left. \left. + \left(\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (1 + 2\Theta_1^n \Theta_2^n) - \Theta_2^n \right) (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla}) + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \bar{\nabla} \right\} \right], \end{aligned}$$

where from (A.43), (A.44), and (C.6) the flexible-price steady-state utility is given as

$$\begin{aligned} \bar{U}^F &= \log(\bar{c}^F) - \frac{\eta}{\eta+1} \left(\frac{N^F}{\bar{N}} \right)^{\frac{\eta+1}{\eta}} \\ &= \frac{1}{\eta+1} \left[\log(\xi^{C,f}) + \frac{\eta+\alpha}{1-\alpha} \log(\xi^{Y,f}) \right] - \frac{\eta}{\eta+1} \left[\alpha^\alpha (1-\alpha)^{1-\alpha} (GA \cdot GN)^\alpha (\xi^{K,f})^{-\alpha} \right]^{\frac{\eta+1}{(1-\alpha)\eta}} (\xi^{Y,f})^{\frac{\eta+\alpha}{(1-\alpha)\eta}} (\xi^{C,f})^{-1}, \end{aligned}$$

where

$$\begin{aligned} \xi^{F,f} &= (1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left[\beta^{-1} \cdot GA \cdot GN - (1-\delta) \right]^\alpha \left(\frac{\eta+1}{\eta+\alpha} \right), \quad \xi^{H,f} = 1 - \gamma_L \left(\frac{R^K}{R^S} - 1 \right), \quad \xi^{Y,f} = \frac{\xi^{H,f}}{\xi^{F,f}}, \\ \xi^{K,f} &= \alpha(1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot GA \cdot GN \cdot \left[\beta^{-1} \cdot GA \cdot GN - (1-\delta) \right]^{-\frac{\eta(1-\alpha)}{\eta+\alpha}} \xi^{Y,f}, \\ \xi^{C,f} &= 1 - \zeta^G - \xi^{K,f} \left(1 - \frac{1-\delta}{GA \cdot GN} \right). \end{aligned}$$

Note that $\xi^{C,f}$, $\xi^{Y,f}$, and $\xi^{K,f}$ are levels of ξ^C , ξ^Y , and ξ^K when $\theta = 0$ —corresponding to the flexible price steady state. With

$$\log \frac{\xi^Y}{\xi^{Y,f}} = \log \left(\frac{1 - \theta \beta \Pi^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)}}{1 - \theta \beta \Pi^{\epsilon-1}} \right) + \frac{1}{\epsilon-1} \left[1 + \epsilon \left(\frac{1-\alpha}{\eta+\alpha} \right) \right] \log \left(\frac{1-\theta}{1 - \theta \Pi^{\epsilon-1}} \right),$$

and

$$\begin{aligned}\log \frac{\xi^K}{\xi^{K,f}} &= \log \left(\frac{\xi^Y}{\xi^{Y,f}} \right) + \log \Delta, \\ \log \frac{\xi^C}{\xi^{C,f}} &= \log \frac{1 - \zeta^G - \xi^K \left(1 - \frac{1 - \delta}{GA \cdot GN} \right)}{1 - \zeta^G - \xi^{K,f} \left(1 - \frac{1 - \delta}{GA \cdot GN} \right)},\end{aligned}$$

where Δ at the steady state with trend inflation is defined in (A.37), which gives

$$\log \Delta = \log \left(\frac{1 - \theta}{1 - \theta \Pi^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)}} \right) + \frac{\epsilon}{\epsilon - 1} \left(\frac{\eta + 1}{\eta + \alpha} \right) \log \left(\frac{1 - \theta \Pi^{\epsilon-1}}{1 - \theta} \right).$$

If we define $\log X_c = \tilde{c}_t - \hat{c}_t$ as the log-difference in consumption between our steady state (with trend-inflation) and the flexible price steady state, we obtain

$$\log X_c \equiv \log \bar{c} - \log \bar{c}^F = \frac{1}{\eta + 1} \left[\log \left(\frac{\xi^C}{\xi^{C,f}} \right) + \frac{\eta + \alpha}{1 - \alpha} \log \left(\frac{\xi^Y}{\xi^{Y,f}} \right) \right]. \quad (\text{C.28})$$

For labor, we define $\log X_n$ as the log-difference in labor between our steady state with trend inflation and the flexible price steady state, which is, with the help of (C.6), given by

$$\log X_n \equiv \log \bar{n} - \log \bar{n}^F = -\frac{\alpha}{1 - \alpha} \log \frac{\xi^K}{\xi^{K,f}} + \frac{1}{1 - \alpha} \left(\frac{\eta + \alpha}{\eta + 1} \right) \log \frac{\xi^Y}{\xi^{Y,f}} - \frac{\eta}{\eta + 1} \log \frac{\xi^C}{\xi^{C,f}} + \frac{\eta + \alpha}{(\eta + 1)(1 - \alpha)} \log \Delta. \quad (\text{C.29})$$

With $\tilde{c}_t = \hat{c}_t + \log X_c$, $\tilde{n}_t = \hat{n}_t + \log X_n$, and the stationarity assumption (Coibion et al., 2012; Dordal i Carreras et al., 2016), we can get

$$\mathbb{E} \left[\tilde{c}_t - \Phi \left(1 - \left(\frac{\eta + 1}{\eta} \right) \Theta_1^n \right) \tilde{n}_t \right] = \log X_c - \Phi \left(1 - \left(\frac{\eta + 1}{\eta} \right) \Theta_1^n \right) \log X_n.$$

Second order terms: With $\tilde{n}_t = \hat{n}_t + \log X_n$, second-order terms can be collected as

$$\begin{aligned}& -\Phi \left[\frac{\eta + 1}{2\eta} \mathbb{E} (\hat{n}_t^2) + \frac{\eta + 1}{2\eta} (\Theta_2^n)^2 \mathbb{E} \left((\text{Var}_\nu(\hat{n}_t(\nu)) - \bar{\nabla})^2 \right) - \frac{\eta + 1}{\eta} \Theta_2^n \mathbb{E} (\hat{n}_t (\text{Var}_\nu(\hat{n}_t(\nu)) - \bar{\nabla})) \right. \\ & \left. + \left(\frac{1}{2} \left(\frac{\eta + 1}{\eta} \right) (1 + 2\Theta_1^n \Theta_2^n) - \Theta_2^n - \frac{\eta + 1}{\eta} \Theta_2^n \log X_n \right) \mathbb{E} (\text{Var}_\nu(\hat{n}_t(\nu)) - \bar{\nabla}) \right],\end{aligned}$$

which, after we can plug (C.27) into, becomes

$$\begin{aligned}
 & -\Phi \left[\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \text{Var}(\hat{n}_t) + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_2^n)^2 \left(\frac{\epsilon}{1+\frac{\alpha}{\eta}} \right)^4 \mathbb{E}(D_t - \bar{D})^2 \right. \\
 & \quad \left. - \frac{\eta+1}{\eta} \Theta_2^n \left(\frac{\epsilon}{1+\frac{\alpha}{\eta}} \right)^2 \text{Cov}(\hat{n}_t, D_t) \right. \\
 & \quad \left. + \left(\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (1 + 2\Theta_1^n \Theta_2^n) - \Theta_2^n \left(1 + \frac{\eta+1}{\eta} \log X_n \right) \right) \left(\frac{\epsilon}{1+\frac{\alpha}{\eta}} \right)^2 \mathbb{E}(D_t - \bar{D}) \right]. \quad (\text{C.30})
 \end{aligned}$$

Finally, by plugging (C.19) into (C.30), we get the following proposition. Since κ_D is of the same order as shock processes, up to a second-order, we can ignore covariance terms and the square term of D_t . Taking expectations of (C.19) under stationarity then gives

$$\mathbb{E}(D_t - \bar{D}) = F_D \mathbb{E}(D_t - \bar{D}) + Z_D \text{Var}(\hat{\pi}_t),$$

so that

$$\mathbb{E}(D_t - \bar{D}) = \frac{Z_D}{1 - F_D} \text{Var}(\hat{\pi}_t).$$

Therefore, a 2nd-order approximation to the expected per-period welfare would be given as

$$\mathbb{E}U_t - \bar{U}^F = \Omega_0 + \Omega_n \text{Var}(\hat{n}_t) + \Omega_\pi \text{Var}(\hat{\pi}_t) + h.o.t.,$$

with

$$\begin{aligned}
 \Omega_0 &= \log X_c - \Phi \left(1 - \left(\frac{\eta+1}{\eta} \right) \Theta_1^n \right) \log X_n + \Phi \left(\Theta_1^n - \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_1^n)^2 \right) - \Phi \frac{1}{2} \frac{\eta+1}{\eta} (\log X_n)^2 \\
 & \quad - \Phi \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \left(\frac{\epsilon}{1+\frac{\alpha}{\eta}} \right)^2 \bar{D}, \\
 \Omega_n &= -\Phi \frac{1}{2} \left(\frac{\eta+1}{\eta} \right), \\
 \Omega_\pi &= -\Phi \left[\left(\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (1 + 2\Theta_1^n \Theta_2^n) - \Theta_2^n \left(1 + \frac{\eta+1}{\eta} \log X_n \right) \right) \left(\frac{\epsilon}{1+\frac{\alpha}{\eta}} \right)^2 \frac{Z_D}{1 - F_D} \right],
 \end{aligned}$$

where $\log X_c$ and $\log X_n$ are defined in (C.28) and (C.29), respectively; the coefficients Θ_1^n and Θ_2^n are provided in (C.22); and $\bar{D}$ is determined by jointly solving (C.12)—which describes the definition of Θ_1^p —and (C.20). The parameters κ_D , Z_D , F_D , G_D , and H_D are specified in (C.20). Higher-order terms are denoted by *h.o.t.* Finally, the coefficient Ω_0 is part of the policy-independent terms (*t.i.p.*) that appear in Proposition 1.

D Summary of Equilibrium Equations

D.1 Equilibrium Equations: Conventional Policy (CP)

$$(i). \quad \frac{C_t}{A_t \bar{N}_t} = (1 - \zeta_t^G) \left(\frac{Y_t}{A_t \bar{N}_t} \right) + \left(\frac{1 - \delta}{GA_t \cdot GN} \right) \left(\frac{K_t}{A_{t-1} \bar{N}_{t-1}} \right) - \left(\frac{K_{t+1}}{A_t \bar{N}_t} \right), \quad (D.1)$$

$$(ii). \quad 1 = \beta \cdot \mathbb{E}_t \left[\frac{R_{t+1}^S}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right)} \right], \quad (D.2)$$

$$(iii). \quad \lambda_t^{HB,1} = 1 - \sum_{f=2}^F \lambda_t^{HB,f}, \quad (D.3)$$

$$(iv). \quad - \left(\frac{B_t^{G,f}}{A_t \bar{N}_t P_t} + \frac{\overline{B^{CB,f}}}{ANP} \right) \cdot \left(\lambda_t^{HB,f} \right)^{-1} = \gamma_L \cdot (1 + \zeta^F) \cdot \left(\frac{1 - \lambda_t^K}{\lambda_t^K} \right) \left(\frac{Y_t}{A_t \bar{N}_t} \right), \quad \forall f > 1, \quad (D.4)$$

$$(v). \quad YD_t^1 = \max \{ YD_t^{1*}, 1 \}, \quad (D.5)$$

$$(vi). \quad \frac{YD_t^{1*}}{YD^1} = \left(\frac{YD_{t-1}^{1*}}{YD^1} \right)^{\rho_1} \left(\frac{YD_{t-2}^{1*}}{YD^1} \right)^{\rho_2} \left[\left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\gamma_\pi} \left(\frac{Y_t}{\bar{Y}_t} \right)^{\gamma_y} \exp \left(\tilde{\epsilon}_t^{YD^1} \right) \right]^{1 - (\rho_1 + \rho_2)}, \quad (D.6)$$

$$(vii). \quad \lambda_t^{HB,f} = \left(\frac{\mathbb{E}_t \left[\frac{\beta z_t^f}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \cdot \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right)} \frac{(YD_{t+1}^{f-1})^{-(f-1)}}{(YD_t^f)^{-f}} \right]}{\Phi_t^B} \right)^{\kappa_B}, \quad \forall f \quad (D.7)$$

$$(viii). \quad \Phi_t^B = \left[\sum_{j=1}^F \mathbb{E}_t \left[\frac{\beta z_t^j}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \cdot \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right)} \frac{(YD_{t+1}^{j-1})^{-(j-1)}}{(YD_t^j)^{-j}} \right] \right]^{\kappa_B \frac{1}{\kappa_S}}, \quad (D.8)$$

$$(ix). \quad \lambda_t^K = \left(\frac{z_t^K \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K]}{\Phi_t^S} \right)^{\kappa_S}, \quad (D.9)$$

$$(x). \quad \Phi_t^S = \left[\left(\mathbb{E}_t [Q_{t,t+1} R_{t+1}^{HB}] \right)^{\kappa_S} + \left(z_t^K \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K] \right)^{\kappa_S} \right]^{\frac{1}{\kappa_S}}, \quad (D.10)$$

$$(xi). \quad R_t^j = \sum_{f=0}^{F-1} \lambda_{t-1}^{j,f+1} \frac{(YD_t^f)^{-f}}{(YD_{t-1}^{f+1})^{-(f+1)}}, \quad j \in \{HB, G, CB\} \quad (D.11)$$

$$(xii). \quad R_t^S = (1 - \lambda_{t-1}^K) R_t^{HB} + \lambda_{t-1}^K R_t^K, \quad (D.12)$$

$$(xiii). \quad 1 = \mathbb{E}_t \left[Q_{t,t+1} \Pi_{t+1} \left[(1 - \delta) + \frac{P_{t+1}^K}{P_{t+1}} \right] \right], \quad (D.13)$$

$$(xiv). \quad F_t = (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{(1 + \varsigma_F)^{-1} \epsilon}{\epsilon - 1} \right) \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} \left(\frac{Y_t}{A_t \bar{N}_t} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t} \right)^{\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} \\ + \theta \beta \mathbb{E}_t \left[\Pi_{t+1}^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} F_{t+1} \right], \quad (D.14)$$

$$(xv). \quad H_t = \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-1} \frac{Y_t}{A_t \bar{N}_t} \left[1 - \gamma_L \cdot \left(\tilde{R}_{t+1}^K - 1 \right) \right] + \theta \beta \mathbb{E}_t \left[\Pi_{t+1}^{\epsilon-1} H_{t+1} \right], \quad (D.15)$$

$$(xvi). \quad \frac{F_t}{H_t} = \left(\frac{1 - \theta}{1 - \theta \Pi_t^{\epsilon-1}} \right)^{\left(\frac{1}{\epsilon-1} \right) \left[1 + \epsilon \left(\frac{1-\alpha}{\eta+\alpha} \right) \right]}, \quad (D.16)$$

$$(xvii). \quad \Delta_t = (1 - \theta) \left(\frac{1 - \theta \Pi_t^{\epsilon-1}}{1 - \theta} \right)^{\left(\frac{\epsilon}{\epsilon-1} \right) \left(\frac{\eta+1}{\eta+\alpha} \right)} + \theta \Pi_t^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} \Delta_{t-1}, \quad (D.17)$$

$$(xviii). \quad \frac{N_t}{\bar{N}_t} = (1 - \alpha)^{\left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-\alpha \left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{Y_t}{A_t \bar{N}_t} \right)^{\left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{P_t^K}{P_t} \right)^{\alpha \left(\frac{\eta}{\eta+\alpha} \right)} \Delta_t^{\frac{\eta}{\eta+1}}, \quad (D.18)$$

$$(xix). \quad \frac{K_t}{A_{t-1} \bar{N}_{t-1}} = \alpha (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot G A_t \cdot G N \cdot \left(\frac{C_t}{A_t \bar{N}_t} \right)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} \left(\frac{Y_t}{A_t \bar{N}_t} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t} \right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha} \right)} \Delta_t, \quad (D.19)$$

$$(xx). \quad \frac{B_t^G}{P_t A_t \bar{N}_t} = \frac{R_t^G}{\Pi_t \cdot G A_t \cdot G N} \cdot \frac{B_{t-1}^G}{P_{t-1} A_{t-1} \bar{N}_{t-1}} - [\zeta_t^G + \zeta^F - \zeta_t^T] \left(\frac{Y_t}{A_t \bar{N}_t} \right), \quad (D.20)$$

$$(xxi). \quad \lambda_t^{G,1} = \frac{1}{1 + \sum_{l=2}^F a^{B,l} \exp(\tilde{u}_t^{B,l})}, \quad \lambda_t^{G,f} = \frac{a^{B,f} \exp(\tilde{u}_t^{B,f})}{1 + \sum_{l=2}^F a^{B,l} \exp(\tilde{u}_t^{B,l})}, \quad \forall f > 1, \quad (D.21)$$

$$(xxii). \quad \tilde{u}_t^{B,f} = \sum_{j=2}^J \tau_{fj}^B u_t^{B,j}, \quad (D.22)$$

$$(xxiii). \quad u_t^{B,j} = \rho_{B,j} u_{t-1}^{B,j} + \varepsilon_t^{B,j}, \quad \forall j = 1, \dots, J \quad (D.23)$$

$$(xxiv). \quad B_t^{G,f} = \lambda_t^{G,f} B_t^G, \quad \forall f = 1, \dots, F, \quad (D.24)$$

$$(xxv). \quad G A_t = \exp(\mu + \varepsilon_t^A), \quad (D.25)$$

$$(xxvi). \quad \zeta_t^G = \frac{1}{1 + a^G \exp(-u_t^G)}, \quad (D.26)$$

$$(xxvii). \quad \zeta_t^T = \frac{1}{1 + a^T \exp(-u_t^T)}, \quad (D.27)$$

$$(xxviii). \quad u_t^G = \rho_G \cdot u_{t-1}^G + \varepsilon_t^G, \quad (D.28)$$

$$(xxix). \quad u_t^T = \rho_T \cdot u_{t-1}^T + \varepsilon_t^T, \quad (D.29)$$

$$(xxx). \quad Q_{t,t+1} = \frac{\beta}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \frac{\left(\frac{C_t}{A_t \bar{N}_t}\right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}}\right)}, \quad (D.30)$$

$$(xxxi). \quad \tilde{R}_{t+1}^K \equiv \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K], \quad (D.31)$$

$$(xxxii). \quad YD_t^f \equiv \left(Q_t^f\right)^{-1/f}, \quad \forall f, \quad (D.32)$$

$$(xxxiii). \quad \frac{B_t^{CB,f}}{A_t \bar{N}_t P_t} = \frac{\overline{B^{CB,f}}}{\overline{ANP}}, \quad \forall f = 2, \dots, F. \quad (D.33)$$

D.2 Equilibrium Equations: Yield-Curve-Control Policy (YCC)

Summary of relevant equilibrium conditions:

$$(i). \quad \frac{C_t}{A_t \bar{N}_t} = (1 - \zeta_t^G) \left(\frac{Y_t}{A_t \bar{N}_t}\right) + \left(\frac{1 - \delta}{GA_t \cdot GN}\right) \left(\frac{K_t}{A_{t-1} \bar{N}_{t-1}}\right) - \left(\frac{K_{t+1}}{A_t \bar{N}_t}\right), \quad (D.34)$$

$$(ii). \quad 1 = \beta \mathbb{E}_t \left[\frac{R_{t+1}^S}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \frac{\left(\frac{C_t}{A_t \bar{N}_t}\right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}}\right)} \right], \quad (D.35)$$

$$(iii). \quad - \left(\frac{\lambda_t^{CB,1} - \lambda_t^{G,1}}{\lambda_t^{CB,1} - \lambda_t^{HB,1}} \right) \cdot \frac{B_t^G}{A_t \bar{N}_t P_t} = \gamma_L \cdot (1 + \zeta^F) \cdot \left(\frac{1 - \lambda_t^K}{\lambda_t^K} \right) \left(\frac{Y_t}{A_t \bar{N}_t} \right), \quad (D.36)$$

$$(iv). \quad \lambda_t^{CB,f} = \frac{\lambda_t^{HB,f} \cdot \left(1 - \sum_{i \neq \{f,1\}} \lambda_t^{CB,i} - \lambda_t^{G,1}\right) - \lambda_t^{G,f} \cdot \left(1 - \sum_{i \neq \{f,1\}} \lambda_t^{CB,i} - \lambda_t^{HB,1}\right)}{\left(\lambda_t^{HB,1} + \lambda_t^{HB,f}\right) - \left(\lambda_t^{G,1} + \lambda_t^{G,f}\right)}, \quad f = 2, \dots, F, \quad (D.37)$$

$$(v). \quad YD_t^1 = \max \{YD_t^{1*}, 1\}, \quad (D.38)$$

$$(vi). \quad \frac{YD_t^{f*}}{\overline{YD}^f} = \left(\frac{YD_{t-1}^{f*}}{\overline{YD}^f}\right)^{\rho_1} \left(\frac{YD_{t-2}^{f*}}{\overline{YD}^f}\right)^{\rho_2} \left[\left(\frac{\Pi_t}{\bar{\Pi}}\right)^{\gamma_\pi^f} \left(\frac{Y_t}{\bar{Y}_t}\right)^{\gamma_y^f} \exp(\varepsilon_t^{YD^f}) \right]^{1 - (\rho_1 + \rho_2)}, \quad (D.39)$$

$$(vii). \quad YD_t^{YCC,f} = \overline{YD}^f \left(\frac{YD_t^{CP,f}}{\overline{YD}^f}\right)^{\gamma_{CP}^f} \left[\frac{YD_t^{f*}}{\overline{YD}^f}\right]^{1 - \gamma_{CP}^f}, \quad f \geq 2, \quad (D.40)$$

$$(viii). \quad \lambda_t^{HB,f} = \left(\frac{\mathbb{E}_t \left[\frac{\beta z_t^f}{\Pi_{t+1} \cdot G A_{t+1} \cdot GN} \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right) \left(Y D_{t+1}^{f-1} \right)^{-(f-1)}}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right) \left(Y D_t^f \right)^{-f}} \right]}{\Phi_t^B} \right)^{\kappa_B}, \quad \forall f, \quad (\text{D.41})$$

$$(ix). \quad \Phi_t^B = \left[\sum_{j=1}^F \mathbb{E}_t \left[\frac{\beta z_t^j}{\Pi_{t+1} \cdot G A_{t+1} \cdot GN} \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right) \left(Y D_{t+1}^{j-1} \right)^{-(j-1)}}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right) \left(Y D_t^j \right)^{-j}} \right] \right]^{\kappa_B \frac{1}{\kappa_B}}, \quad (\text{D.42})$$

$$(x). \quad \lambda_t^K = \left(\frac{z_t^K \cdot \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K]}{\Phi_t^S} \right)^{\kappa_S}, \quad (\text{D.43})$$

$$(xi). \quad \Phi_t^S = \left[\left(\mathbb{E}_t [Q_{t,t+1} R_{t+1}^{HB}] \right)^{\kappa_S} + \left(z_t^K \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K] \right)^{\kappa_S} \right]^{\frac{1}{\kappa_S}}, \quad (\text{D.44})$$

$$(xii). \quad R_t^j = \sum_{f=0}^{F-1} \lambda_{t-1}^{j,f+1} \frac{\left(Y D_t^f \right)^{-f}}{\left(Y D_{t-1}^{f+1} \right)^{-(f+1)}} \quad j \in \{HB, G, CB\}, \quad (\text{D.45})$$

$$(xiii). \quad R_t^S = (1 - \lambda_{t-1}^K) R_t^{HB} + \lambda_{t-1}^K R_t^K, \quad (\text{D.46})$$

$$(xiv). \quad 1 = \mathbb{E}_t \left[Q_{t,t+1} \Pi_{t+1} \left[(1 - \delta) + \frac{P_{t+1}^K}{P_{t+1}} \right] \right], \quad (\text{D.47})$$

$$(xv). \quad F_t = (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{(1 + \varsigma_F)^{-1} \epsilon}{\epsilon - 1} \right) \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} \left(\frac{Y_t}{A_t \bar{N}_t} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t} \right)^{\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} + \theta \beta \mathbb{E}_t \left[\Pi_{t+1}^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} F_{t+1} \right], \quad (\text{D.48})$$

$$(xvi). \quad H_t = \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-1} \frac{Y_t}{A_t \bar{N}_t} \left[1 - \gamma_L \cdot \left(\tilde{R}_{t+1}^K - 1 \right) \right] + \theta \beta \mathbb{E}_t \left[\Pi_{t+1}^{\epsilon-1} H_{t+1} \right], \quad (\text{D.49})$$

$$(xvii). \quad \frac{F_t}{H_t} = \left(\frac{1 - \theta}{1 - \theta \Pi_t^{\epsilon-1}} \right)^{\left(\frac{1}{\epsilon-1} \right) \left[1 + \epsilon \left(\frac{1-\alpha}{\eta+\alpha} \right) \right]}, \quad (\text{D.50})$$

$$(xviii). \quad \Delta_t = (1 - \theta) \left(\frac{1 - \theta \Pi_t^{\epsilon-1}}{1 - \theta} \right)^{\left(\frac{\epsilon}{\epsilon-1} \right) \left(\frac{\eta+1}{\eta+\alpha} \right)} + \theta \Pi_t^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} \Delta_{t-1}, \quad (\text{D.51})$$

$$(xix). \quad \frac{N_t}{\bar{N}_t} = (1 - \alpha)^{\left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-\alpha \left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{Y_t}{A_t \bar{N}_t} \right)^{\left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{P_t^K}{P_t} \right)^{\alpha \left(\frac{\eta}{\eta+\alpha} \right)} \Delta_t^{\frac{\eta}{\eta+1}}, \quad (\text{D.52})$$

$$(xx). \quad \frac{K_t}{A_{t-1}\bar{N}_{t-1}} = \alpha(1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot GA_t \cdot GN \cdot \left(\frac{C_t}{A_t\bar{N}_t}\right)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} \left(\frac{Y_t}{A_t\bar{N}_t}\right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t}\right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha}\right)} \Delta_t, \quad (\text{D.53})$$

$$(xxi). \quad \frac{B_t^G}{P_t A_t \bar{N}_t} = \frac{R_t^G}{\Pi_t \cdot GA_t \cdot GN} \frac{B_{t-1}^G}{P_{t-1} A_{t-1} \bar{N}_{t-1}} - [\zeta_t^G + \zeta^F - \zeta_t^T] \left(\frac{Y_t}{A_t \bar{N}_t}\right), \quad (\text{D.54})$$

$$(xxii). \quad \lambda_t^{G,1} = \frac{1}{1 + \sum_{l=2}^F a^{B,l} \exp(\tilde{u}_t^{B,l})}, \quad \lambda_t^{G,f} = \frac{a^{B,f} \exp(\tilde{u}_t^{B,f})}{1 + \sum_{l=2}^F a^{B,l} \exp(\tilde{u}_t^{B,l})}, \quad \forall f > 1, \quad (\text{D.55})$$

$$(xxiii). \quad \tilde{u}_t^{B,f} = \sum_{j=2}^J \tau_{fj}^B u_t^{B,j}, \quad (\text{D.56})$$

$$(xxiv). \quad u_t^{B,j} = \rho_{B,j} u_{t-1}^{B,j} + \varepsilon_t^{B,j}, \quad \forall j = 1, \dots, J \quad (\text{D.57})$$

$$(xxv). \quad B_t^{G,f} = \lambda_t^{G,f} B_t^G, \quad \forall f = 1, \dots, F, \quad (\text{D.58})$$

$$(xxvi). \quad \hat{\varepsilon}_t^{YD,f} = \sum_{l=1}^L \tau_{fl}^{YD} \varepsilon_t^{YD,l}, \quad (\text{D.59})$$

$$(xxvii). \quad Q_{t,t+1} = \frac{\beta}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \frac{\left(\frac{C_t}{A_t \bar{N}_t}\right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}}\right)}, \quad (\text{D.60})$$

$$(xxviii). \quad \tilde{R}_{t+1}^K \equiv \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K], \quad (\text{D.61})$$

$$(xxix). \quad YD_t^f \equiv (Q_t^f)^{-1/f}, \quad \forall f. \quad (\text{D.62})$$

D.3 Summary of Conventional Policy Linearized Equations

Those are the essential equations to solve the model, other variables can be found on the equations above. The composite coefficients used below are introduced earlier in this appendix: $\Theta^{14}-\Theta^{20}$ in equation (A.83), Ξ in equation (A.52), $\Psi^{G,4}-\Psi^{G,6}$ in equation (A.58), $\Psi^{16}-\Psi^{24}$ in equation (A.68), $\Psi^{26}-\Psi^{36}$ in equation (A.69), $\Psi^{37}-\Psi^{45}$ in equation (A.74), and $\Psi^{46}-\Psi^{55}$ in equation (A.75).

$$(i). \quad \hat{y}_t = \mathbb{E}_t \left[\hat{y}_{t+1} + \Psi^{46} \cdot \hat{\pi}_{t+1} - \Psi^{47} \cdot \vec{\hat{z}}_t - \Psi^{48} \cdot \hat{z}_t^K - \Psi^{49} \cdot \vec{\hat{y}}_t^d - \Psi^{50} \cdot \mathbb{E}_t \left[\vec{\hat{y}}_{t+1}^d \right] \right. \\ \left. - \Psi^{51} \cdot \hat{r}_{t+1}^K - \Psi^{52} \cdot (\hat{k}_t - \hat{\varepsilon}_t^A) + \Psi^{53} \cdot \hat{k}_{t+1} - \Psi^{54} \cdot \hat{k}_{t+2} + \Psi^{55} \cdot \hat{u}_t^G \right] \quad (\text{D.63})$$

$$(ii). \quad \overrightarrow{\hat{y}d_t} = \Theta^{14} \cdot \hat{b}_t^G - \Theta^{15} \cdot \hat{y}_t + \Theta^{16} \cdot \hat{r}_{t+1}^K - \Theta^{17} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}} \right] - \Theta^{18} \cdot \overrightarrow{\hat{z}_t} + \Theta^{19} \cdot \hat{z}_t^K + \Theta^{20} \cdot \overrightarrow{\hat{u}_t^B} \quad (D.64)$$

$$(iii). \quad \hat{y}d_t^1 = \max \left\{ \hat{y}d_t^{1*}, 0 \right\} \quad (D.65)$$

$$(iv). \quad \hat{y}d_t^{1*} = \rho_1 \hat{y}d_{t-1}^{1*} + \rho_2 \hat{y}d_{t-2}^{1*} + (1 - \rho_1 - \rho_2) \left[\gamma_\pi \hat{\pi}_t + \gamma_y \hat{y}_t + \hat{\varepsilon}_t^{YD^1} \right] \quad (D.66)$$

$$(v). \quad \hat{\varepsilon}_t^{YD^f} = \sum_{l=1}^L \tau_{f,l}^{YD} \varepsilon_t^{YD^l}, \quad \forall f = 1, \dots, F \quad (D.67)$$

$$(vi). \quad \begin{aligned} \hat{r}_{t+1}^K = & -\Psi^{37} \cdot \overrightarrow{\hat{z}_t} - \Psi^{38} \cdot \hat{z}_t^K - \Psi^{39} \cdot \overrightarrow{\hat{y}d_t} - \Psi^{40} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}} \right] + \Psi^{41} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{42} \cdot \mathbb{E}_t [\hat{y}_{t+1}] \\ & + \Psi^{43} \cdot \hat{k}_{t+1} - \Psi^{44} \cdot \mathbb{E}_t [\hat{k}_{t+2}] - \Psi^{45} \cdot \hat{u}_t^G \end{aligned} \quad (D.68)$$

$$(vii). \quad \begin{aligned} \hat{b}_t^G = & \frac{R^G}{\Pi \cdot GA \cdot GN} \cdot \left[\Psi^{G,4} \overrightarrow{\hat{u}_{t-1}^B} - \Psi^{G,5} \overrightarrow{\hat{y}d_t} + \Psi^{G,6} \overrightarrow{\hat{y}d_{t-1}} - \hat{\pi}_t - \hat{\varepsilon}_t^A + \hat{b}_{t-1}^G \right] \\ & + \left(1 - \frac{R^G}{\Pi \cdot GA \cdot GN} \right) \left[\hat{y}_t + \left(\frac{\zeta^G}{\zeta^G + \zeta^F - \zeta^T} \right) \frac{a^G}{1 + a^G} \hat{u}_t^G - \left(\frac{\zeta^T}{\zeta^G + \zeta^F - \zeta^T} \right) \frac{a^T}{1 + a^T} \hat{u}_t^T \right] \end{aligned} \quad (D.69)$$

$$(viii). \quad \begin{aligned} \hat{f}_t = & -\Psi^{16} \cdot \overrightarrow{\hat{z}_t} - \Psi^{17} \cdot \hat{z}_t^K - \Psi^{18} \cdot \left[\hat{k}_t - \hat{\varepsilon}_t^A \right] + \Psi^{19} \cdot \hat{y}_t - \Psi^{20} \cdot \overrightarrow{\hat{y}d_t} - \Psi^{21} \cdot \hat{r}_{t+1}^K - \Psi^{22} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}} \right] \\ & + \Psi^{23} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{24} \cdot \mathbb{E}_t [\hat{f}_{t+1}] \end{aligned} \quad (D.70)$$

$$(ix). \quad \begin{aligned} \hat{h}_t = & \Psi^{26} \cdot \overrightarrow{\hat{z}_t} + \Psi^{27} \cdot \hat{z}_t^K - \Psi^{28} \cdot \left[\hat{k}_t - \hat{\varepsilon}_t^A \right] + \Psi^{29} \cdot \hat{u}_t^G + \Psi^{30} \cdot \hat{y}_t + \Psi^{31} \cdot \overrightarrow{\hat{y}d_t} - \Psi^{32} \cdot \hat{r}_{t+1}^K \\ & + \Psi^{33} \cdot \hat{k}_{t+1} + \Psi^{34} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}} \right] + \Psi^{35} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{36} \cdot \mathbb{E}_t [\hat{h}_{t+1}] \end{aligned} \quad (D.71)$$

$$(x). \quad \hat{f}_t - \hat{h}_t = \left[1 + \epsilon \left(\frac{1 - \alpha}{\eta + \alpha} \right) \right] \left(\frac{\theta \Pi^{\epsilon-1}}{1 - \theta \Pi^{\epsilon-1}} \right) \hat{\pi}_t \quad (D.72)$$

$$(xi). \quad u_t^{B,j} = \rho_{B,j} \cdot u_{t-1}^{B,j} + \varepsilon_t^{B,j}, \quad \forall j = 1, \dots, J \quad (D.73)$$

$$(xii). \quad u_t^G = \rho_G \cdot u_{t-1}^G + \varepsilon_t^G \quad (D.74)$$

$$(xiii). \quad u_t^T = \rho_T \cdot u_{t-1}^T + \varepsilon_t^T \quad (D.75)$$

$$(xiv). \quad \overrightarrow{\hat{z}_t} = \rho_z \overrightarrow{\hat{z}_{t-1}} + \hat{\varepsilon}_t^z \quad (D.76)$$

$$(xv). \quad \hat{z}_t^K = \rho_z^K \hat{z}_{t-1}^K + \hat{\varepsilon}_t^{z,K} \quad (D.77)$$

$$(xvi). \quad \hat{c}_t = \left[(1 - \zeta^G) \frac{Y}{C} \right] \left[\hat{y}_t - \frac{1}{1 + a^G} \hat{u}_t^G \right] + \left[\frac{1 - \delta}{GA \cdot GN} \frac{K}{C} \right] \left(\hat{k}_t - \hat{\varepsilon}_t^A \right) - \frac{K}{C} \hat{k}_{t+1} \quad (D.78)$$

$$(xvii). \quad \hat{n}_t = \left(\frac{\eta}{\eta + \alpha} \right) \left[1 + \left(\frac{\alpha}{1 - \alpha} \right) \left(\frac{\eta + 1}{\eta} \right) \right] \hat{y}_t - \left(\frac{\alpha}{1 - \alpha} \right) [\hat{k}_t - \hat{\varepsilon}_t^A] \quad (D.79)$$

$$(xviii). \quad \hat{\Delta}_t = 0 \quad (D.80)$$

Note that price dispersion has no first-order effect around the symmetric steady state, hence (D.80). Lead identities such as $\tilde{R}_{t+1}^K \equiv \mathbb{E}_t[Q_{t,t+1}R_{t+1}^K]$ and growth definitions $GA_{t+1} \equiv A_{t+1}/A_t$ apply at all dates and are inherited from the non-linear block. The vector notation for bond-preference shocks is kept for the general derivation. In the numerical simulations and determinacy checks, only the first bond-preference component is active, which is equivalent to setting the volatility of the remaining bond-preference components to zero.

D.4 Summary of Yield-Curve Control Policy Linearized Equations

Those are the essential equations to solve the model, other variables can be found on the equations above. The composite coefficients used below are introduced earlier in this appendix: $\Theta^{14}-\Theta^{20}$ in equation (A.83), Ξ in equation (A.52), $\Psi^{G,4}-\Psi^{G,6}$ in equation (A.58), $\Psi^{16}-\Psi^{24}$ in equation (A.68), $\Psi^{26}-\Psi^{36}$ in equation (A.69), $\Psi^{37}-\Psi^{45}$ in equation (A.74), and $\Psi^{46}-\Psi^{55}$ in equation (A.75).

$$(i). \quad \begin{aligned} \hat{y}_t = \mathbb{E}_t \Big[& \hat{y}_{t+1} + \Psi^{46} \cdot \hat{\pi}_{t+1} - \Psi^{47} \cdot \vec{\hat{z}}_t - \Psi^{48} \cdot \hat{z}_t^K - \Psi^{49} \cdot \overrightarrow{\hat{y}d_t^{YCC}} - \Psi^{50} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}^{YCC}} \right] \\ & - \Psi^{51} \cdot \hat{r}_{t+1}^K - \Psi^{52} \cdot (\hat{k}_t - \hat{\varepsilon}_t^A) + \Psi^{53} \cdot \hat{k}_{t+1} - \Psi^{54} \cdot \hat{k}_{t+2} + \Psi^{55} \cdot \hat{u}_t^G \Big] \end{aligned} \quad (D.81)$$

$$(ii). \quad \overrightarrow{\hat{y}d_t^{CP}} = \Theta^{14} \cdot \hat{b}_t^G - \Theta^{15} \cdot \hat{y}_t + \Theta^{16} \cdot \hat{r}_{t+1}^K - \Theta^{17} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}^{YCC}} \right] - \Theta^{18} \cdot \vec{\hat{z}}_t + \Theta^{19} \cdot \hat{z}_t^K + \Theta^{20} \cdot \overrightarrow{\hat{u}_t^B} \quad (D.82)$$

$$(iii). \quad \hat{y}d_t^{YCC,1} = \max \left\{ \hat{y}d_t^{1*}, 0 \right\} = \hat{y}d_t^{CP,1} \quad (D.83)$$

$$(iv). \quad \hat{y}d_t^{1*} = \rho_1 \hat{y}d_{t-1}^{1*} + \rho_2 \hat{y}d_{t-2}^{1*} + (1 - \rho_1 - \rho_2) \left[\gamma_\pi \hat{\pi}_t + \gamma_y \hat{y}_t + \tilde{\varepsilon}_t^{YD^1} \right] \quad (D.84)$$

$$(v). \quad \hat{y}d_t^{YCC,f} = \gamma_{CP}^f \hat{y}d_t^{CP,f} + \left(1 - \gamma_{CP}^f \right) \hat{y}d_t^{*,f}, \quad \forall f \geq 2 \quad (D.85)$$

$$(vi). \quad \hat{y}d_t^{*,f} = \rho_1 \hat{y}d_{t-1}^{*,f} + \rho_2 \hat{y}d_{t-2}^{*,f} + (1 - \rho_1 - \rho_2) \left[\gamma_\pi^f \hat{\pi}_t + \gamma_y^f \hat{y}_t + \tilde{\varepsilon}_t^{YD^f} \right], \quad f \geq 2 \quad (D.86)$$

$$(vii). \quad \tilde{\varepsilon}_t^{YD^f} = \sum_{l=1}^L \tau_{f,l}^{YD} \varepsilon_t^{YD^l}, \quad \forall f = 1, \dots, F \quad (D.87)$$

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$$\begin{aligned}
 (viii). \quad \hat{r}_{t+1}^K = & -\Psi^{37} \cdot \vec{\hat{z}}_t - \Psi^{38} \cdot \hat{z}_t^K - \Psi^{39} \cdot \overrightarrow{\hat{y}d_t^{YCC}} - \Psi^{40} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}^{YCC}} \right] + \Psi^{41} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{42} \cdot \mathbb{E}_t [\hat{y}_{t+1}] \\
 & + \Psi^{43} \cdot \hat{k}_{t+1} - \Psi^{44} \cdot \mathbb{E}_t [\hat{k}_{t+2}] - \Psi^{45} \cdot \hat{u}_t^G
 \end{aligned} \tag{D.88}$$

$$\begin{aligned}
 (ix). \quad \hat{b}_t^G = & \frac{R^G}{\Pi \cdot GA \cdot GN} \cdot \left[\Psi^{G,4} \overrightarrow{\hat{u}_{t-1}^B} - \Psi^{G,5} \overrightarrow{\hat{y}d_t^{YCC}} + \Psi^{G,6} \overrightarrow{\hat{y}d_{t-1}^{YCC}} - \hat{\pi}_t - \hat{\varepsilon}_t^A + \hat{b}_{t-1}^G \right] \\
 & + \left(1 - \frac{R^G}{\Pi \cdot GA \cdot GN} \right) \left[\hat{y}_t + \left(\frac{\zeta^G}{\zeta^G + \zeta^F - \zeta^T} \right) \frac{a^G}{1 + a^G} \hat{u}_t^G - \left(\frac{\zeta^T}{\zeta^G + \zeta^F - \zeta^T} \right) \frac{a^T}{1 + a^T} \hat{u}_t^T \right]
 \end{aligned} \tag{D.89}$$

$$\begin{aligned}
 (x). \quad \hat{f}_t = & -\Psi^{16} \cdot \vec{\hat{z}}_t - \Psi^{17} \cdot \hat{z}_t^K - \Psi^{18} \cdot [\hat{k}_t - \hat{\varepsilon}_t^A] + \Psi^{19} \cdot \hat{y}_t - \Psi^{20} \cdot \overrightarrow{\hat{y}d_t^{YCC}} - \Psi^{21} \cdot \hat{r}_{t+1}^K - \Psi^{22} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}^{YCC}} \right] \\
 & + \Psi^{23} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{24} \cdot \mathbb{E}_t [\hat{f}_{t+1}]
 \end{aligned} \tag{D.90}$$

$$\begin{aligned}
 (xi). \quad \hat{h}_t = & \Psi^{26} \cdot \vec{\hat{z}}_t + \Psi^{27} \cdot \hat{z}_t^K - \Psi^{28} \cdot [\hat{k}_t - \hat{\varepsilon}_t^A] + \Psi^{29} \cdot \hat{u}_t^G + \Psi^{30} \cdot \hat{y}_t + \Psi^{31} \cdot \overrightarrow{\hat{y}d_t^{YCC}} - \Psi^{32} \cdot \hat{r}_{t+1}^K + \Psi^{33} \cdot \hat{k}_{t+1} \\
 & + \Psi^{34} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}^{YCC}} \right] + \Psi^{35} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{36} \cdot \mathbb{E}_t [\hat{h}_{t+1}]
 \end{aligned} \tag{D.91}$$

$$(xii). \quad \hat{f}_t - \hat{h}_t = \left[1 + \epsilon \left(\frac{1 - \alpha}{\eta + \alpha} \right) \right] \left(\frac{\theta \Pi^{\epsilon-1}}{1 - \theta \Pi^{\epsilon-1}} \right) \hat{\pi}_t \tag{D.92}$$

$$(xiii). \quad u_t^{B,j} = \rho_{B,j} \cdot u_{t-1}^{B,j} + \varepsilon_t^{B,j}, \quad \forall j = 1, \dots, J \tag{D.93}$$

$$(xiv). \quad u_t^G = \rho_G \cdot u_{t-1}^G + \varepsilon_t^G \tag{D.94}$$

$$(xv). \quad u_t^T = \rho_T \cdot u_{t-1}^T + \varepsilon_t^T \tag{D.95}$$

$$(xvi). \quad \vec{\hat{z}}_t = \rho_z \vec{\hat{z}}_{t-1} + \vec{\hat{\varepsilon}}_t^z \tag{D.96}$$

$$(xvii). \quad \hat{z}_t^K = \rho_z^K \hat{z}_{t-1}^K + \hat{\varepsilon}_t^{z,K} \tag{D.97}$$

$$(xviii). \quad \hat{c}_t = \left[(1 - \zeta^G) \frac{Y}{C} \right] \left[\hat{y}_t - \frac{1}{1 + a^G} \hat{u}_t^G \right] + \left[\frac{1 - \delta}{GA \cdot GN} \frac{K}{C} \right] (\hat{k}_t - \hat{\varepsilon}_t^A) - \frac{K}{C} \hat{k}_{t+1} \tag{D.98}$$

$$(xix). \quad \hat{n}_t = \left(\frac{\eta}{\eta + \alpha} \right) \left[1 + \left(\frac{\alpha}{1 - \alpha} \right) \left(\frac{\eta + 1}{\eta} \right) \right] \hat{y}_t - \left(\frac{\alpha}{1 - \alpha} \right) [\hat{k}_t - \hat{\varepsilon}_t^A] \tag{D.99}$$

$$(xx). \quad \hat{\Delta}_t = 0 \tag{D.100}$$

Note that price dispersion has no first-order effect around the symmetric steady state, hence (D.100). Lead identities such as $\tilde{R}_{t+1}^K \equiv \mathbb{E}_t[Q_{t,t+1} R_{t+1}^K]$ and growth definitions $GA_{t+1} \equiv A_{t+1}/A_t$ apply at all dates and are inherited from the non-linear block. The vector notation for bond-preference shocks is kept for the general derivation. In the numerical

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simulations and determinacy checks, only the first bond-preference component is active, which is equivalent to setting the volatility of the remaining bond-preference components to zero.

E Additional Results: Steady-State Comparative Statics

Comparative statics with κ_B Figure E.2 illustrates how the steady-state yield curve responds as κ_B increases. When κ_B is higher, household demand for bonds across maturities becomes more elastic, which results in a flatter yield curve. When $\kappa_B \rightarrow \infty$, we revert to the expectations hypothesis case, resulting in a flat yield curve in the steady-state.

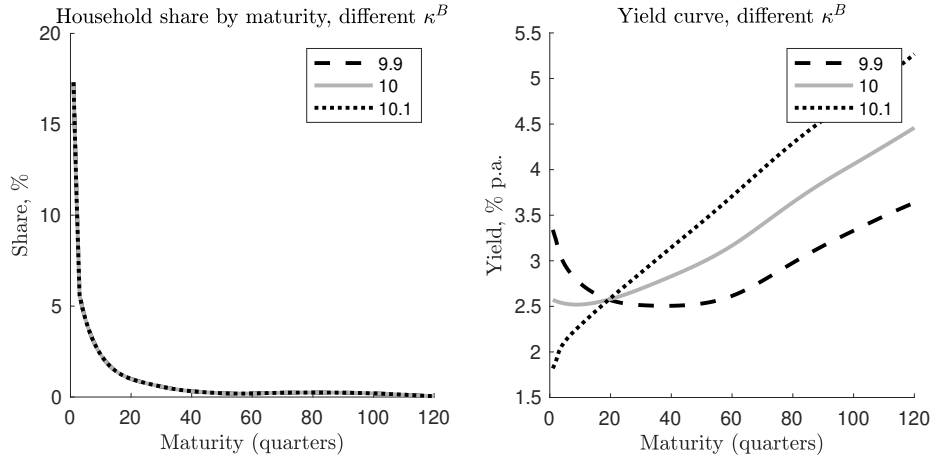


Figure E.2: Variations in κ_B : as $\kappa_B \rightarrow \infty$, the model converges to the expectation hypothesis, wherein all discounted expected returns become equalized.

Comparative statics with z^K Figure E.3 presents comparative statics with respect to z^K , which affects the allocation of savings between the bond and loan markets. Holding the calibrated $\{z^f\}_{f=1}^F$ fixed and varying $z^K \in [0.8, 1.3]$, a higher z^K indicates that households supply more loans to firms rather than invest in the bond market, thereby increasing λ^K . This reallocation raises steady-state capital, output, and consumption.

As households tilt their portfolios toward loans, the average marginal propensity to consume (MPC) and the equilibrium loan rate R^K both decline. Through portfolio rebalancing, this shifts the entire yield curve downward and, somewhat paradoxically, increases credit spreads. At the same time, the fall in R^G reduces the ratio of government bond holdings to GDP.

Comparative statics with κ_S Figure E.4 presents comparative statics for the shape parameter κ_S , which affects the savings allocation condition between the bond and loan markets. Holding the calibrated values $\{z^f\}_{f=1}^F$ and z^K fixed, and varying κ_S over the range

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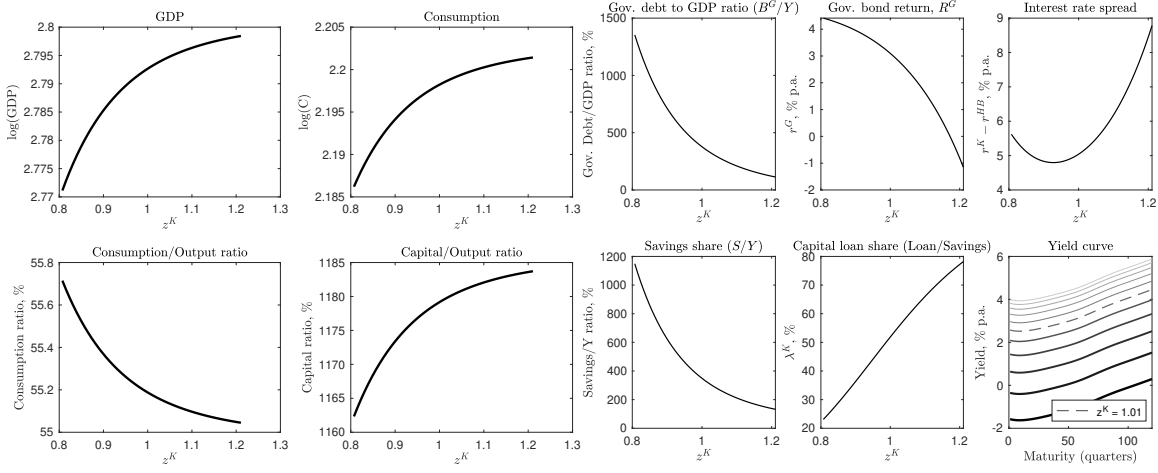


Figure E.3: Variations in z^K

$[0, 25]$, we find that an increase in κ_S leads to a higher λ^K . Intuitively, since $R^K > R^{HB}$ in the steady state, a larger κ_S makes households' demand for loans more elastic.

This heightened cross-substitution towards loans pushes down the loan interest rate R^K . As borrowing becomes cheaper, firms expand capital demand, which in turn raises output and consumption. At the same time, the average marginal propensity to consume declines as before. Credit spreads widen, and the fall in R^K also reduces the government's effective bond return R^G via portfolio rebalancing, pushing down the entire yield curve further. As a result, the government's debt-to-GDP ratio decreases.

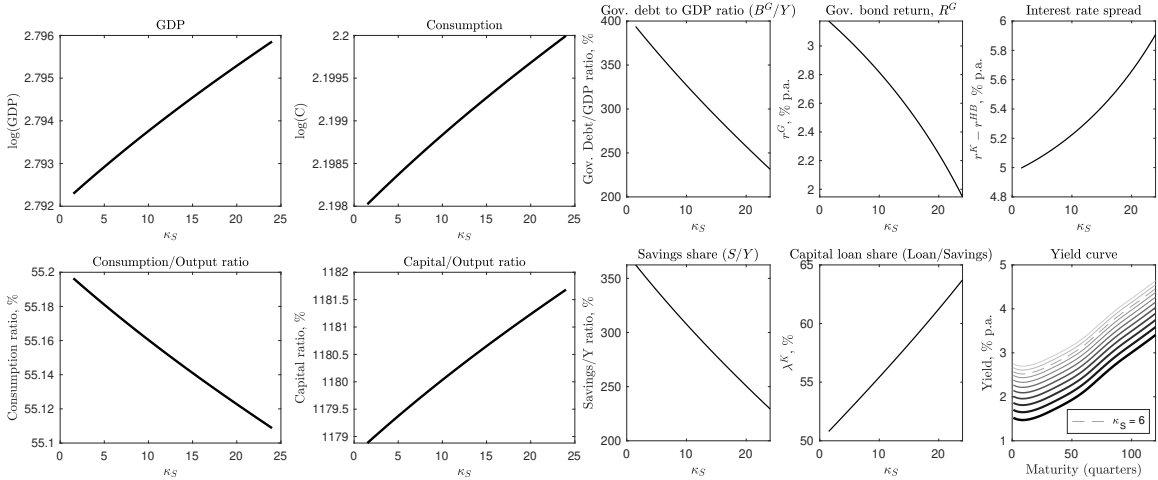


Figure E.4: Variations in κ^S

F Additional Results: Impulse-Responses

F.1 Without ZLB

Technology shock, ε_t^A Figure F.5 displays the impulse-responses to a ε_t^A shock. A positive shock in technology growth GA_t yields similar effects as documented in prior literature under conventional policy,⁹ with output increasing¹⁰ as inflation decreases. Under the yield-curve-control regime, the normalized output decreases less: since inflation falls, bond yields shift downwards, boosting consumption and reducing both the return on capital and the wage.

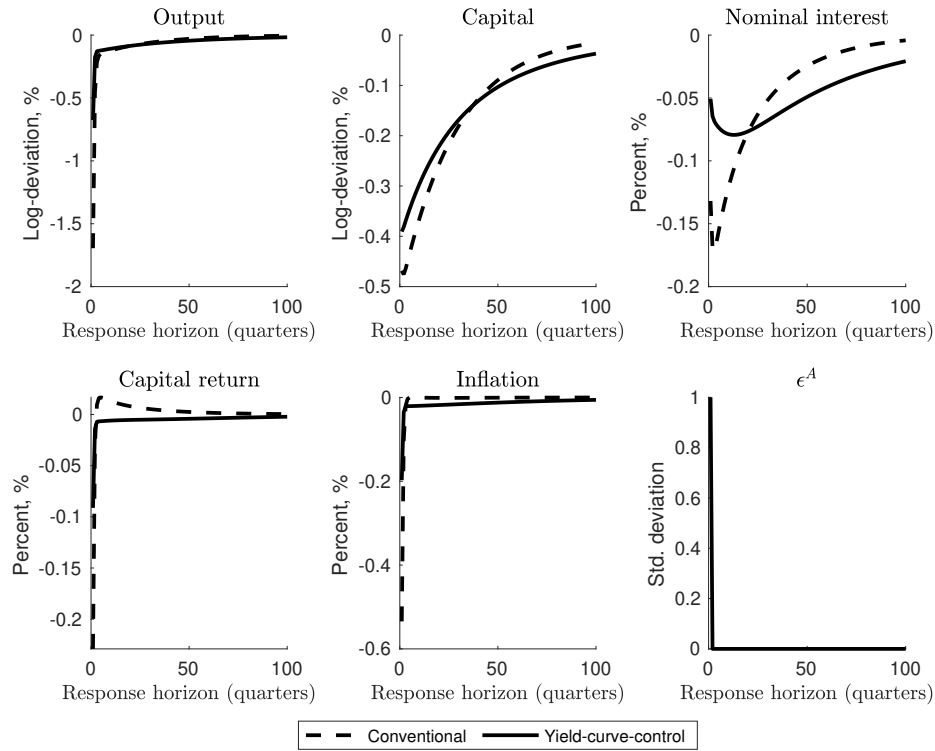


Figure F.5: Impulse response to a ε_t^A shock: A positive technology growth shock generates effects in line with the existing literature, resulting in an increase in output and a decline in inflation. As inflation decreases, all yields undergo a downward shift, which in turn lowers both the capital return and wage relative to the conventional scenario.

⁹For the effects of technology shocks in a standard New Keynesian framework, see e.g., Ireland (2004).

¹⁰Even if the normalized—by trend technology—output drops under our calibration, the output level rises.

Monetary policy shock, $\varepsilon_t^{YD^1}$ Figure F.6 presents the impulse-response to a monetary policy shock $\varepsilon_t^{YD^1}$. Under conventional policy, a contractionary policy shock paradoxically results in an increase in output, inflation, and capital. As explained in Section 2, a contractionary monetary policy shock induces households to reallocate their portfolios toward short-term bonds. As a result, the effective savings rate may even decline, potentially leading to an expansion in aggregate demand and output. At the same time, as the aggregate bond return R^B decreases, households shift more strongly toward loans, prompting more capital accumulation and output increase.

The yield-curve-control policy insulates the economy as in the previous cases: in response to the shock, the central bank moves up the entire yield curve, mitigating the degree of portfolio rebalancing and raising the effective monetary responsiveness. It prevents input prices—loan rates and wages—from rising too high, and thereby mitigates changes in output and inflation.

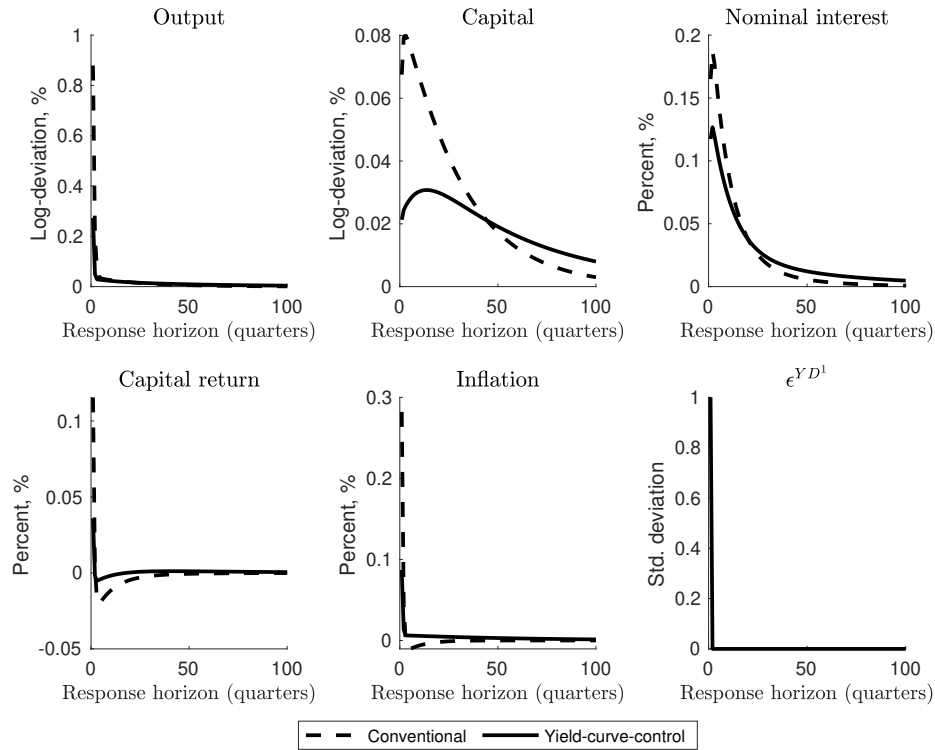


Figure F.6: Impulse response to a $\varepsilon_t^{YD^1}$ shock: A contractionary monetary policy shock paradoxically raises output, inflation, and capital due to household portfolio reallocation towards short-term bonds. The yield-curve-control policy better insulates the economy.

F.2 With ZLB

With mixed policy, z_t^K : In Figure F.7, the mixed policy actually features the longest ZLB spell.

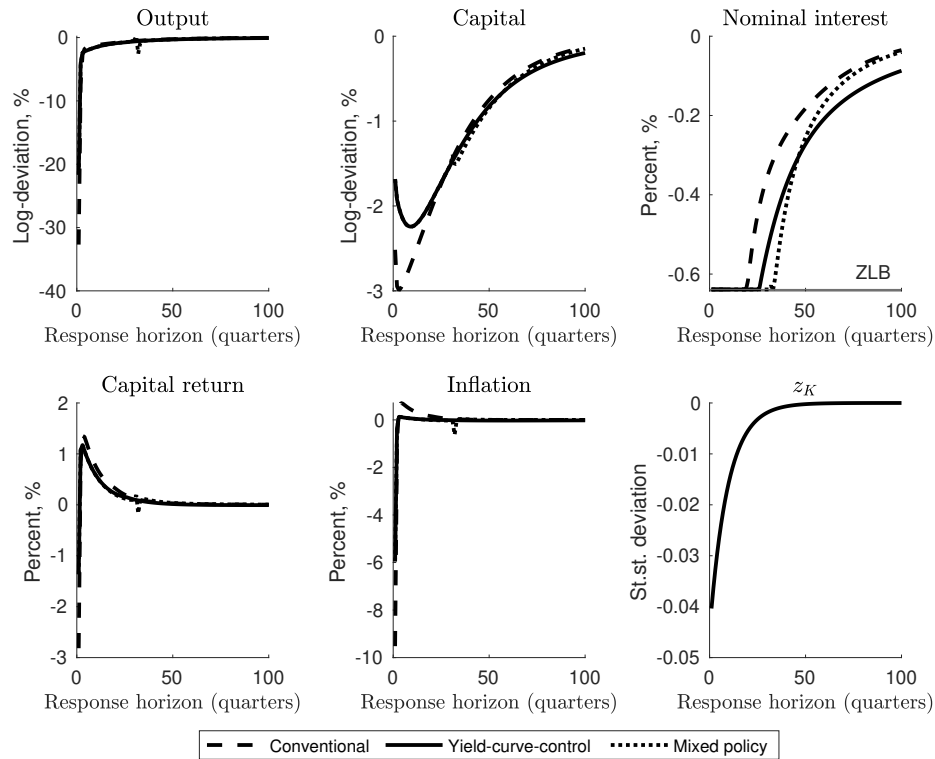


Figure F.7: Impulse-response to z^K shock with ZLB and mixed policy

G Additional Results: Model Determinacy

Under the conventional policy (37) and the yield-curve control policy (38), we derive conditions for model determinacy in terms of the Taylor coefficients— (γ_π, γ_y) for the conventional policy and $(\{\gamma_\pi^f\}_{f=1}^F, \{\gamma_y^f\}_{f=1}^F)$ for the yield-curve control policy. For illustration, we assume that $\gamma_\pi^f = \gamma_\pi$ and $\gamma_y^f = \gamma_y$ for all $f = 1, \dots, F$, which can be easily generalized.

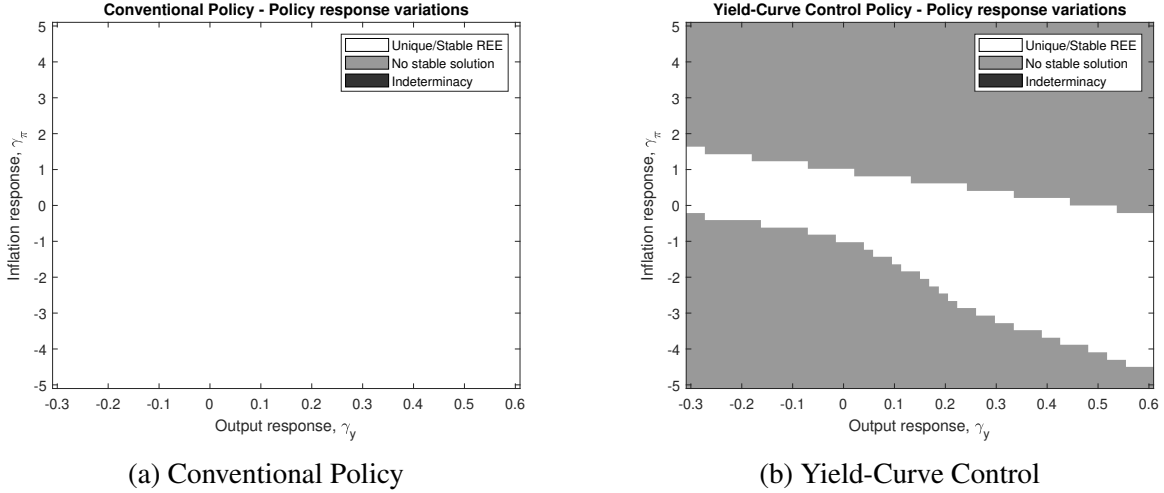


Figure G.8: Determinacy Regions

Figures G.8a and G.8b show the determinacy regions in the (γ_π, γ_y) plane for the conventional and yield-curve control policies, respectively. First, under the conventional policy, determinacy is guaranteed for most values of the Taylor coefficients.¹¹ When the short-term policy rate increases in response to inflation, households reallocate their portfolios toward short-term bonds, reducing the aggregate bond return R^B . At the same time, households shift more strongly toward loans, raising the effective savings rate. As a result, the overall transmission of monetary policy is amplified, making the effective Taylor coefficient substantially larger.

In contrast, the yield-curve control policy's determinacy depends on both coefficients. Notably, when output targeting is zero ($\gamma_y = 0$), inflation-targeting cannot be too aggressive in order for the model to yield a unique, stable equilibrium under certain calibrations. These findings differ from those of the traditional New Keynesian model, which typically requires

¹¹Note that it deviates from the traditional New Keynesian model, which typically requires an aggressive inflation-targeting response (or a combination of responses to inflation and output) to ensure a unique, stable equilibrium. It is from that our model introduces a portfolio rebalancing channel on the households' side and an input substitution channel on the firms' side.

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an aggressive inflation-targeting response (or a combination of responses to inflation and output) to ensure a unique and stable equilibrium. Again, this difference arises from the portfolio rebalancing channel on the households' side and the presence of physical capital: with a very high inflation-targeting response γ_π , a negative inflation shock leads the central bank to push down the entire yield curve, prompting households to reallocate funds toward loans and increasing the effective savings rate, which in turn dampens aggregate demand and inflation, causing a negative feedback loop and instability.

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